A GRADUATE LECTURE

Concept Bottleneck Models

From black-box predictions to interpretable, editable concepts

Machine Learning Concepts

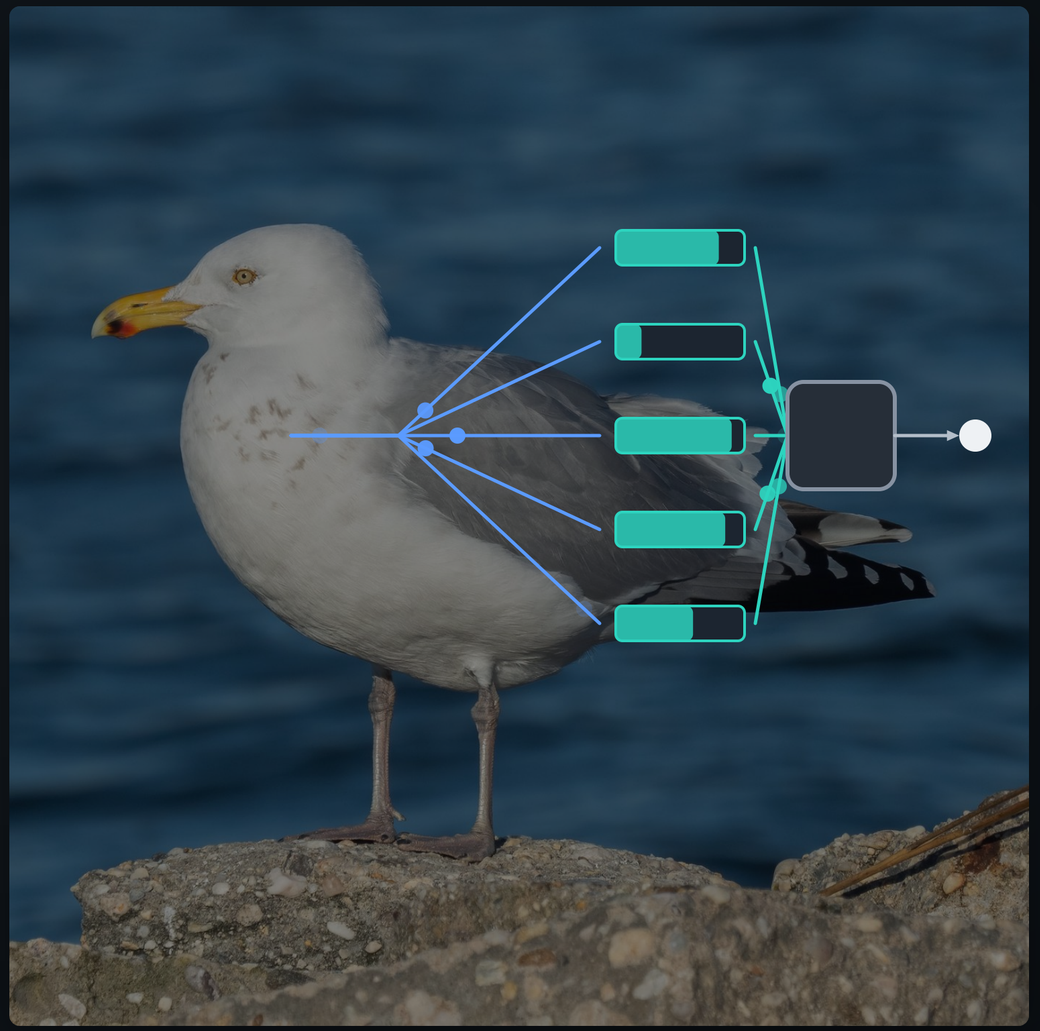
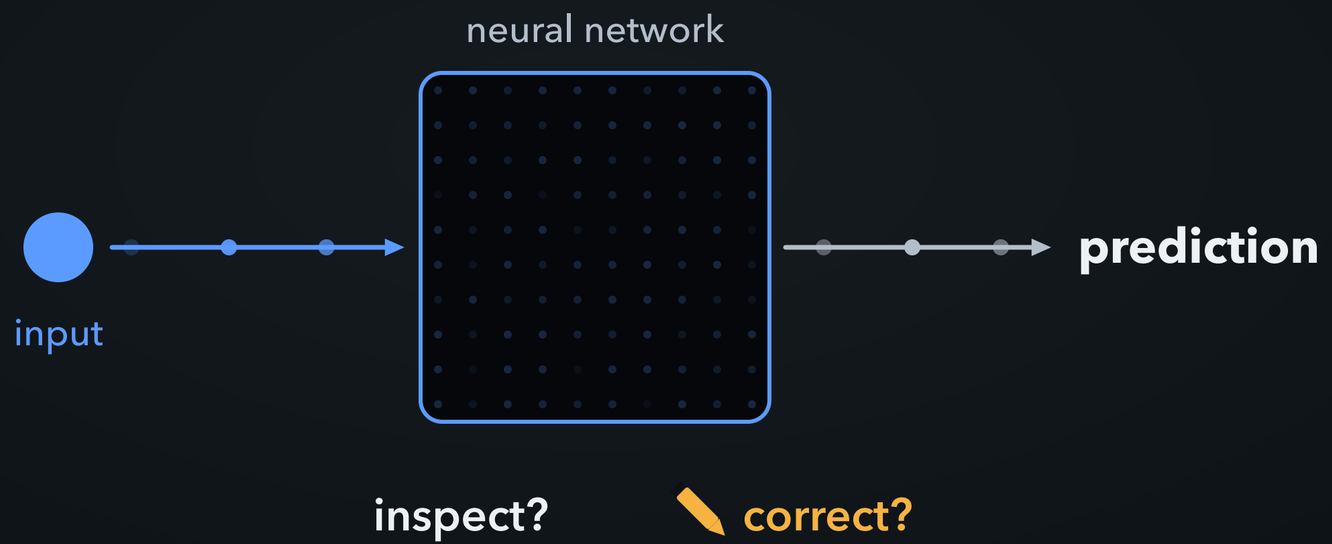


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

PART 1

The black box

Can a person inspect, and correct, the evidence a network used?



The black box

A real photograph and a real model, CLIP

Contrastive Language-Image Pre-training (CLIP)

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

a real, trained model

COMPUTED · REAL MODEL · CLIP ViT-B/32, zero-shot over 6 species names

The black box

CLIP compares the photo with one sentence per species

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

"a photo of a **{species}**, a type of bird."

a photo of a **American herring gull**, a type of bird.

a photo of a **glaucous-winged gull**, a type of bird.

a photo of a **ring-billed gull**, a type of bird.

a photo of a **great black-backed gull**, a type of bird.

a photo of a **laughing gull**, a type of bird.

a photo of a **glaucous gull**, a type of bird.

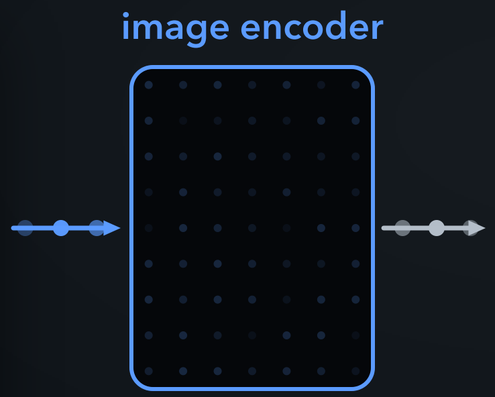
the image is compared with each sentence

PART 1

The black box

The photo goes in, a score for every species comes out

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)



COMPUTED · REAL MODEL · CLIP ViT-B/32, softmax over 6 species names

PART 1

The black box

The answer is right. What made the model prefer it?

image encoder

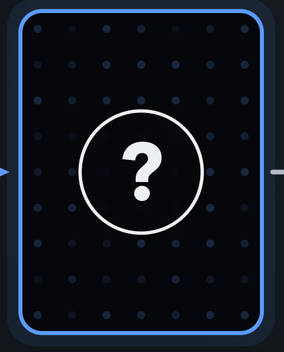


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

COMPUTED · REAL MODEL · CLIP ViT-B/32, softmax over 6 species names

The black box

Occlusion: grey out one patch, measure the score drop

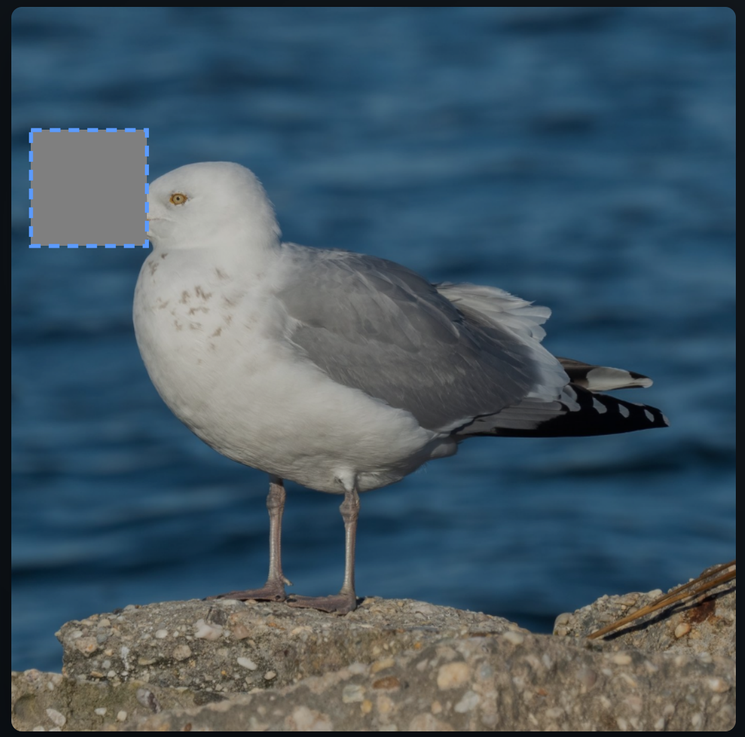


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

grey out one patch, run the model again

drop in the **herring gull** log-probability

0.92

COMPUTED · REAL MODEL · CLIP, one grey patch at each of 14 x 14 positions

The black box

The largest drops sit on the bill and the head

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

brighter: a larger drop in the **score**

`COMPUTED · REAL MODEL · occlusion sensitivity, 14 x 14 grey patches`

The black box

Which property did the model recognise there?

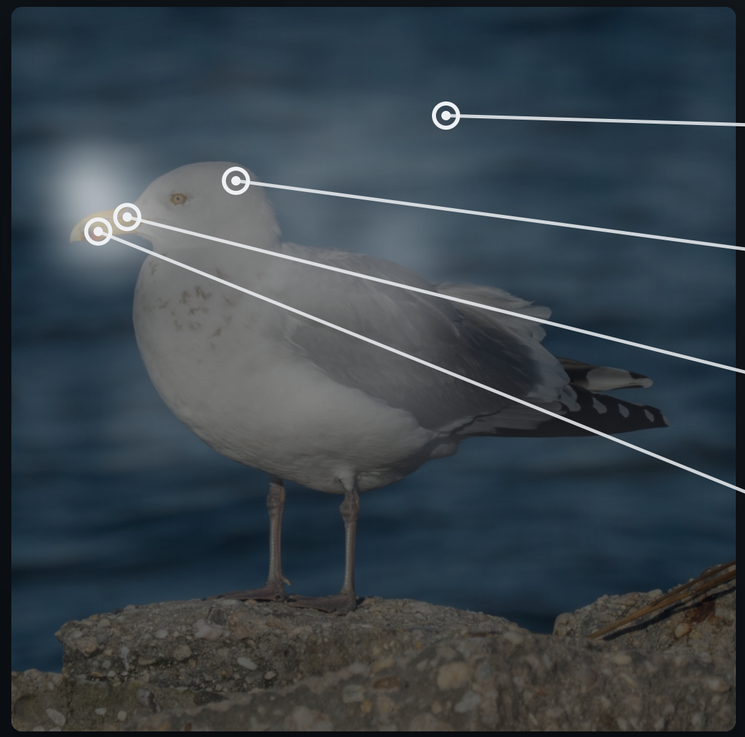


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

The black box

The map shows where the score is sensitive, not which property

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

- ✓ where the **score** is sensitive
- ✗ which property the model detected
- ✗ what a person could correct

The black box

What we want: evidence stated as named properties

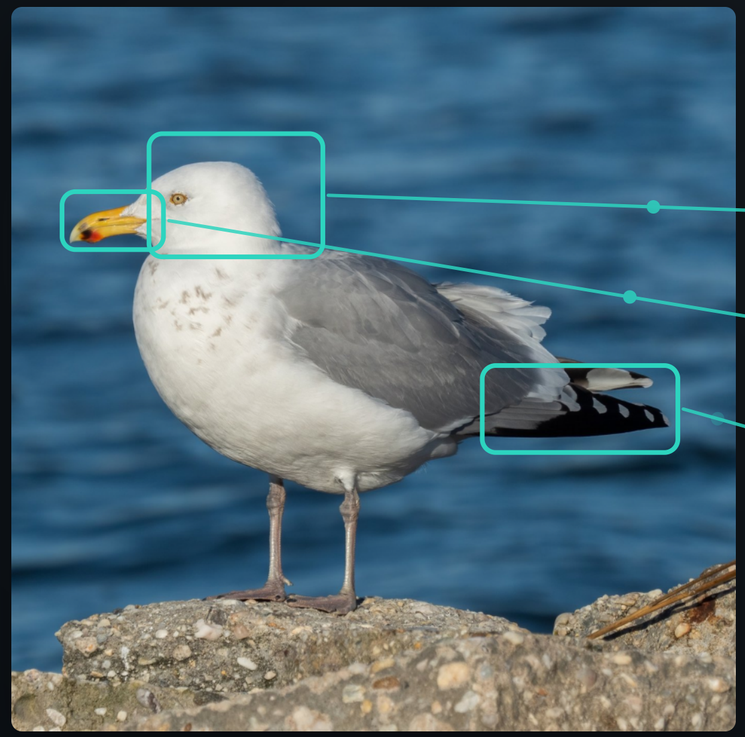


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

named properties

- head colour
- bill shape
- wingtip colour

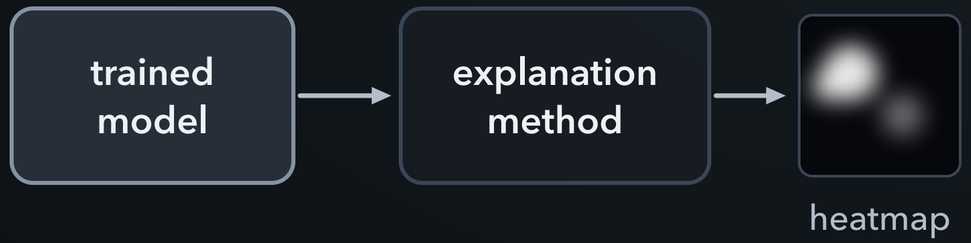
each claim can be checked

Post-hoc or interpretable by design?

Two routes: explain a model afterwards, or build one you can read

post-hoc

explain it afterwards



ante-hoc

interpretable by design

Post-hoc or interpretable by design?

Post-hoc: a method explains a trained model on one input

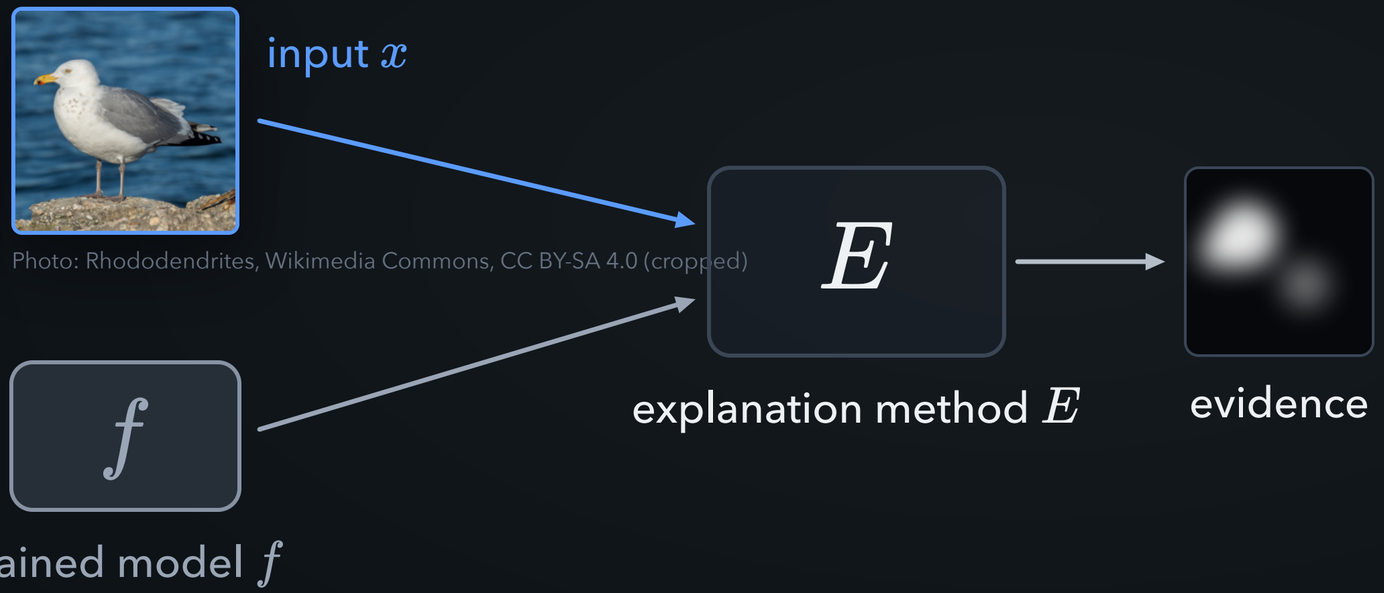


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

$$\text{evidence} = E(f, x)$$

Post-hoc or interpretable by design?

Saliency: nudge one pixel, watch one class score

input photo

Photo: Yathin S Krishnappa, Wikimedia Commons, CC BY-SA 3.0; centre crop

red value: +0.001

ResNet-50
`trained, weights fixed`

Residual Network with 50 layers

chosen score: **zebra**

before softmax

+0.00014

tiny change in the score

sensitivity = size of **score change** ÷ size of pixel change

$$S_i^{(k)} = \left| \frac{\partial s_k(x)}{\partial x_i} \right| \approx \left| \frac{\Delta s_k}{\Delta x_i} \right| \qquad \frac{0.00014}{0.001} \approx 0.14$$

`COMPUTED · REAL MODEL · real photo, fixed ImageNet model, one pixel channel changed`

Post-hoc or interpretable by design?

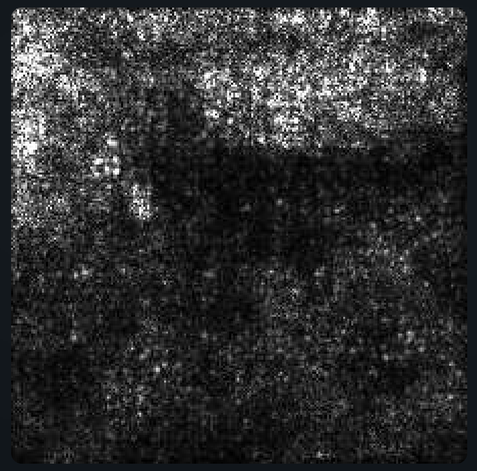
Same photo, same model, a different class to explain

same input photo

same ResNet-50, fixed weights

zebra score

probability: 37.7% (top class)



saliency for the zebra score

tiger score

probability: 0.08% (not predicted)

saliency for the tiger score

Same brightness scale: brighter = more sensitive

Low class probability does not mean zero pixel sensitivity

COMPUTED · REAL MODEL · raw score gradients; max over RGB; shared scale, brightest 1% clipped

Post-hoc or interpretable by design?

Occlusion: grey out a patch and measure how far the score drops

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

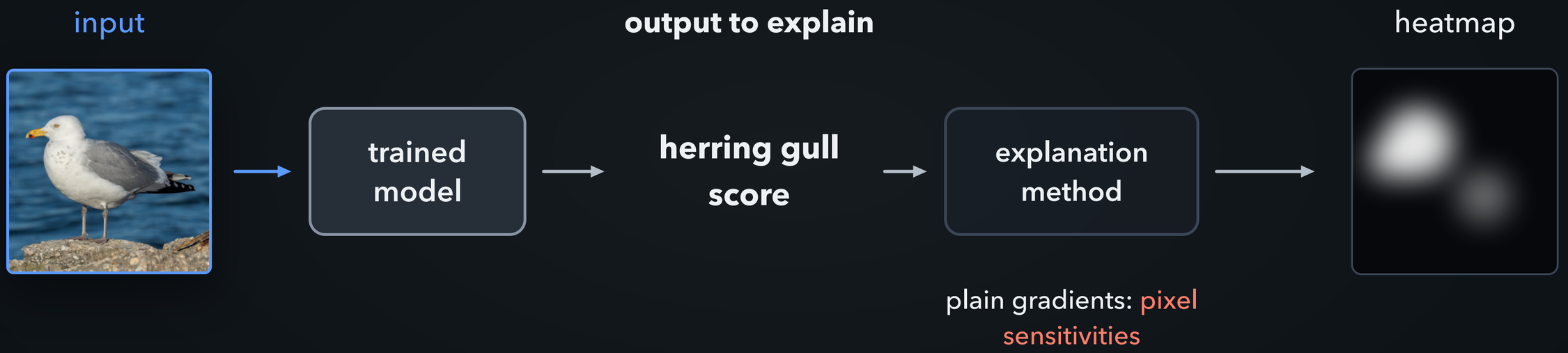
COMPUTED · REAL MODEL · CLIP occlusion map of the gull, Part 1

score on the photo minus **score** with patch p greyed

$$\Delta_p = \hat{y}_k(x) - \hat{y}_k(x \text{ with } p \text{ greyed})$$

Post-hoc or interpretable by design?

From an input photo and a class score to a heatmap



The method uses this photo, the model and the chosen **class**

Brighter pixels: the chosen score is more sensitive there

`SCHEMATIC · workflow and heatmap illustration, not computed for this photo`

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

Post-hoc or interpretable by design?

Does the heatmap depend on what the model learned?

Same photo, same **herring gull class**, same method

input photo

trained weights

replace learned weights

almost unchanged

random weights

reported example: guided backpropagation

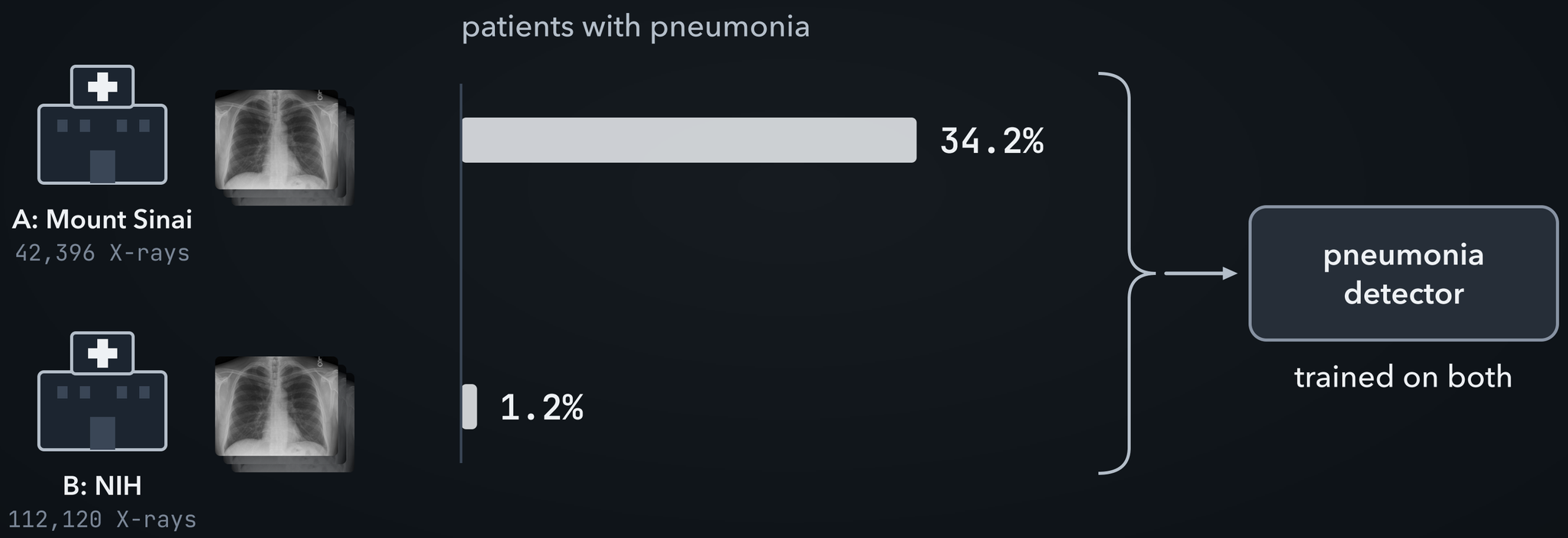
Before trusting a heatmap, check that it depends on the learned weights

`SCHEMATIC · drawn maps illustrating a reported failure, Adebayo et al., NeurIPS 2018`

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

Post-hoc or interpretable by design?

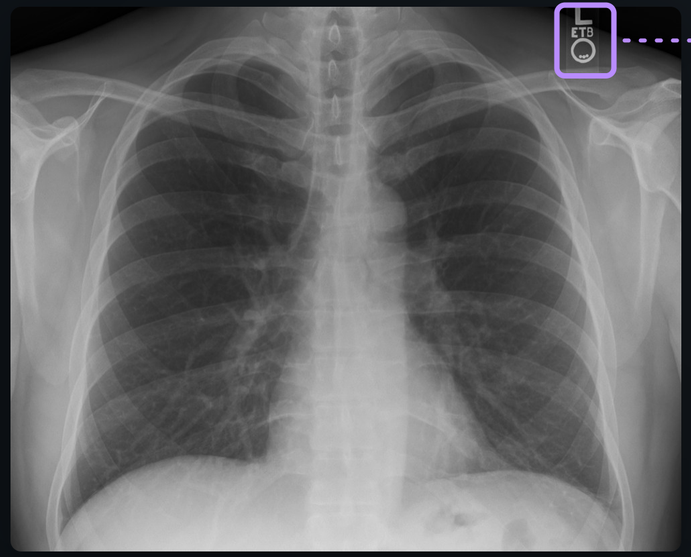
Limit 2: a model can learn a shortcut



REPORTED · Zech et al., PLOS Medicine 2018

Post-hoc or interpretable by design?

The network could tell which hospital an X-ray came from

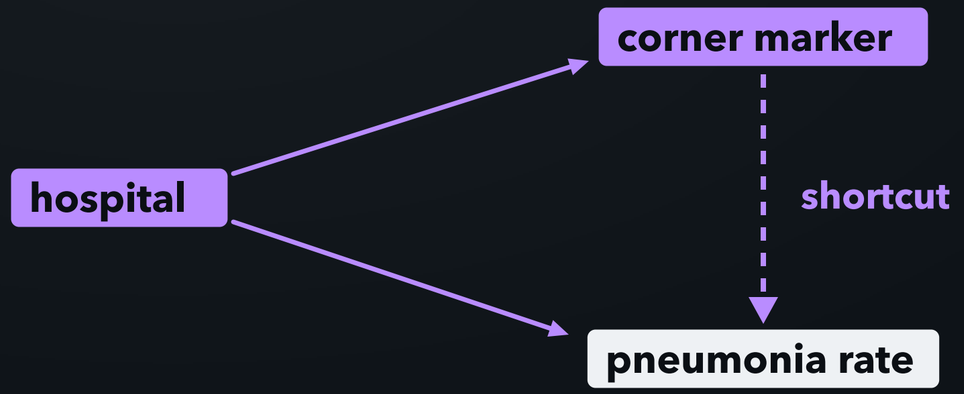


Chest X-ray: Stillwaterising, Wikimedia Commons, CC0

REPORTED · Zech et al., PLOS Medicine 2018

which hospital?

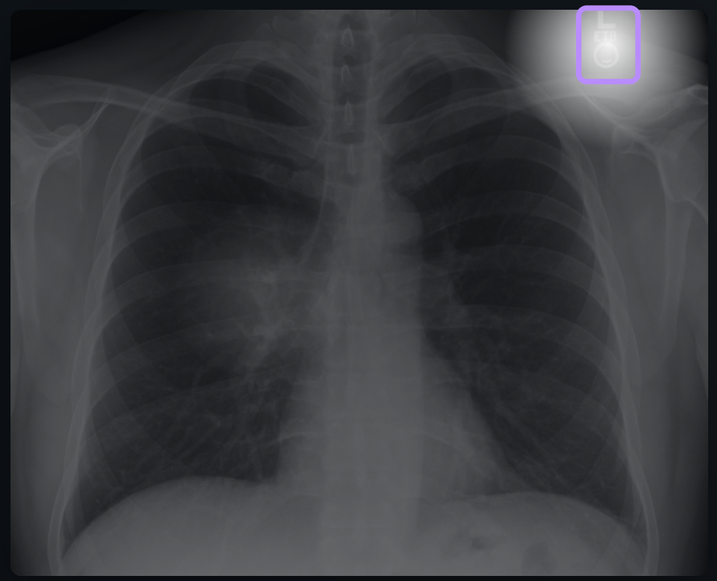
hospital named correctly for 99.95% of NIH X-rays



it can predict pneumonia without looking at the lungs

Post-hoc or interpretable by design?

An explanation can expose the shortcut, but the fix is new data



Chest X-ray: Stillwaterising, Wikimedia Commons, CC0

REPORTED · Zech et al., PLOS Medicine 2018

✓ exposes the shortcut

✗ cannot remove it

pneumonia AUC

same hospitals	0.931
a new hospital	0.815

fix: **new data** and retraining

Post-hoc or interpretable by design?

Limit 3: a heatmap says where, not what

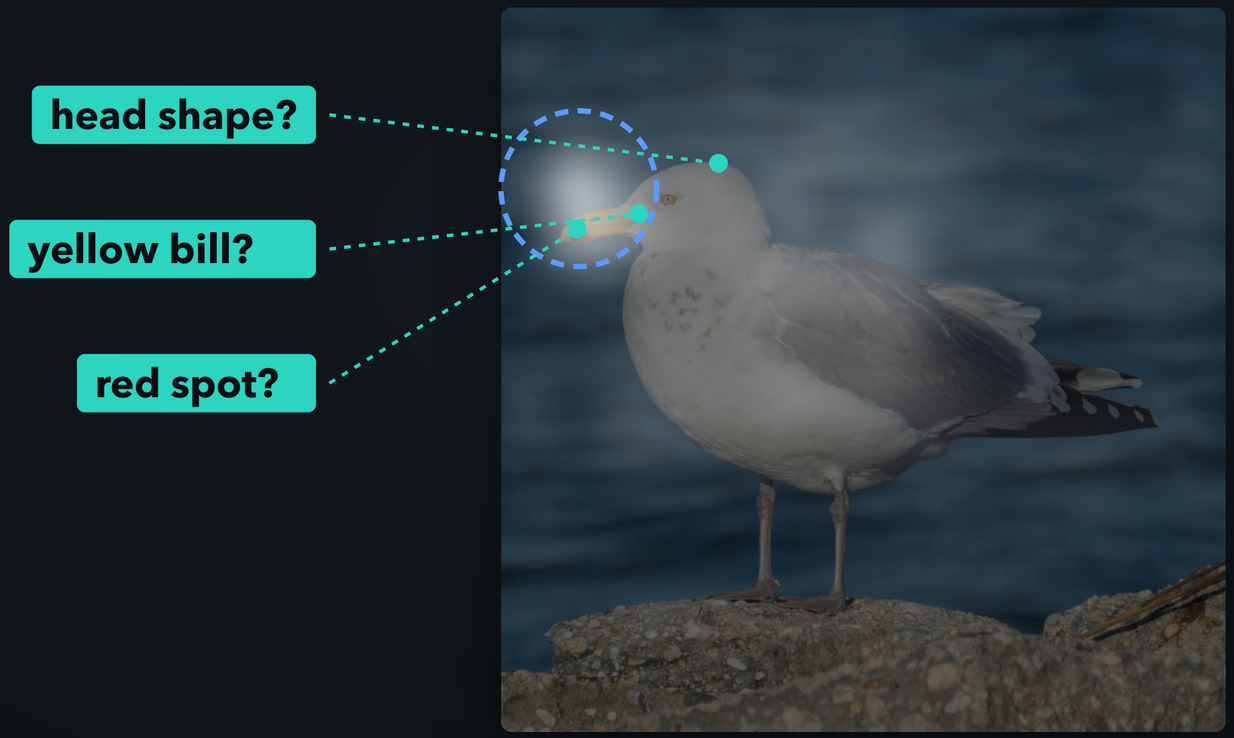


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

COMPUTED · REAL MODEL · CLIP occlusion map of the gull, Part 1

where: the score was sensitive here

what the model saw there: not said

Post-hoc or interpretable by design?

Experts speak in findings, and the model has no knob for them

findings

- narrowed joint space
- bone spur
- black wingtips

expert

black-box model

no named part to correct

next: a concept bottleneck adds that knob (Part 2)

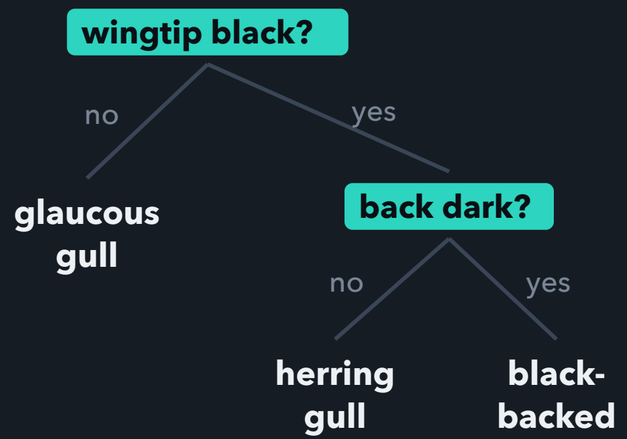
Post-hoc or interpretable by design?

Ante-hoc: the prediction is computed from quantities a person can read

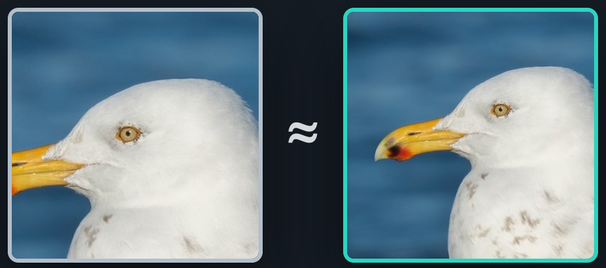
sparse linear model

$$\hat{y} = 2.1\,\text{wingtip black} - 1.3\,\text{bill hooked} + \dots$$

short decision tree



prototype network



this part ≈ looks like that

Chen et al. 2019

ILLUSTRATIVE VALUES · hand-written examples

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

Post-hoc or interpretable by design?

The case for models that are interpretable by design

> For high-stakes decisions, prefer models people can read over black boxes explained afterwards.
>
> Rudin, Nature Machine Intelligence 2019 (paraphrase)

Post-hoc or interpretable by design?

A concept bottleneck predicts concepts first, then the answer

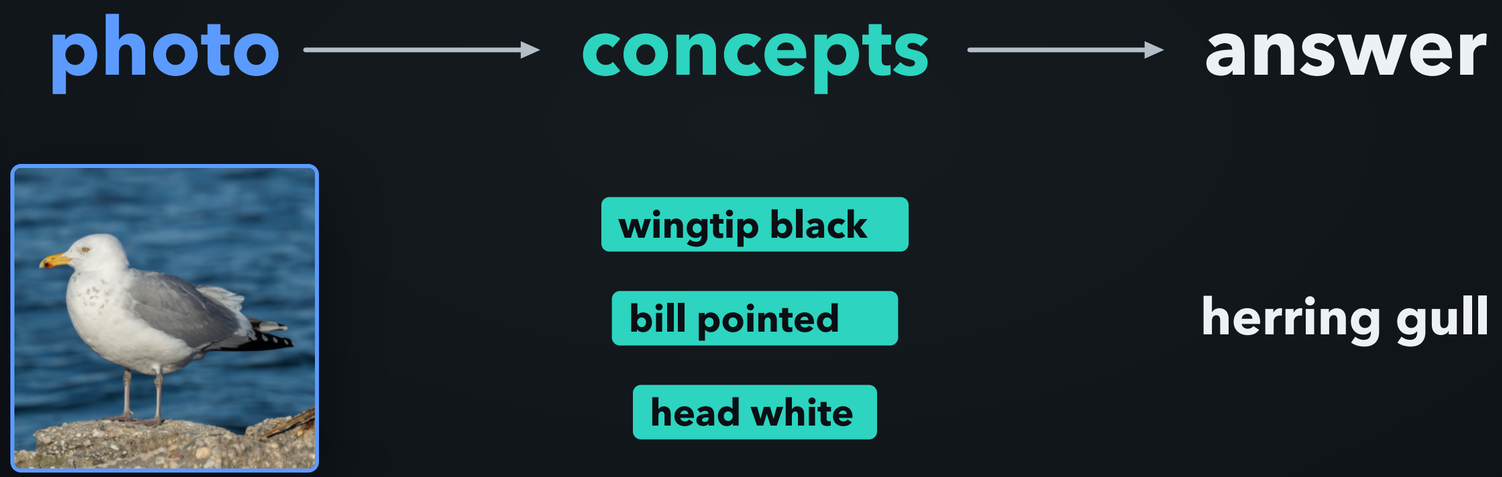
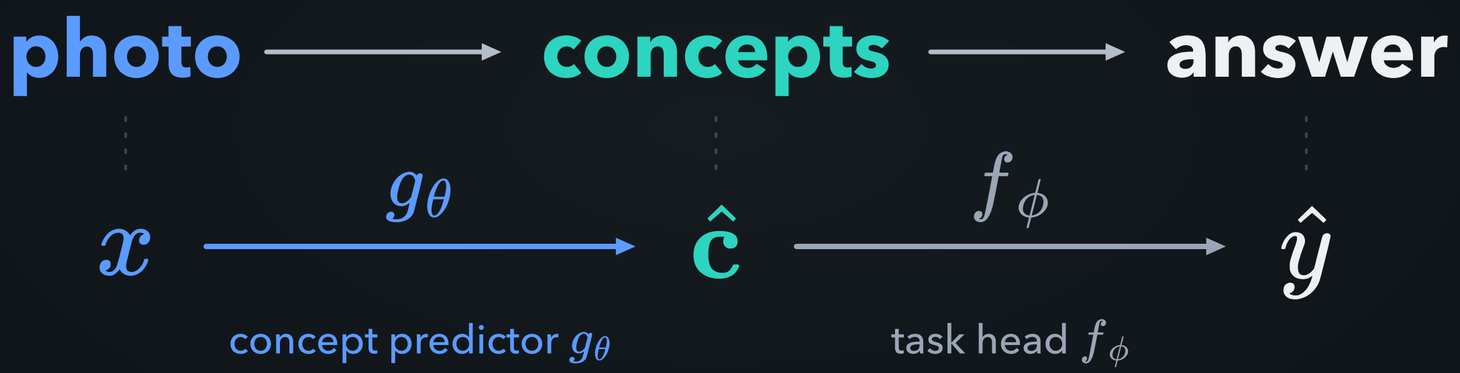


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

the **answer** is computed from the concepts alone

Post-hoc or interpretable by design?

The same path in symbols



$\hat{\mathbf{c}}$: one predicted value per concept

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Here is that idea on a real photograph

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

a layer of named variables

Introducing the concept bottleneck

A concept predictor fills the layer, a task head reads it

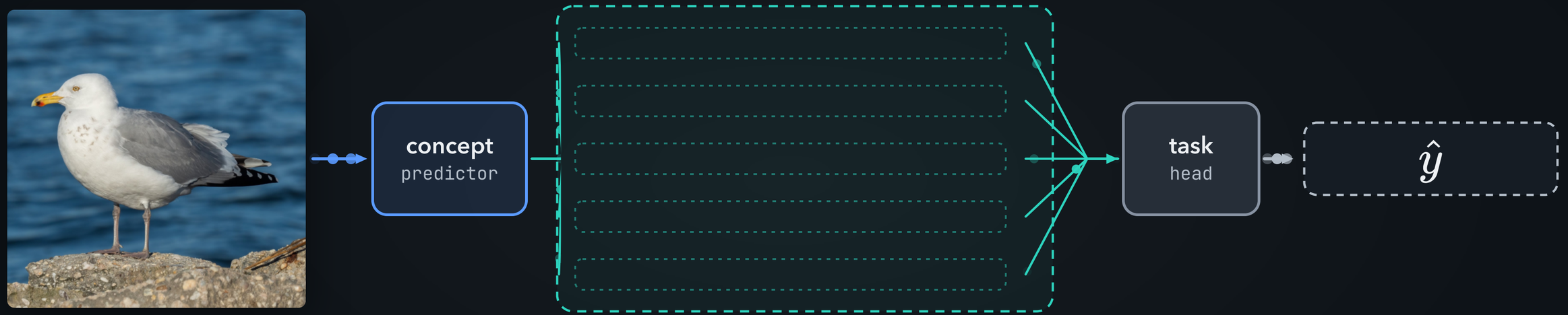


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

$$x \xrightarrow{g_\theta} \hat{\mathbf{c}} \xrightarrow{f_\phi} \hat{y}$$

Introducing the concept bottleneck

The concept vector has five named entries

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

g_θ

$$\hat{\mathbf{c}} = \begin{bmatrix} 0.80 \\ 0.20 \\ 0.90 \\ 0.85 \\ 0.60 \end{bmatrix}$$

- 0.80 — wingtip grey
- 0.20 — wingtip black
- 0.90 — bill pointed
- 0.85 — head white
- 0.60 — tail rounded

each entry: a value between 0 and 1

ILLUSTRATIVE VALUES · hand-written concept values, for teaching

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The rule: the task head reads only these five numbers

concept predictor

wingtip grey	0.80
wingtip black	0.20
bill pointed	0.90
head white	0.85
tail rounded	0.60

task head

$\hat{y}$

the bottleneck: the only input to the task head

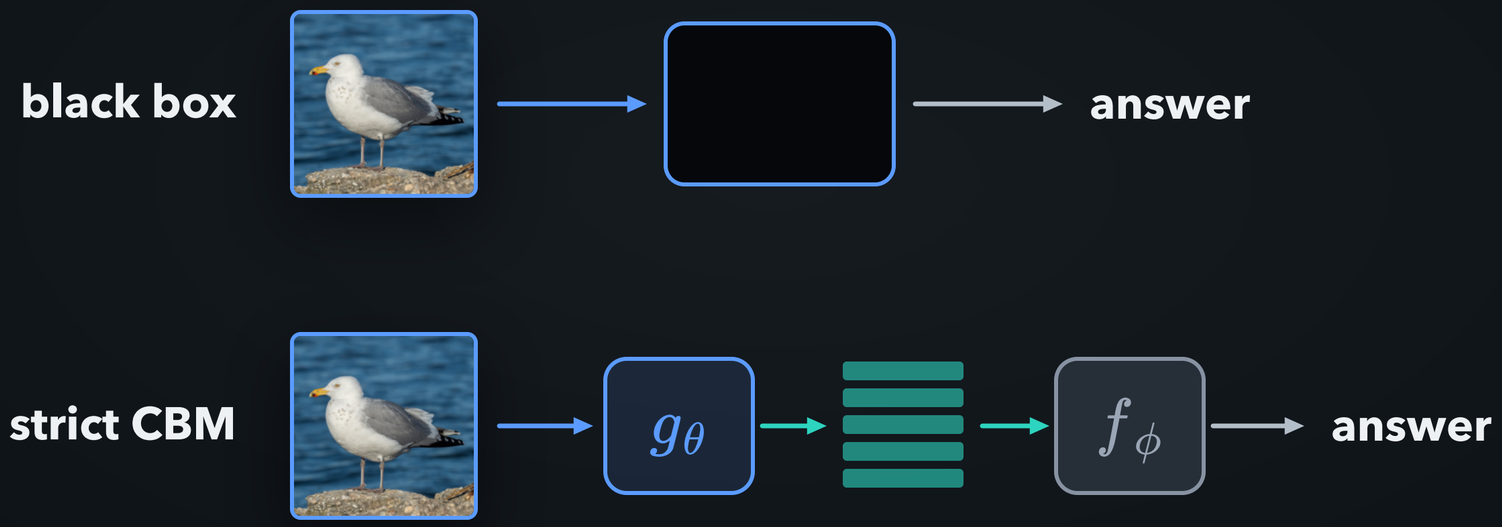
no other path from the image

ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A strict CBM predicts the concepts first, then the answer from them alone



the **answer** from the concepts alone

Introducing the concept bottleneck

The values and the head on these slides are written by hand

	weight toward American herring gull
wingtip grey	-2.0
wingtip black	+2.0
bill pointed	+0.4
head white	-0.2
tail rounded	+0.2

the head adds up weight × value, then turns the total into two **shares**

ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The model believes the wingtips are grey, so it says glaucous-winged gull

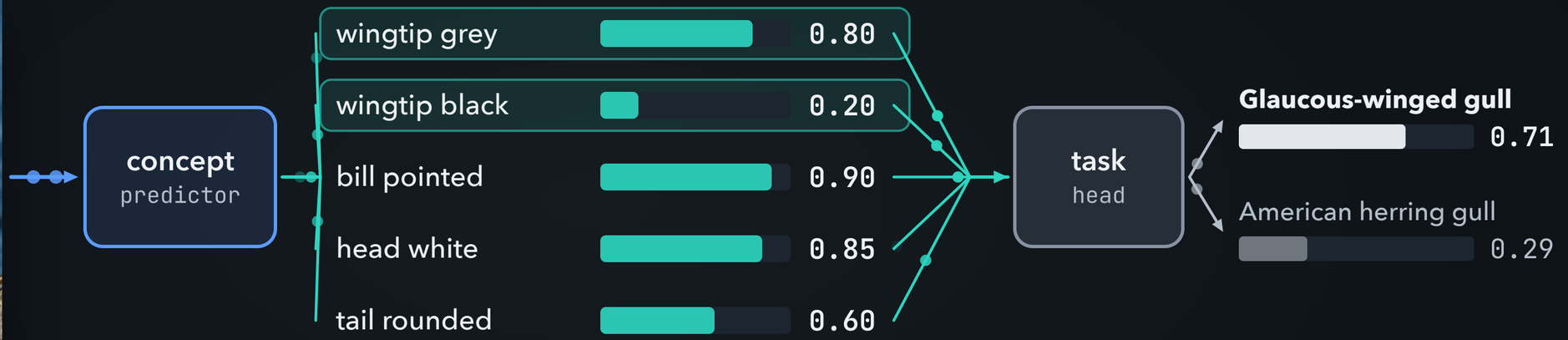


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

grey high, black low: the head favours **glaucous-winged gull**

ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The wingtips are black: an observer corrects the wing colour

the wingtips are black

concept predictor

wingtip grey	0.00
wingtip black	1.00
bill pointed	0.90
head white	0.85
tail rounded	0.60

task head

Glaucous-winged gull 0.71

American herring gull 0.29

wing colour: corrected as one group

ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The same task head recomputes, and herring gull now leads

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

unchanged

concept predictor

- wingtip grey 0.00
- wingtip black 1.00
- bill pointed 0.90
- head white 0.85
- tail rounded 0.60

task head

unchanged

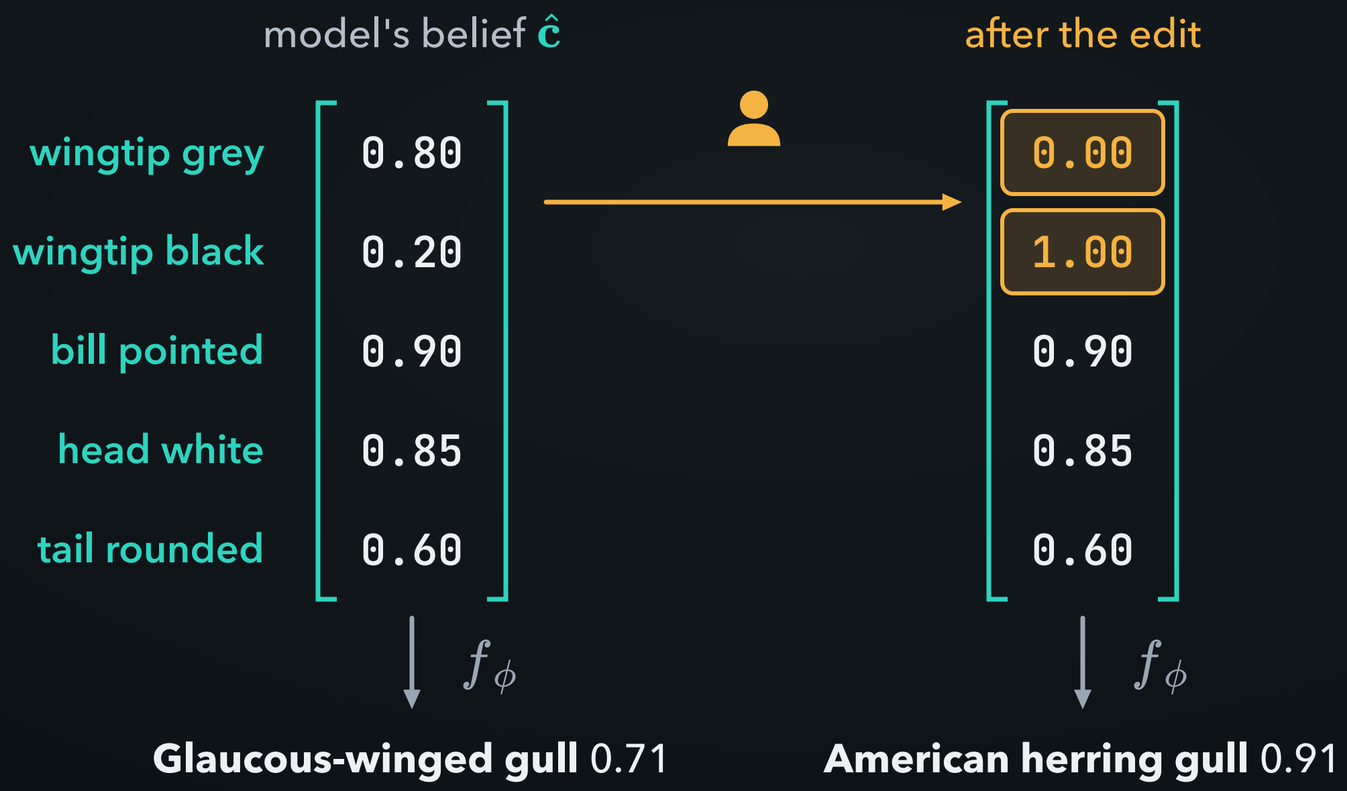
- Glaucous-winged gull 0.09
- **American herring gull** 0.91

same image, same head, edited concepts: a new **answer**

ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

Introducing the concept bottleneck

An edited belief, a new prediction



ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Multi-task learning puts a concept head beside the class head

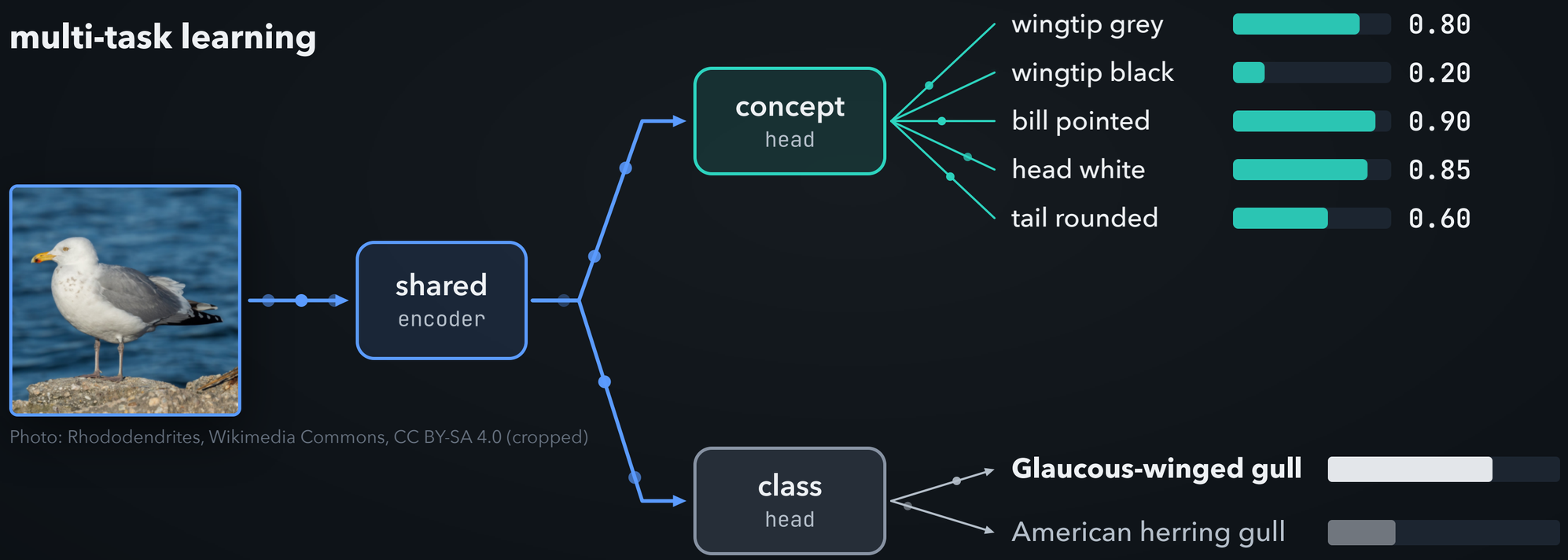


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

There, an edit to the concepts changes nothing

multi-task learning

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

shared encoder

concept head

wingtip grey	0.00
wingtip black	1.00
bill pointed	0.90
head white	0.85
tail rounded	0.60

class head

Glaucous-winged gull

American herring gull

no change

the class head never reads the concept head

Introducing the concept bottleneck

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

In a CBM, the concepts lie on the path to the prediction

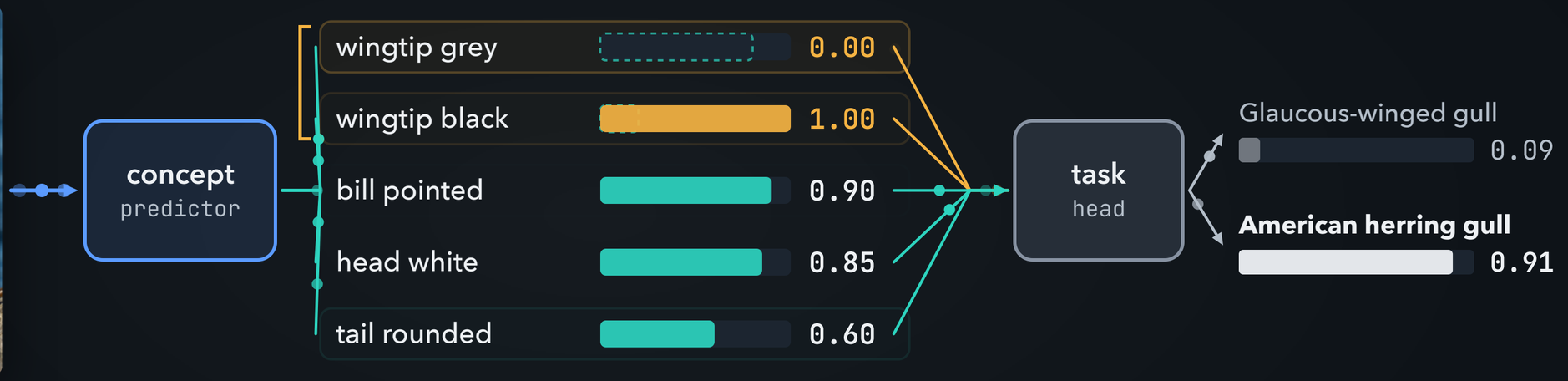


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

every **prediction** passes through the concepts

ILLUSTRATIVE VALUES · hand-written concept values and head, for teaching

The CUB benchmark

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The standard benchmark for concept bottlenecks is a bird dataset

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

our photo, not from CUB

Caltech-UCSD Birds-200-2011

CUB

Wah et al., Caltech 2011

The CUB benchmark

Many photographs of many similar species

11,788
photographs

200
species

8 of the species are gulls

California · Glaucous-winged · Heermann · Herring · Ivory · Ring-billed · Slaty-backed · Western

DATASET · CUB-200-2011, counted

The CUB benchmark

Every photograph is described by the same attributes

312 yes-or-no attributes, in 28 groups

back colour	back pattern	belly colour	belly pattern
bill colour	bill length	**bill shape**	breast colour
breast pattern	crown colour	**eye colour**	forehead colour
head pattern	leg colour	nape colour	primary colour
shape	size	tail pattern	tail shape
throat colour	under tail colour	underparts colour	upper tail colour
upperparts colour	**wing colour**	**wing pattern**	wing shape

The CUB benchmark

Each attribute is a yes-or-no question

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

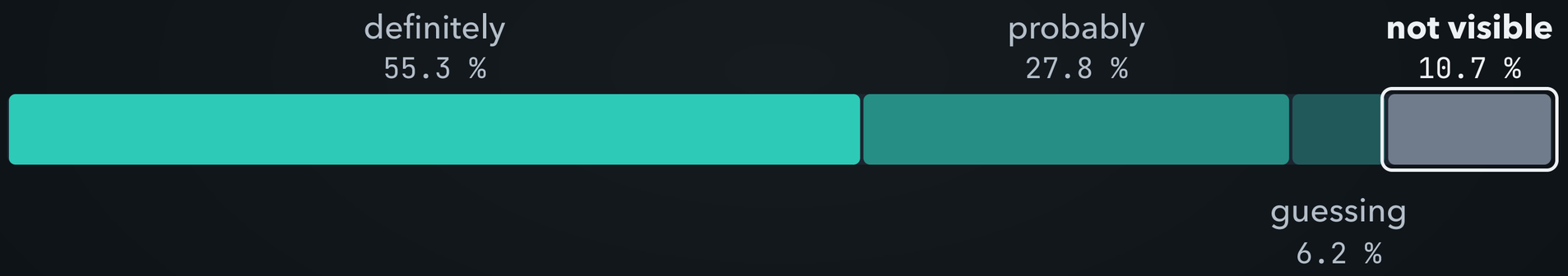
Does it have a hooked bill?	yes →	1
Is its wing grey?	yes →	1
Is its wing yellow?	no →	0

the answer to question j: yes is 1, no is 0

$$c_j \in \{0, 1\}$$

The CUB benchmark

Annotators also recorded how sure they were

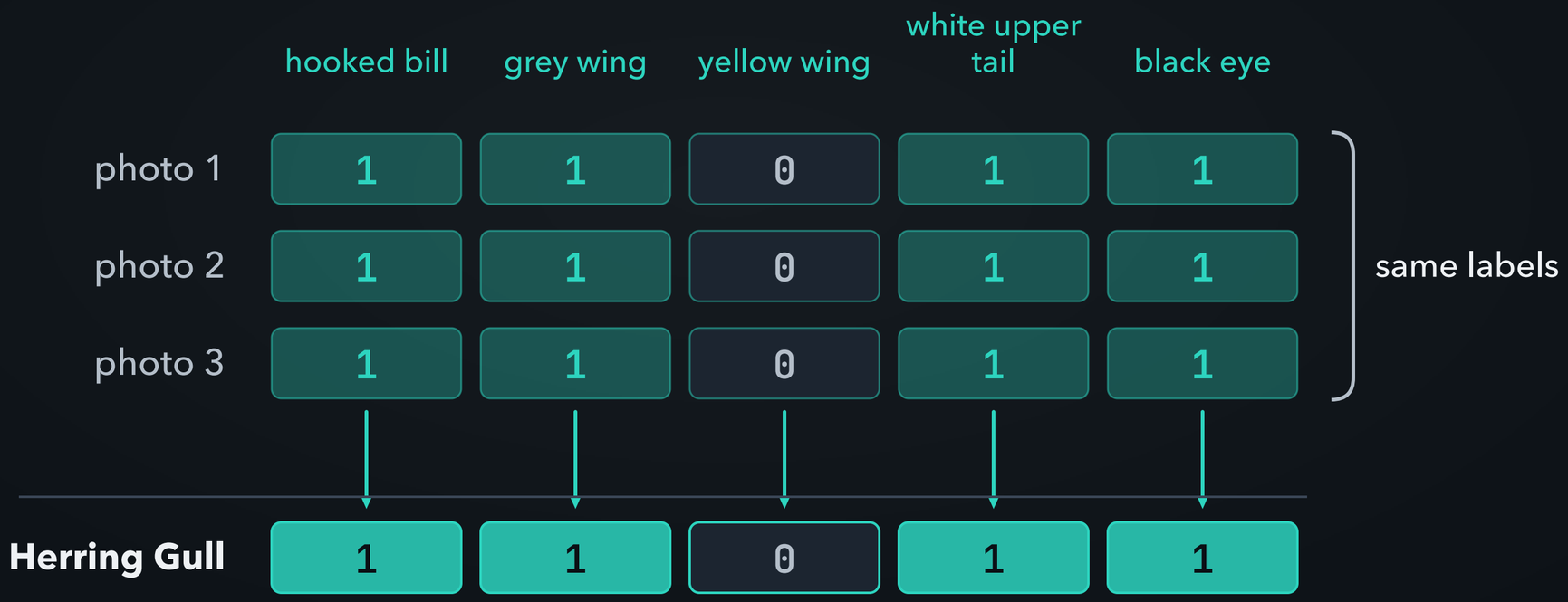


about one label in ten: not visible

DATASET · CUB annotations, counted

The CUB benchmark

Single photos are noisy: take a vote within each species

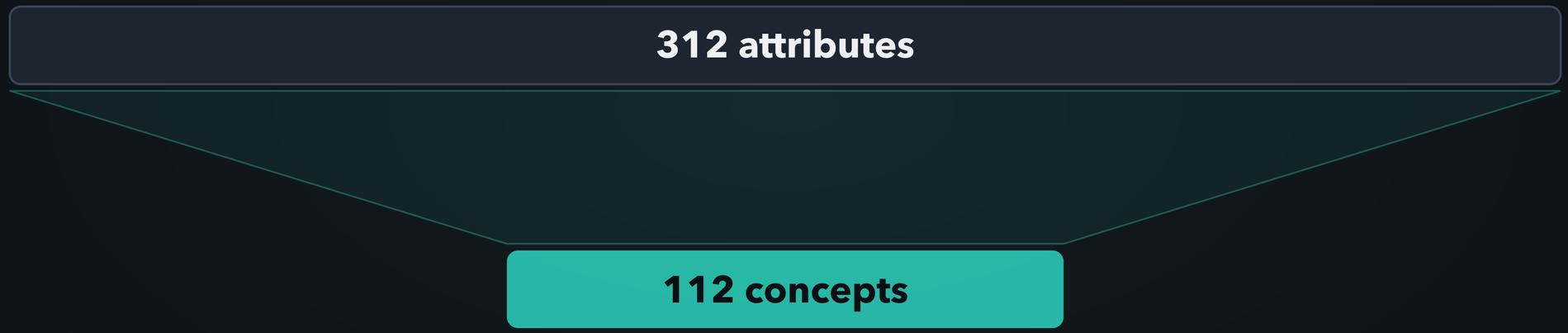


Koh et al., 2020

ILLUSTRATIVE VALUES · photo rows; the voted row is Herring Gull's in CUB

The CUB benchmark

Keep only the attributes that many species share



present in at least 10 species

DATASET · CUB, filtered as in Koh et al. 2020

The CUB benchmark

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

One training record has three parts

Photo: Rhododendrites, CC BY-SA 4.0

DATASET · Herring Gull's labels in CUB, first 20 of 112; our photo

The CUB benchmark

The same record in symbols

concept labels, each 0 or 1

$$\mathbf{c} \in \{0,1\}^{112}$$

species label

$$y \in \{1, \ldots, 200\}$$

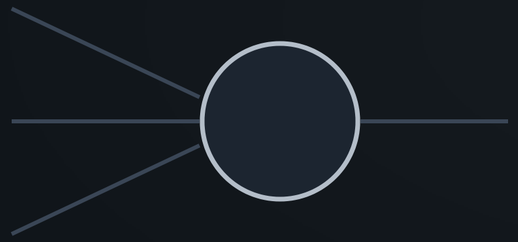
predicted concepts — concept predictor — one value per concept

$$\hat{\mathbf{c}} = g_\theta(x) \in [0,1]^{112}$$

What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A concept is a claim that a person can check



a name tag on a neuron

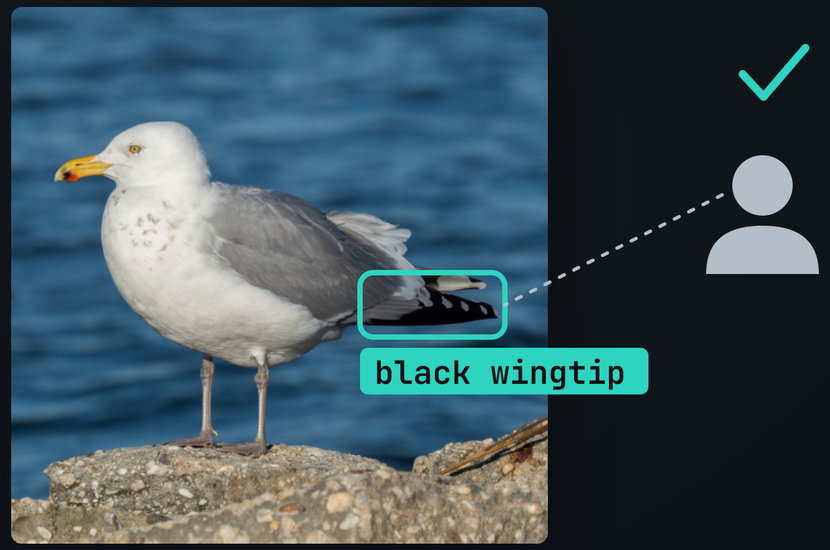


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

a claim a person checks in the input

What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Meaning: a definition that an expert can apply

1 **meaning** 2 grounding 3 representation

expert

`grey wingtip`

the tips of the folded flight feathers are **grey**, not black

What is a concept?

Grounding: the evidence sits in one region of the photo

1 meaning 2 **grounding** 3 representation

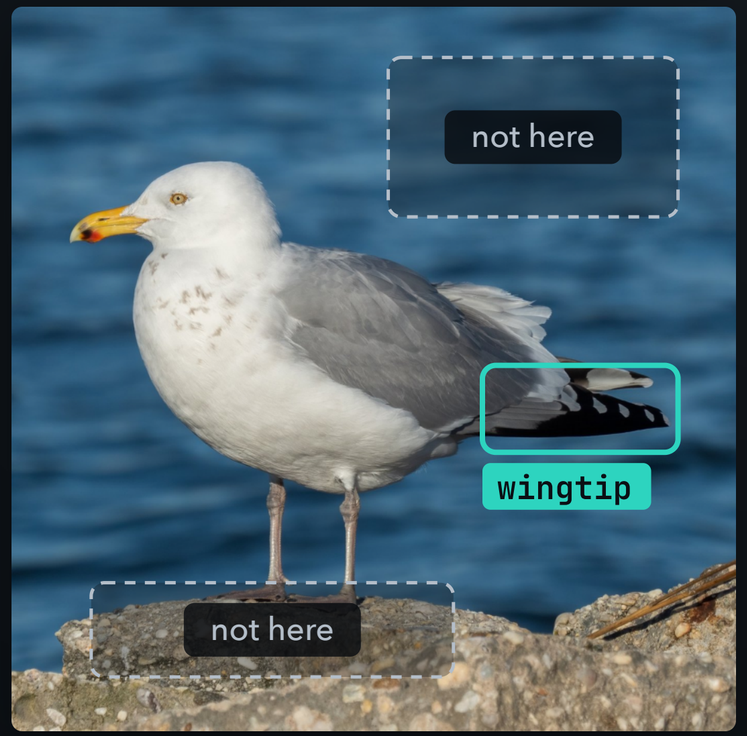


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

the wingtip colour estimate should depend on these feathers

not on the water or the rock

What is a concept?

Representation: the simplest concept is binary

1 meaning 2 grounding 3 **representation**

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

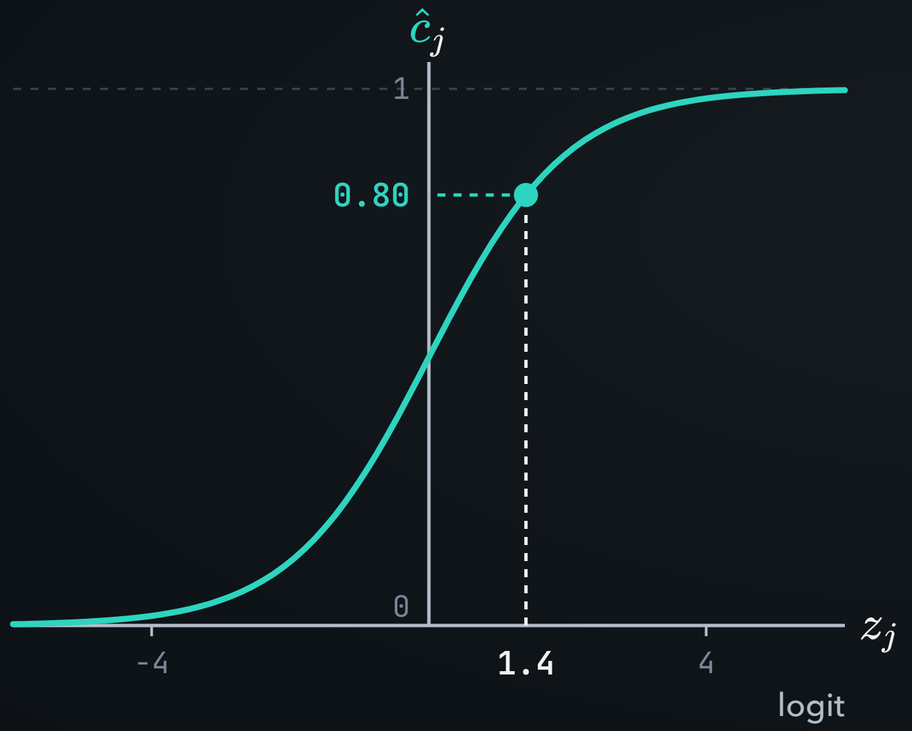
the value of concept j: 0 or 1

$$c_j \in \{0, 1\}$$

What is a concept?

A soft concept turns a logit into a value between 0 and 1

representation ● ● ○ ○ ○ ○



the soft score is the sigmoid of a logit

$$\hat{c}_j = \sigma(z_j) = \frac{1}{1 + e^{-z_j}}$$

example: $z_j = 1.4$ gives $\hat{c}_j = 0.80$

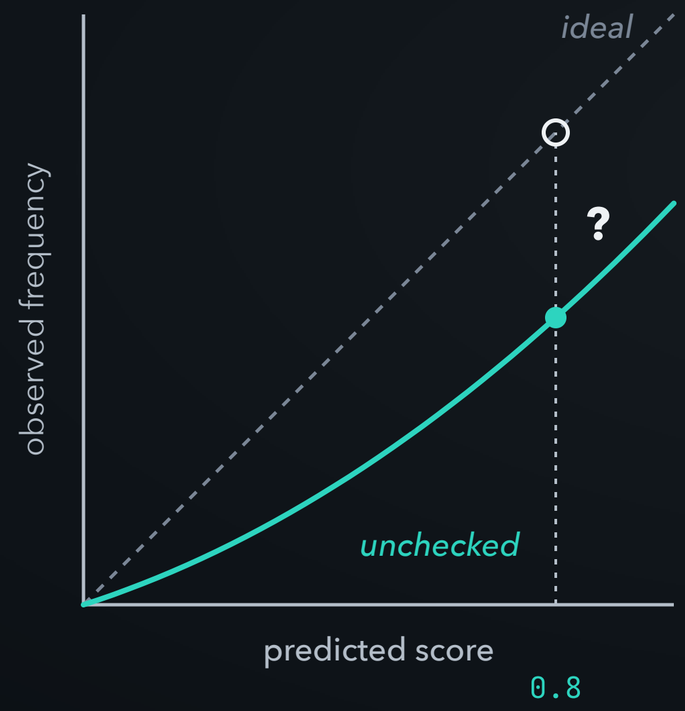
ILLUSTRATIVE VALUES · a worked logit

What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A soft score is not automatically a probability

representation ●●○○○○



0.8 is a score, not automatically a probability

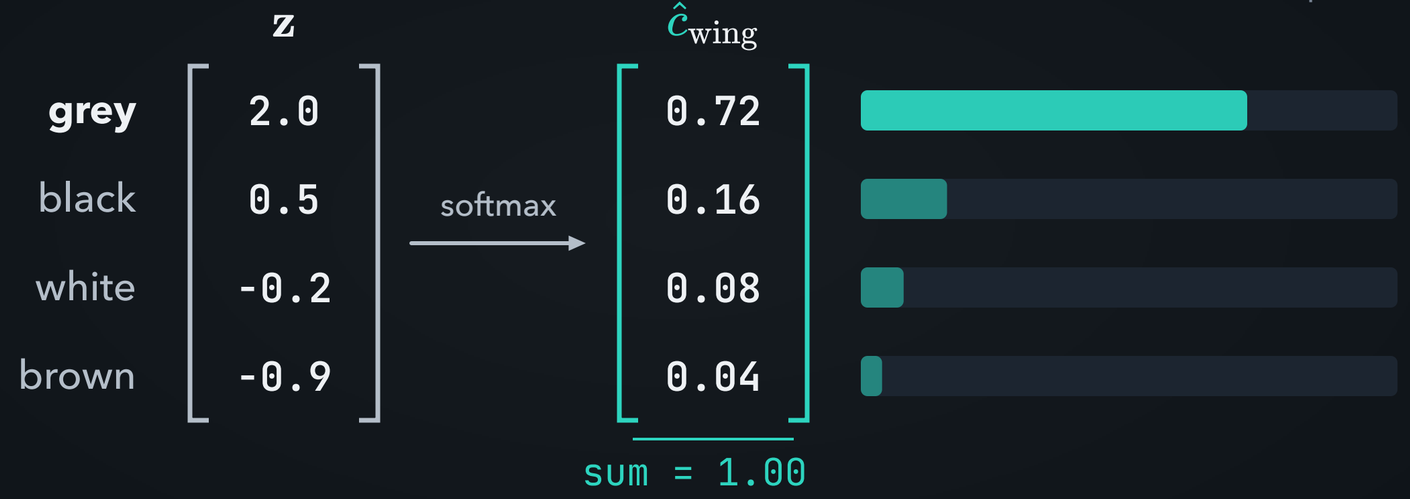
check on data: do 80 % of the cases scored 0.8 have grey wingtips?

SCHEMATIC · a sketch, no data

What is a concept?

Categorical: one value from a list, with scores that sum to 1

representation



the wing colour scores are a softmax of the logits, and they sum to 1

$$\hat{c}_{\text{wing}} = \text{softmax}(\mathbf{z})$$

ILLUSTRATIVE VALUES · logits made up for the example

What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Continuous: a measurement, such as bill length

representation ●●●●○○

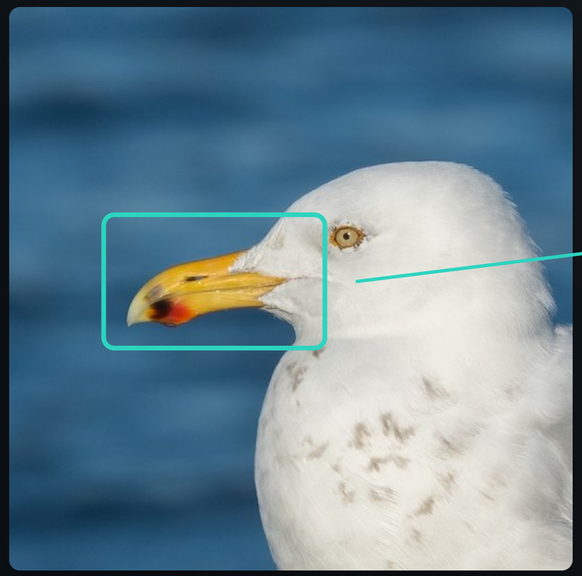
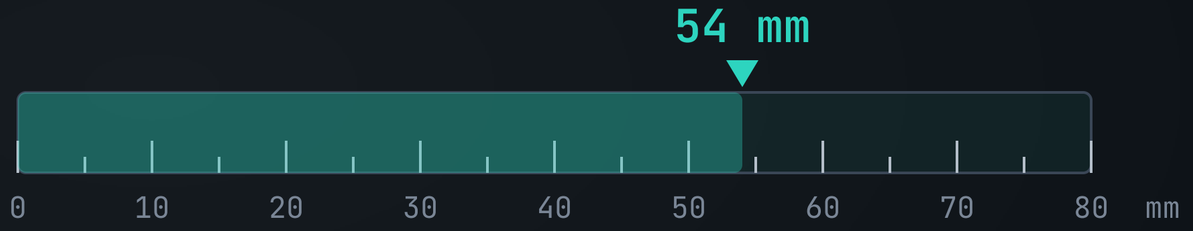


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)



the bill length is a real number, in mm

$$\hat{c}_{\text{bill}} \in \mathbb{R}$$

predicted by regression

ILLUSTRATIVE VALUES · value

What is a concept?

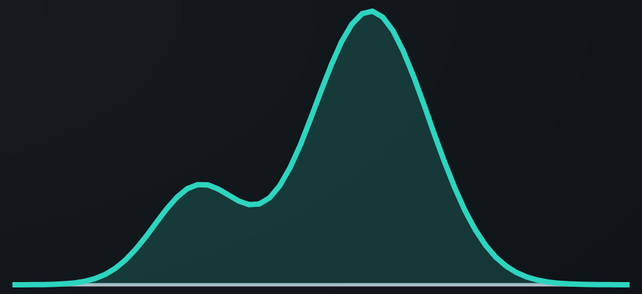
Richer concepts: a vector or a whole distribution

representation ●●●●●○

a vector (embedding)

$$\begin{bmatrix} 0.31 \\ -1.20 \\ 0.85 \\ 0.02 \\ -0.47 \end{bmatrix}$$

a distribution



possible values of the concept

$\hat{c}_j \in \mathbb{R}^m$ or $p(c_j \mid x)$

we meet these in Part 10

ILLUSTRATIVE VALUES · values

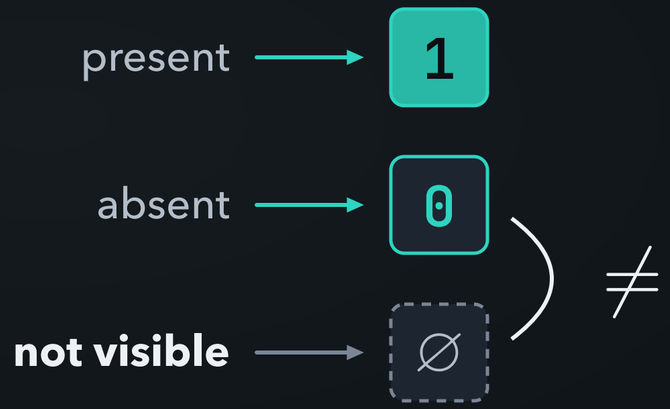
What is a concept?

Unknown: not visible is different from absent

CONSTRUCTED EXAMPLE · wingtip covered for the example

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

representation



CUB label not visible: unknown, not absent

$$c_j = \varnothing \neq 0$$

What is a concept?

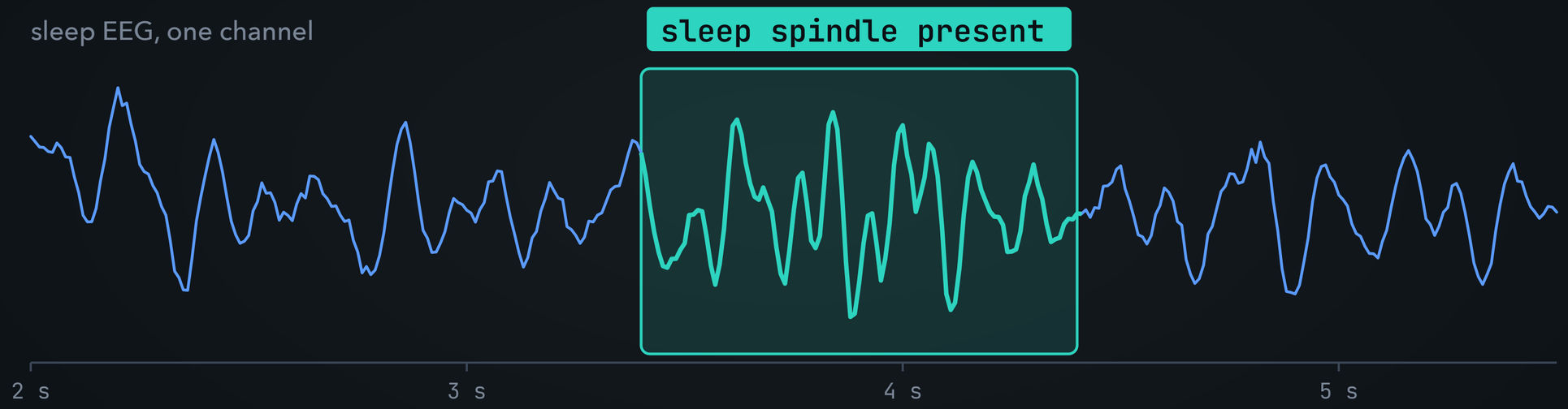
Six ways to store a concept as a number

type	in symbols	example
binary	$c_j \in \{0,\ 1\}$	grey wingtip: 1
soft	$\hat{c}_j = \sigma(z_j)$	logit 1.4 gives 0.80
categorical	$\hat{c}_{\text{wing}} = \text{softmax}(\mathbf{z})$	grey 0.72, scores sum to 1
continuous	$\hat{c}_{\text{bill}} \in \mathbb{R}$	bill length 54 mm
vector or distribution	$\hat{\boldsymbol{c}}_j \in \mathbb{R}^m,\ \ p(c_j \mid \boldsymbol{x})$	more in Part 10
unknown	$c_j = \varnothing$	not visible, not absent

What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The same pattern in a sleep EEG recording



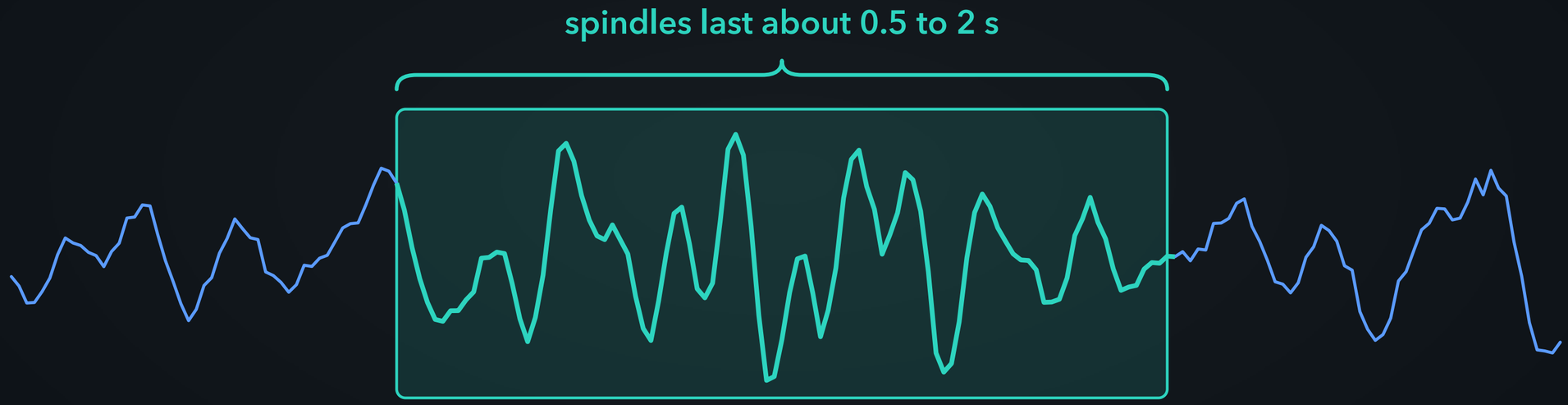
EEG trace → spindle present → sleep stage

SCHEMATIC · seeded sinusoids, not a recording

What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A spindle is a brief burst of brain waves



a burst of 11 to 16 cycles per second (Hz)

sleep scorers: a sign of **stage 2 sleep**

SCHEMATIC · this spindle: 13 Hz, 1 s; not a recording

What is a concept?

The same pattern in a customer message

CUSTOMER MESSAGE

The app works, but I was charged twice. Please refund the extra payment.

↓

`duplicate charge reported`

the customer says one purchase was billed twice

message → duplicate charge → support queue

ILLUSTRATIVE VALUES · an example message

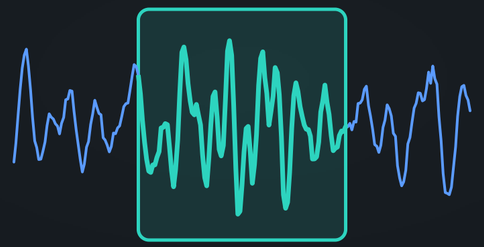
What is a concept?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Each **concept** is a named claim that a specialist can check

`black wingtip`

bird expert ✓



`spindle present`

sleep scorer ✓

“…I was charged twice…”

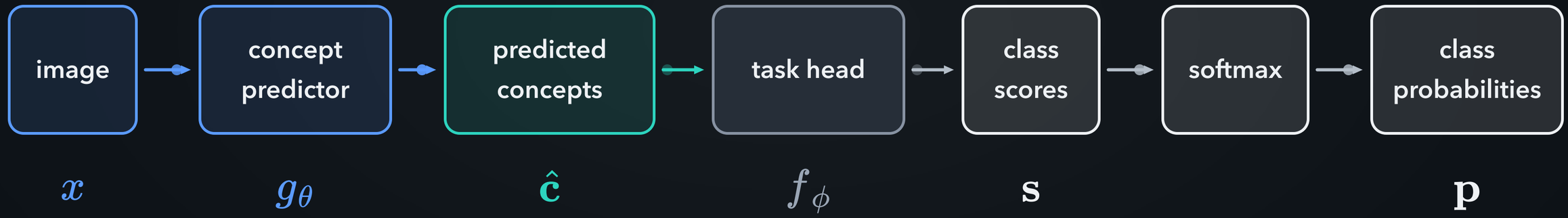
`duplicate charge`

support agent ✓

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

From concepts to predictions: the arithmetic

Follow the image all the way to class probabilities



task head: a linear layer or a multilayer perceptron (MLP)

our worked example uses one linear layer

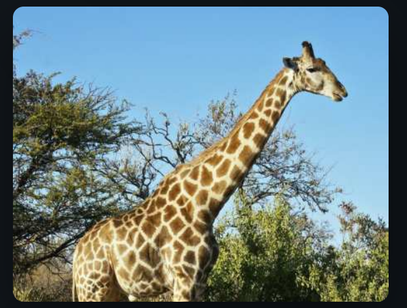
From concepts to predictions: the arithmetic

Three classes and three concepts

classes

zebra

horse



giraffe

concepts

stripes

horse-like shape

long neck

Photo: Yathin S Krishnappa, Wikimedia Commons, CC BY-SA 3.0
Photo: Dietmar Rabich, Wikimedia Commons, CC BY-SA 4.0
Photo: Hans Hillewaert, Wikimedia Commons, CC BY-SA 4.0

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

x g_θ $\hat{c}$ f_ϕ $\hat{y}$

The concept predictor turns this image into three estimates

input image

concept predictor

g_θ

$\hat{\mathbf{c}}$

stripes	0.1
horse-like shape	0.9
long neck	0.1

assumed outputs for this photo

$$\hat{\mathbf{c}} = g_\theta(x)$$

the hat means predicted

Photo: Yathin S Krishnappa, Wikimedia Commons, CC BY-SA 3.0

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

The task head turns predicted concepts into class scores

$\hat{\mathbf{c}}$

stripes	0.1
horse-like shape	0.9
long neck	0.1

predicted concepts

task head

f_ϕ

zebra score

horse score

giraffe score

class scores

The task head reads only the concepts, not the image

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

The concept values describe this image; the weights belong to the layer

values for this image

stripes	0.1
horse-like shape	0.9
long neck	0.1

$\hat{c}$

change with the image

weights inside the linear layer

	stripes	horse-like shape	long neck
zebra	3	1	0
horse	-1	2	-1
giraffe	-1	0	3

W

fixed during prediction

this example has no bias

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

Inside the layer: multiply each concept value by its weight, then add

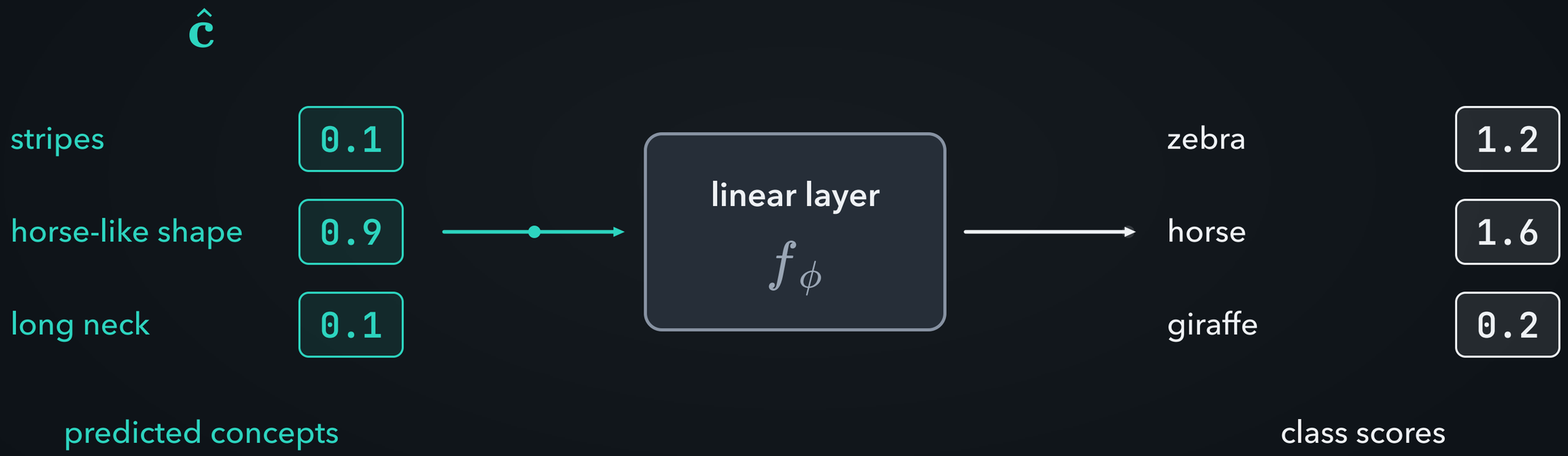
	stripes		horse-like shape		long neck		
zebra weights	3	×	1	×	0		
this image's values	0.1		0.9		0.1		
contributions	0.3	+	0.9	+	0	=	1.2
							zebra score

$$\text{zebra score} = 3 \times 0.1 + 1 \times 0.9 + 0 \times 0.1 = 1.2$$

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

The same layer produces all three class scores



scores = layer weights × predicted concept values

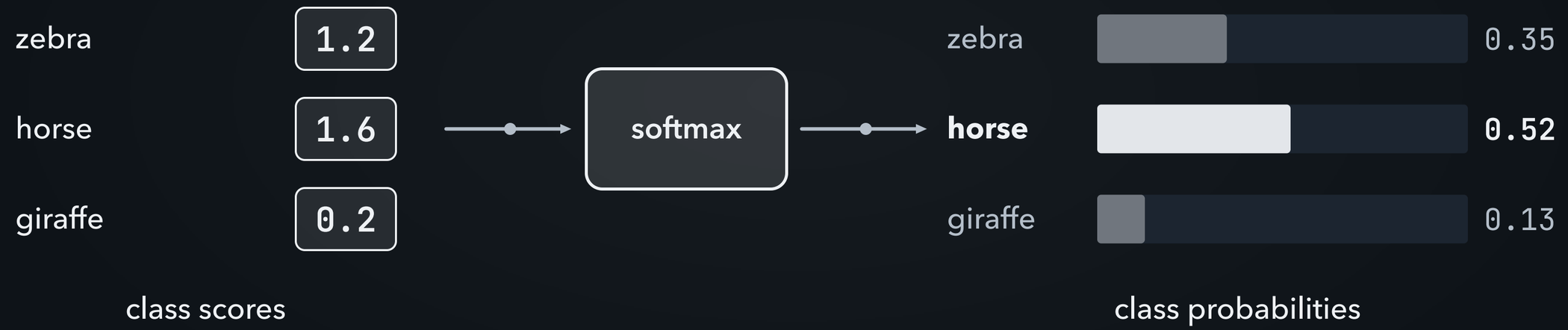
$$\mathbf{s} = f_\phi(\hat{\mathbf{c}}) = W\hat{\mathbf{c}}$$

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Softmax turns the scores into class probabilities



$$p_k = \frac{e^{s_k}}{\sum_j e^{s_j}}$$

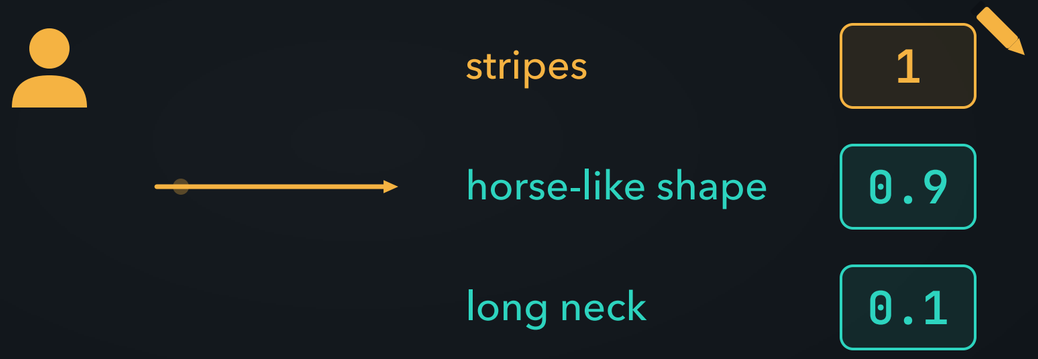
probabilities sum to one

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A person corrects one predicted concept



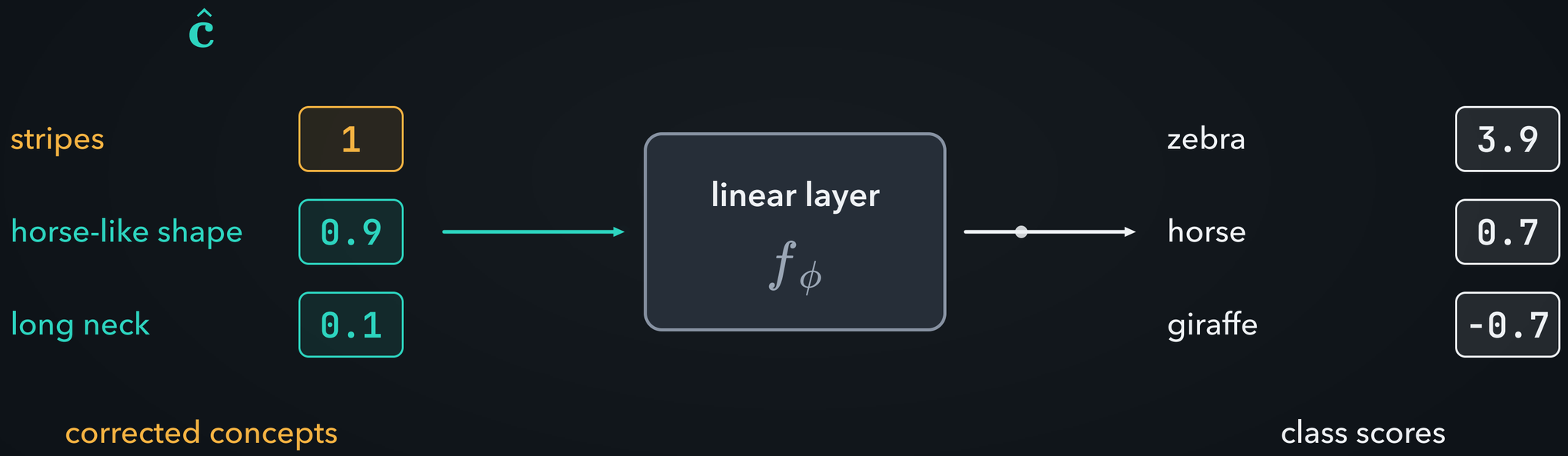
$$\text{stripes: } 0.1 \rightarrow 1$$

Photo: Yathin S Krishnappa, Wikimedia Commons, CC BY-SA 3.0

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

Pass the corrected concepts through the same layer



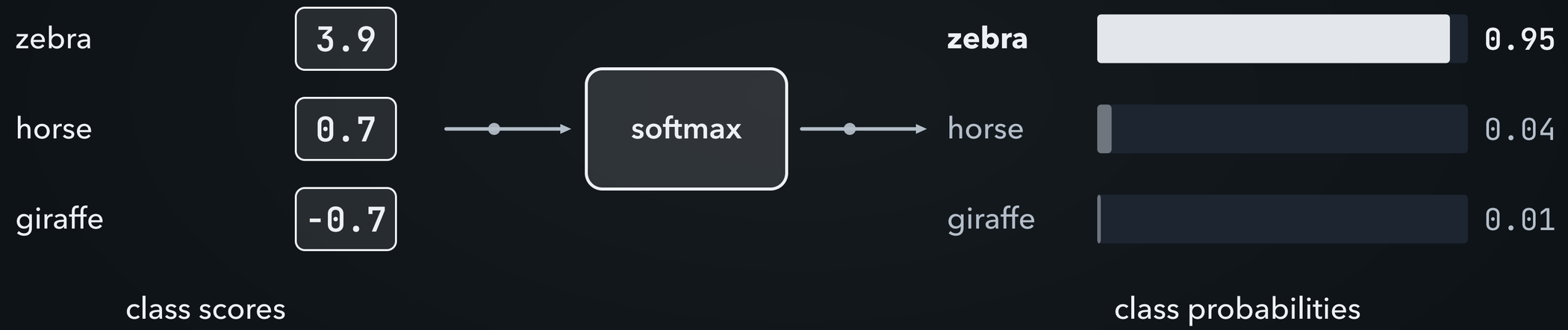
same image, same predictor, same weights; one corrected concept

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

x g_θ $\hat{c}$ f_ϕ $\hat{y}$

After softmax, zebra has the highest probability



before: horse 0.52

after: zebra 0.95

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

Why did the zebra score rise?

change in stripes		zebra's stripes weight		change in zebra score
0.9	×	3	=	$+2.7$

$$\text{zebra score: } 1.2 + 2.7 = 3.9$$

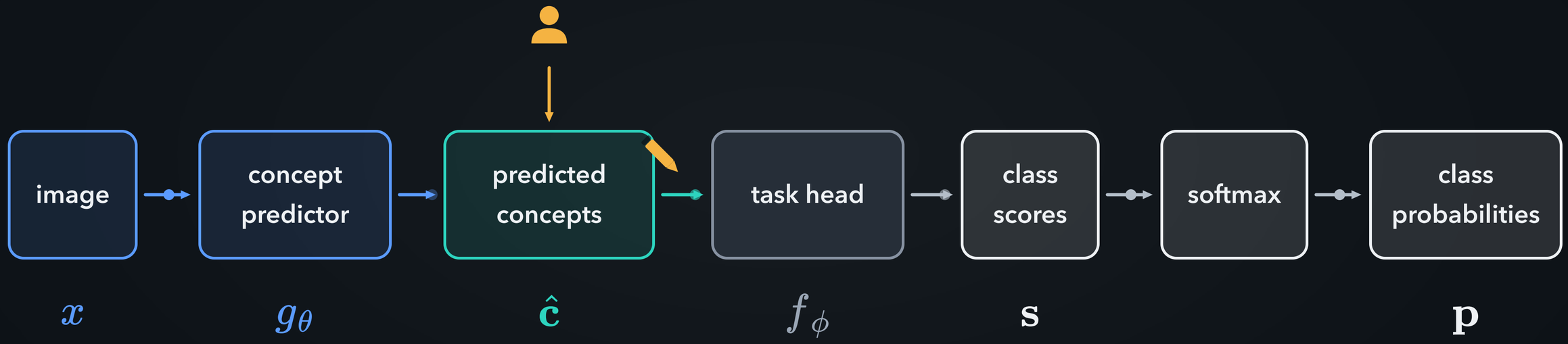
all other contributions stay the same

ILLUSTRATIVE VALUES · hand-chosen concept outputs and weights; arithmetic computed from these values

From concepts to predictions: the arithmetic

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The concept vector is the point where a person can intervene



correct a concept, then run the same task head again

no retraining

How does the model learn the concepts?

Where do the concept values come from?

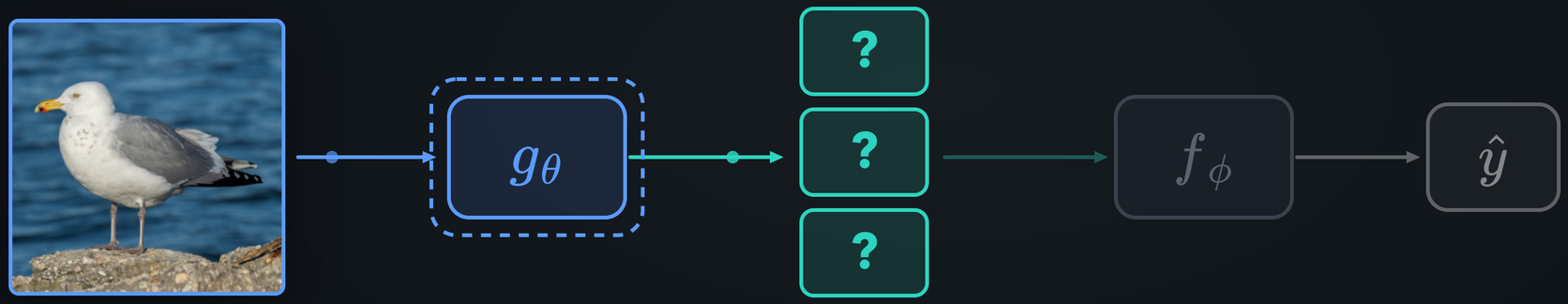


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

photo concept predictor concept values

learned from labelled examples

How does the model learn the concepts?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Each training example carries three things

hooked bill	1
grey wing	1
yellow wing	0
white throat	1

... 112 labels in all

Herring Gull

input x_i

concept labels, 0 or 1 $\mathbf{c}_i$

task label y_i

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

`DATASET · Herring Gull's labels in CUB, 4 of 112; our photo`

How does the model learn the concepts?

The concept loss compares each predicted concept with its label

for each concept, compare the prediction $\hat{c}_k$ with the label c_k

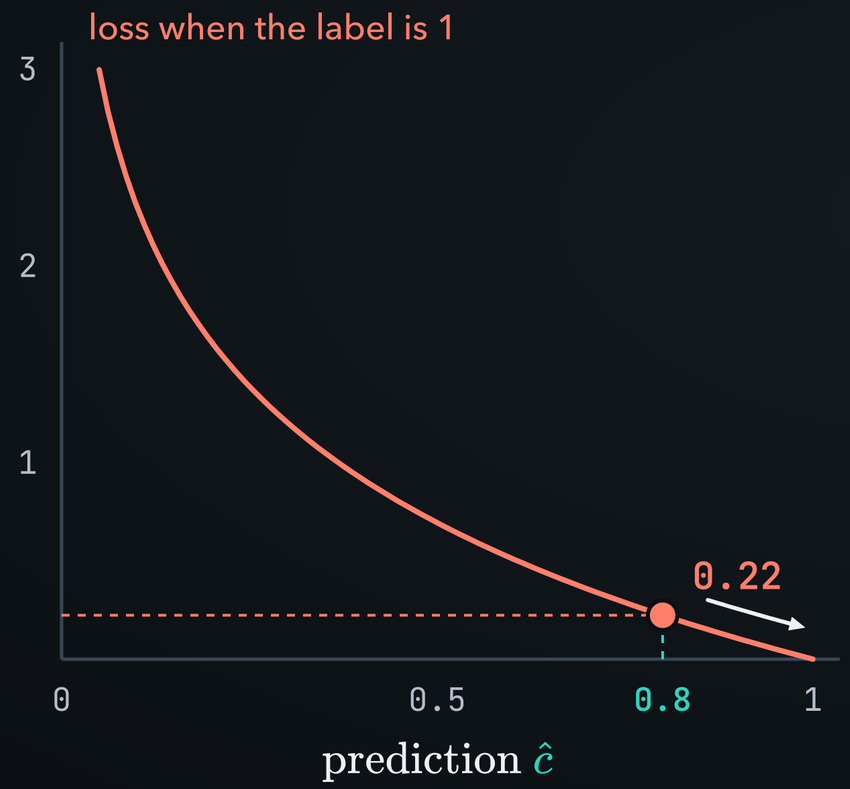
loss for concept k — label 1: a low prediction costs — label 0: a high prediction costs

$$\ell_k = -c_k \log \hat{c}_k - (1 - c_k) \log(1 - \hat{c}_k)$$

binary cross-entropy, averaged over the concepts: $L_c = \frac{1}{K} \sum_k \ell_k$

How does the model learn the concepts?

One term of the concept loss, worked out



grey wing: label 1, prediction 0.8

$$\ell = -\left[1 \cdot \log 0.8 + 0 \cdot \log(1 - 0.8)\right]$$

$$= -\log 0.8 \approx 0.22$$

closer to 1, smaller loss

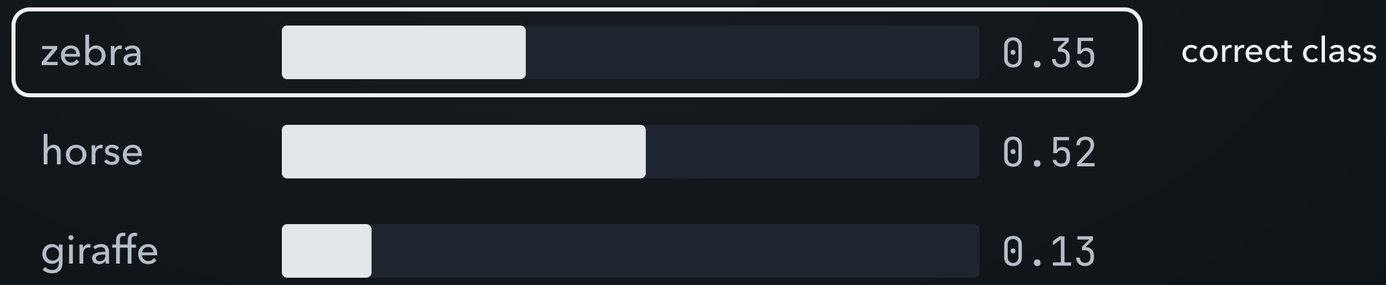
ILLUSTRATIVE VALUES · a hand-chosen prediction

How does the model learn the concepts?

The task loss compares the prediction with the task label

task loss: minus the log of the share given to the correct **class**

$$L_y = -\log \hat{y}_{y_i}$$



the zebra photo of Part 4: $L_y = -\log 0.35 \approx 1.05$

ILLUSTRATIVE VALUES · shares from the Part 4 worksheet

PART 5

How does the model learn the concepts?

A weighted sum combines the two losses

total loss — weight × concept loss — weight × task loss

$$L = \lambda_c L_c + \lambda_y L_y$$

λ_c and λ_y set how much each objective counts

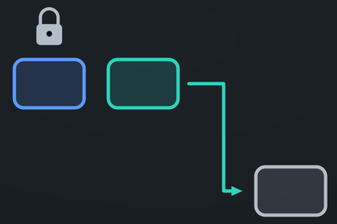
How does the model learn the concepts?

Three ways to train the concept predictor and the task head

INDEPENDENT

each part on its own labels

SEQUENTIAL



predictor first, then the head

JOINT

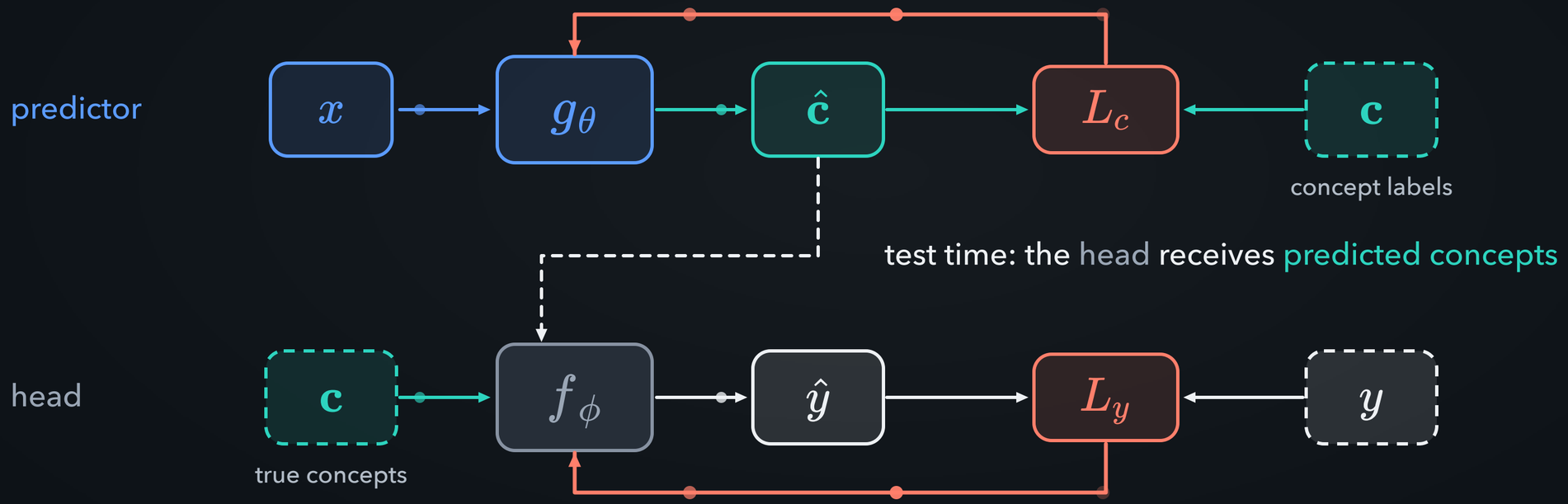
everything at once

Koh et al., ICML 2020

How does the model learn the concepts?

x g_θ $\hat{c}$ f_ϕ $\hat{y}$

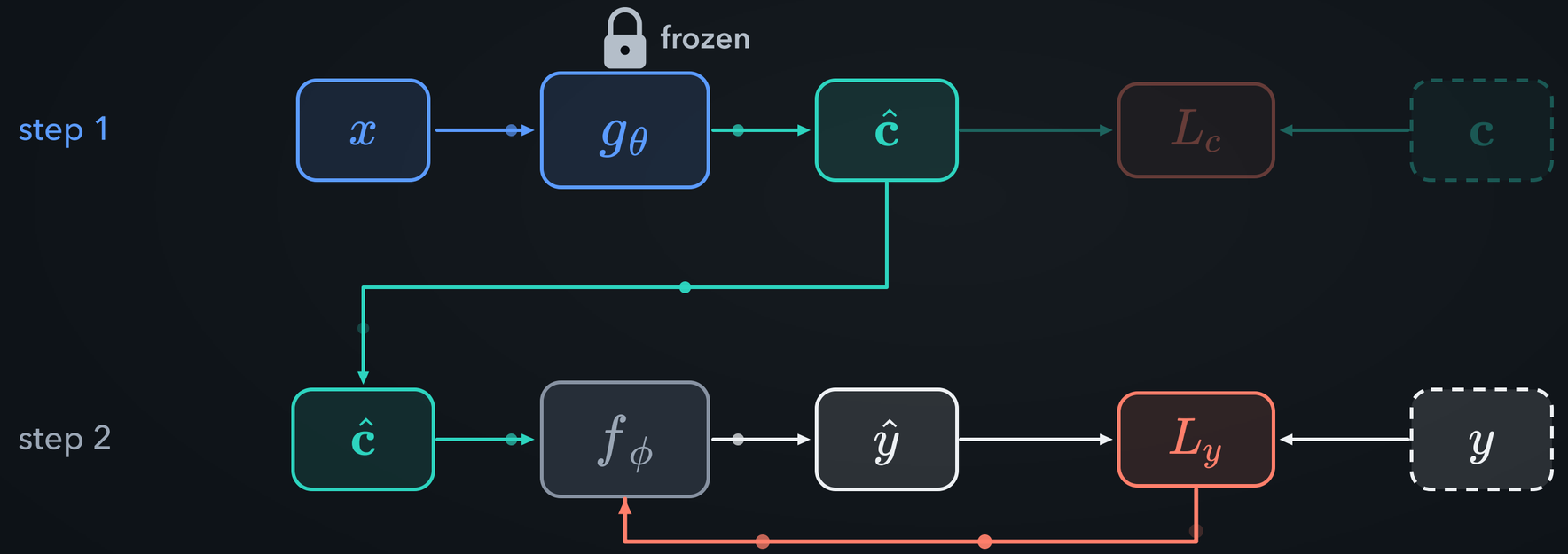
Independent: each part is trained on its own labels



coral: the gradient of each loss, flowing back to what it trains

How does the model learn the concepts?

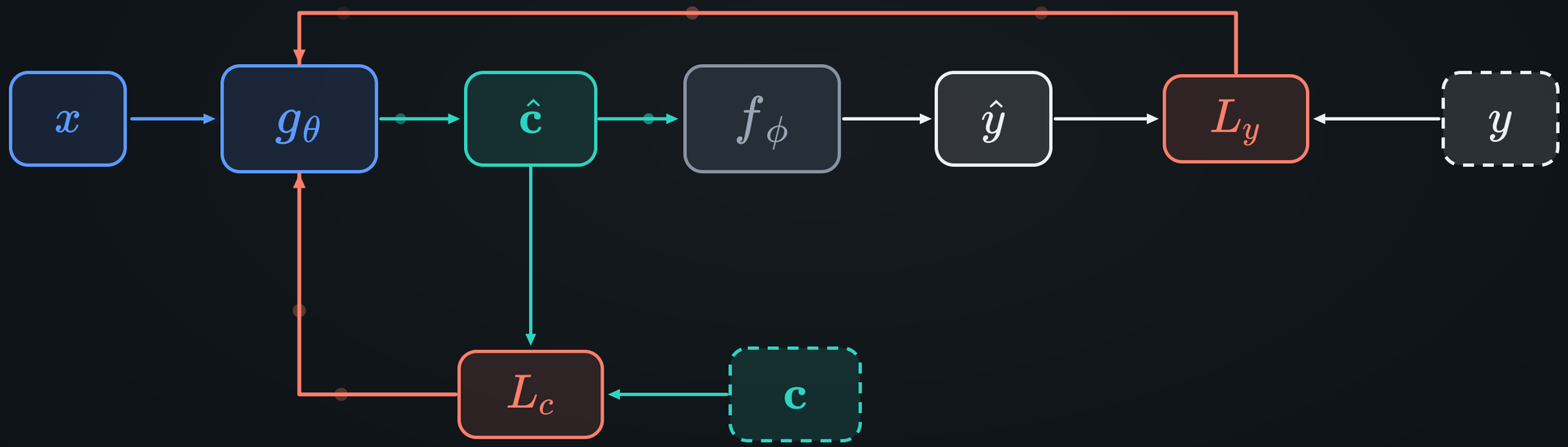
Sequential: freeze the predictor, then train the head on its outputs



the head learns from the predictor's actual mistakes

How does the model learn the concepts?

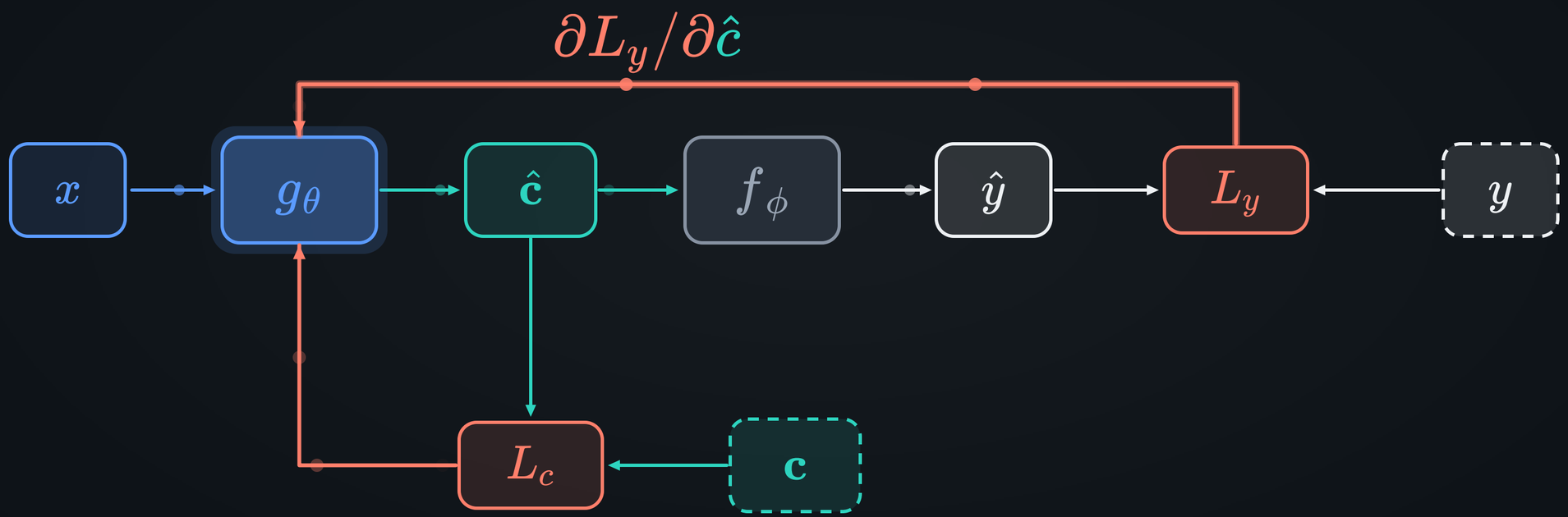
Joint: train the whole system at once, with both losses



both losses train the whole system

How does the model learn the concepts?

In joint training, the task loss also shapes the concept predictor

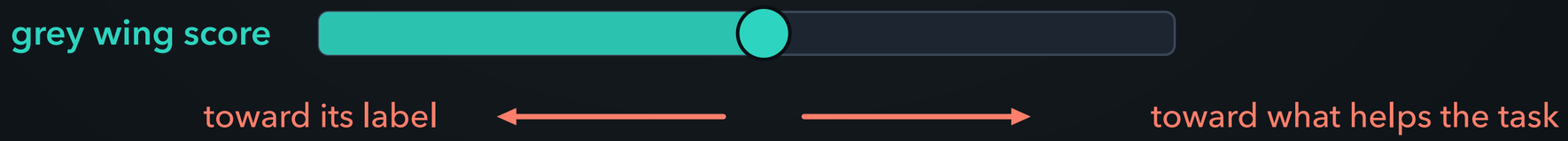


the concept predictor is shaped by both losses

How does the model learn the concepts?

A concept score can be pulled two ways

can improve **task accuracy**, can hurt concept fidelity



whether it happens has to be measured

How does the model learn the concepts?

Our measured run: joint training, $\lambda_c = 2$, $\lambda_y = 1$

concept loss, weight 2 task loss, weight 1

$$L = 2\,L_c + 1\,L_y$$

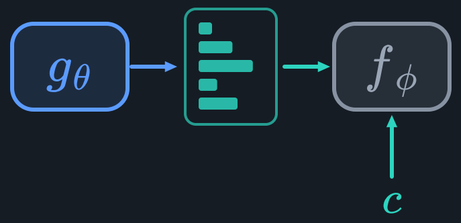
joint training, in the experiment of Part 11

MEASURED · settings of the trained shape model, Part 11

The three schemes on CUB

Does the training scheme matter on real photographs?

independent

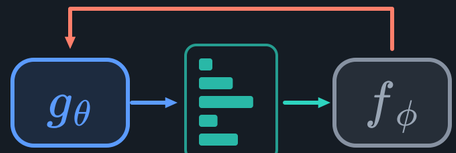


head fitted on true concepts

sequential

predictor frozen, head fitted on its predictions

joint



both trained together on both losses

our small run on CUB: 5,994 training photos, 5,794 test photos

PART 5

The three schemes on CUB

x g_θ $\hat{c}$ f_ϕ $\hat{y}$

Our setup: frozen CLIP features → 112 concepts → 200 species

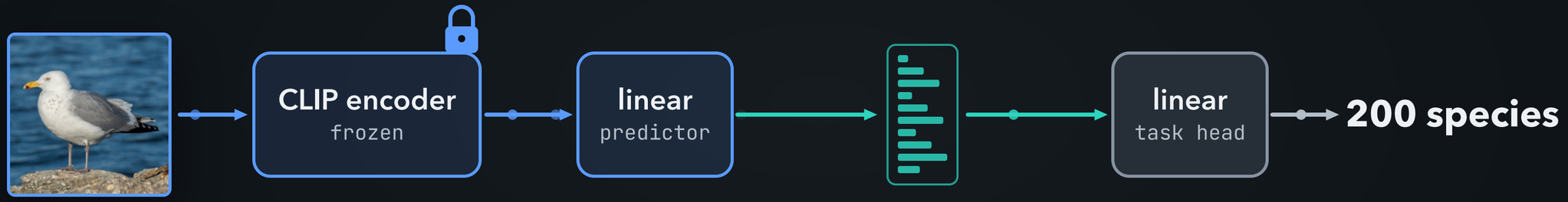


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

x $e(x) \in \mathbb{R}^{512}$ g_θ $\hat{\mathbf{c}} \in [0,1]^{112}$ f_ϕ $\hat{y}$

CUB official split: 5,994 training photos, 5,794 test photos

MEASURED · our run: frozen CLIP ViT-B/32 + linear models, CUB official split, one run

The three schemes on CUB

One photo through the model: features, then concepts, then species

$e(x) \in \mathbb{R}^{512}$

$$\begin{bmatrix} +0.036 \\ +0.003 \\ -0.001 \\ -0.041 \\ \vdots \\ -0.035 \end{bmatrix}$$

$\hat{\mathbf{c}}$

throat: white	0.99
nape: white	0.99
wing: grey	0.98
upperparts: grey	0.97
	⋮

f_ϕ

top 4 of 200 species

Western Gull	0.27
Glaucous-winged Gull	0.22
Slaty-backed Gull	0.13
Herring Gull	0.13

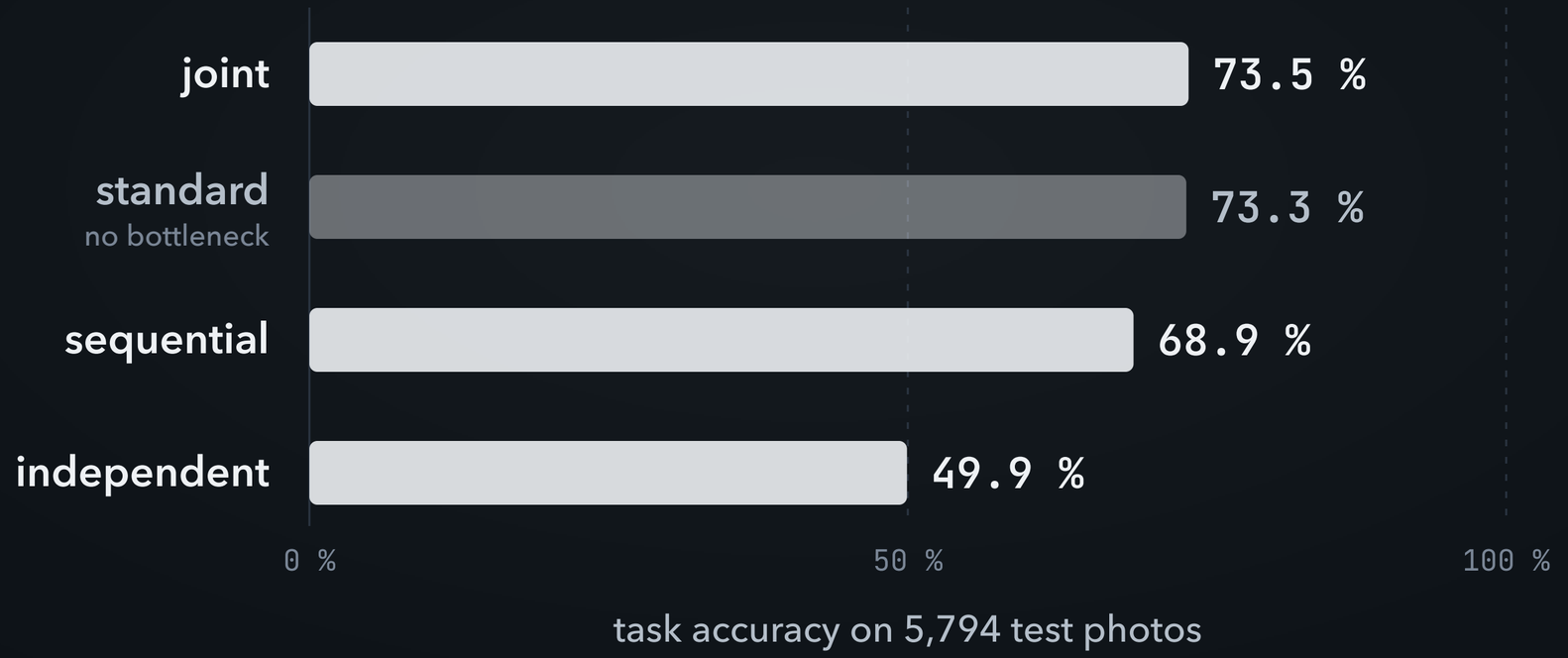
Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

COMPUTED · REAL MODEL · our gull photo: CLIP ViT-B/32 feature, our CUB model

The three schemes on CUB

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Joint training is the most accurate, independent the least

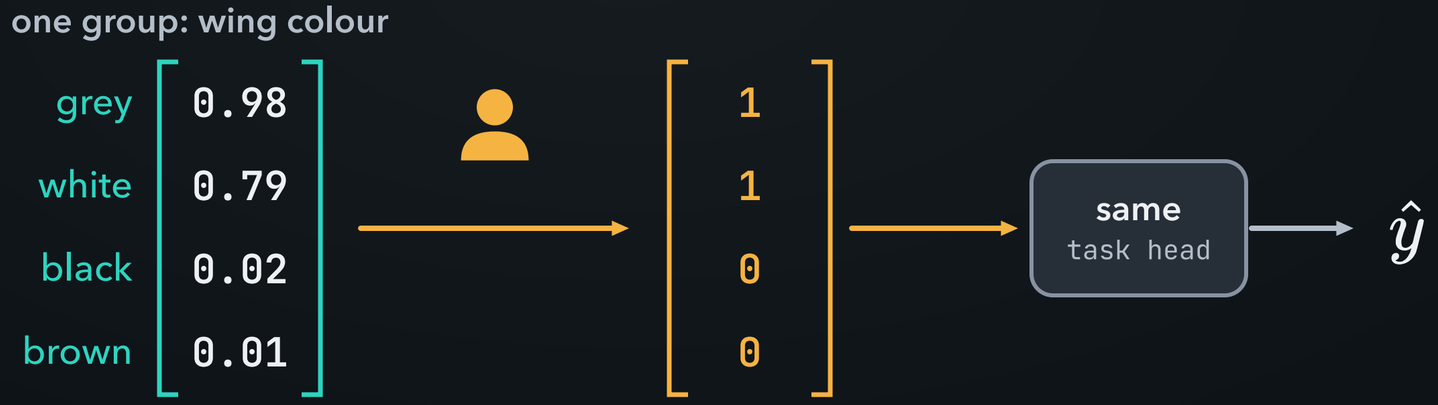


MEASURED · our CUB run, one run

The three schemes on CUB

An intervention replaces predicted values with true values

predicted value $\hat{c}_j$ ← true value c_j

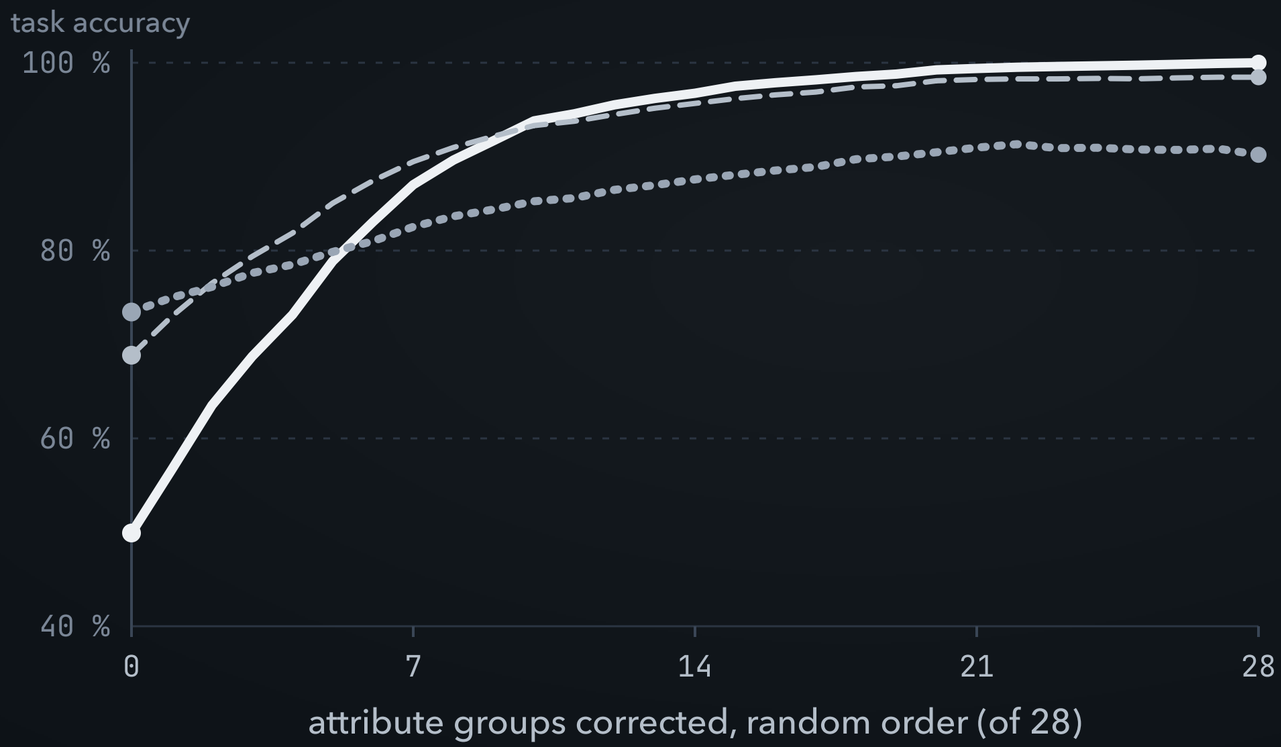


COMPUTED · REAL MODEL · our model on our gull photo; true values: Herring Gull's CUB labels

The three schemes on CUB

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

As more groups are corrected, the independent model gains most



independent	100.0 %
sequential	98.5 %
joint	90.2 %

all 28 groups true: every species has its own pattern, so **independent** reaches 100.0 %

MEASURED · our CUB run, one run, averaged over random orders

Human interventions

A person can fix a concept the model got wrong

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

wingtip black 0.20

the photo and the head stay the same; only a value changes

ILLUSTRATIVE VALUES · hand-written concept value (Part 2)

Human interventions

An intervention has four steps

1. predict concept values
2. a person checks one and corrects it
3. the fixed head recomputes
4. a new **prediction**

$$\hat{\mathbf{c}} = g_\theta(x) \qquad \hat{c}_j \leftarrow c_j \qquad f_\phi \text{ unchanged} \qquad \hat{y}_{\text{new}} = f_\phi(\hat{\mathbf{c}}_{\text{edited}})$$

Human interventions

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Wing colour is one group: correct it together

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

wing colour is one categorical concept: grey → 0, black → 1, together

ILLUSTRATIVE VALUES · hand-written concept values and head

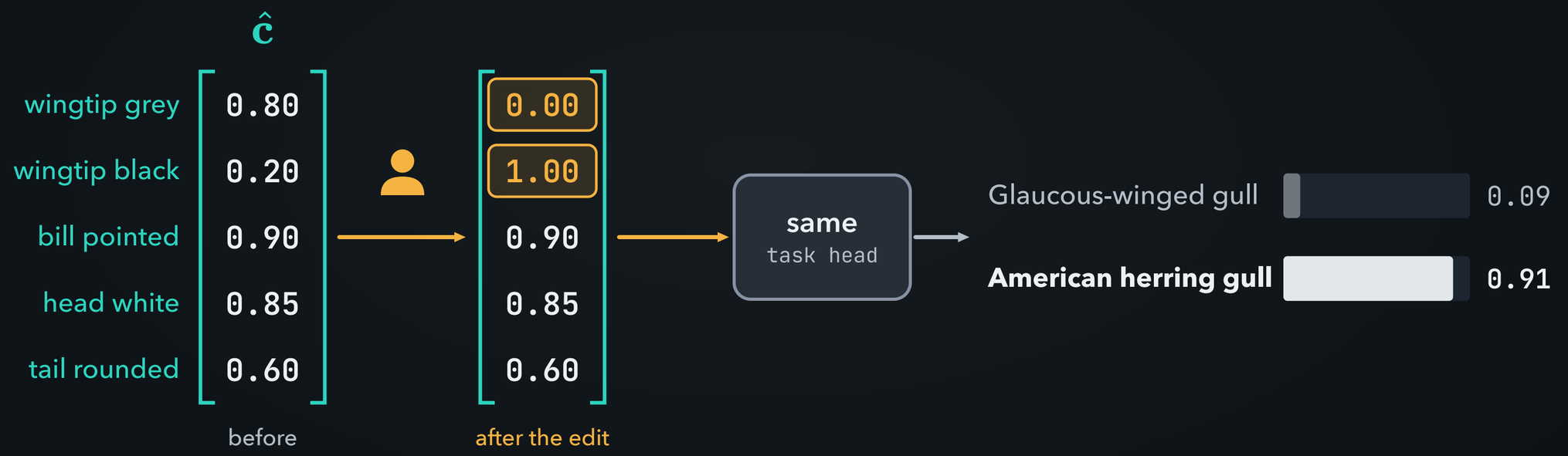
Human interventions

Changing only one value could leave an impossible state

a wingtip has one colour, so the group changes as one

Human interventions

Same head, new values: herring gull rises



ILLUSTRATIVE VALUES · hand-written concept values and head

Human interventions

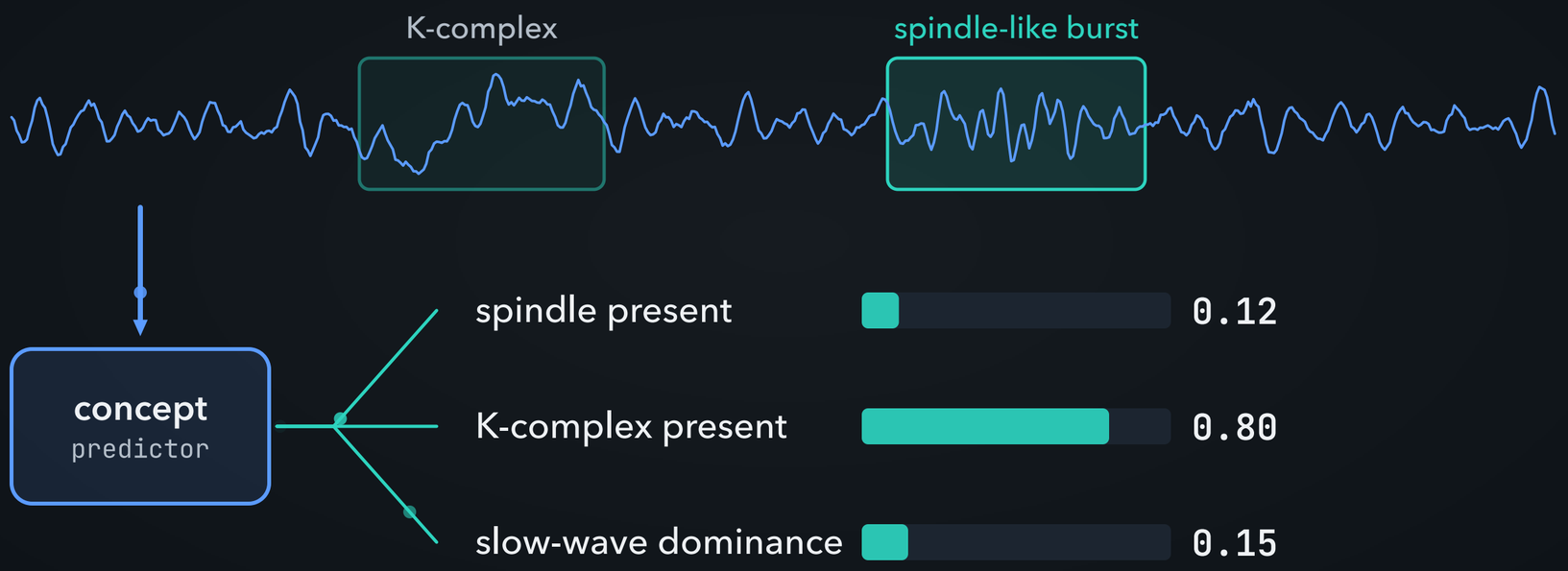
The same mechanism works on a signal

SCHEMATIC · authored trace, not a recording

Human interventions

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

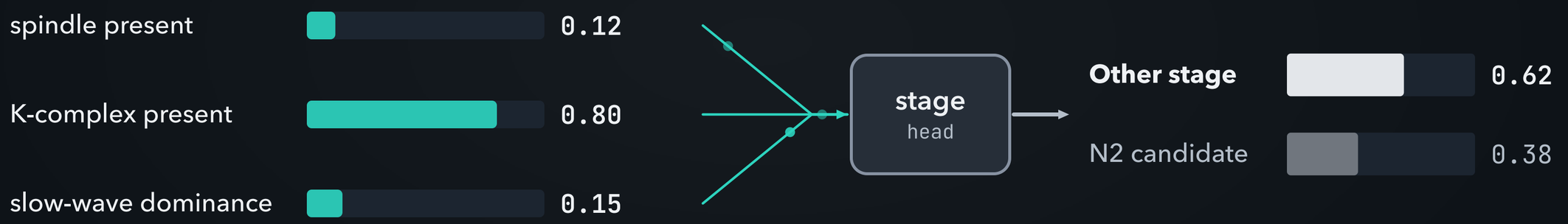
The concept predictor gives spindle a low value



ILLUSTRATIVE VALUES · hand-written concept values; schematic trace

Human interventions

An illustrative stage head favours other stage

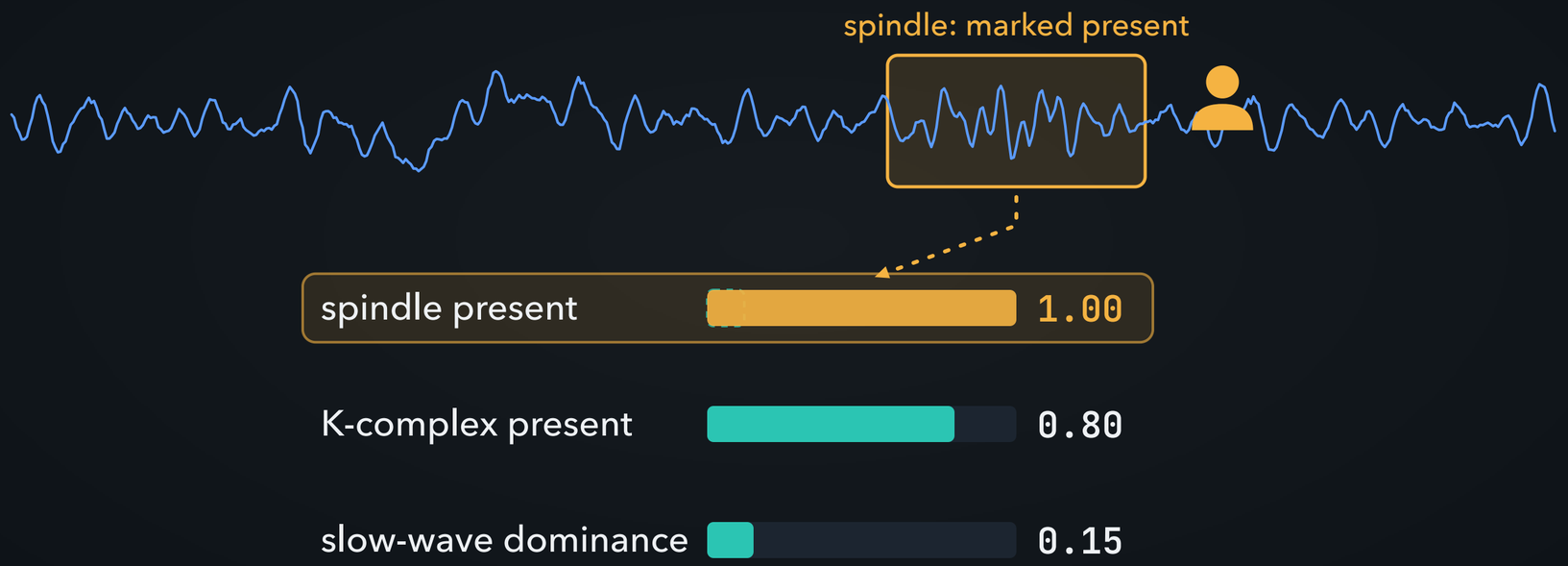


ILLUSTRATIVE VALUES · hand-written stage head

Human interventions

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

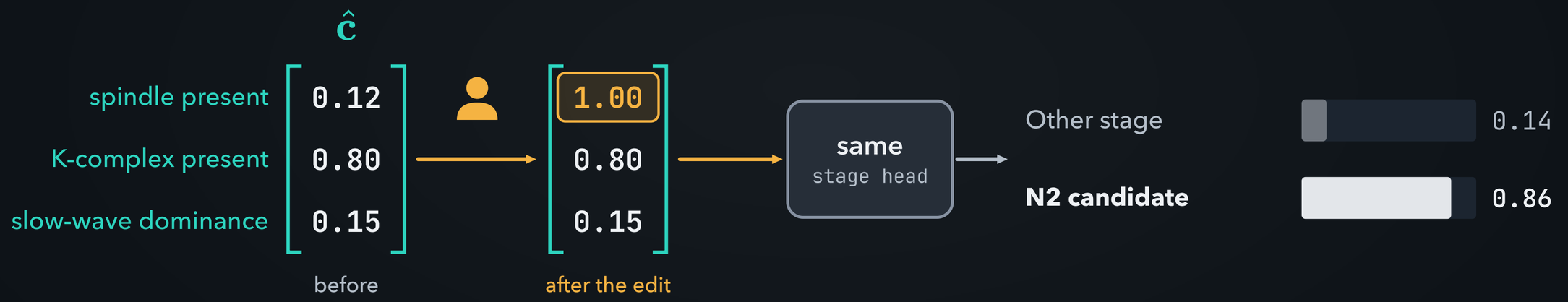
A sleep expert marks the spindle as present



ILLUSTRATIVE VALUES · hand-written concept values; schematic trace

Human interventions

Same head, new input: N2 candidate rises



ILLUSTRATIVE VALUES · hand-written stage head

Human interventions

A demonstration of the mechanism, not a clinical result

real staging: scoring rules applied to 30 s epochs

this stage head was written by hand

SCHEMATIC · authored EEG trace

ILLUSTRATIVE VALUES · hand-written stage head

A correct edit can still hurt

Does a correct edit always help?

gull, Part 2	zebra, Part 4	EEG, Part 6	a correct edit
herring gull 0.29 → 0.91	**zebra** 0.35 → 0.95	**N2 candidate** 0.38 → 0.86	?
✓	✓	✓	✗

so far, every correction helped

PART 7

A correct edit can still hurt

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A support message that belongs in billing

SUPPORT MESSAGE

The app works, but I was charged twice.
Please refund the extra payment.

BILLING QUEUE

where it belongs

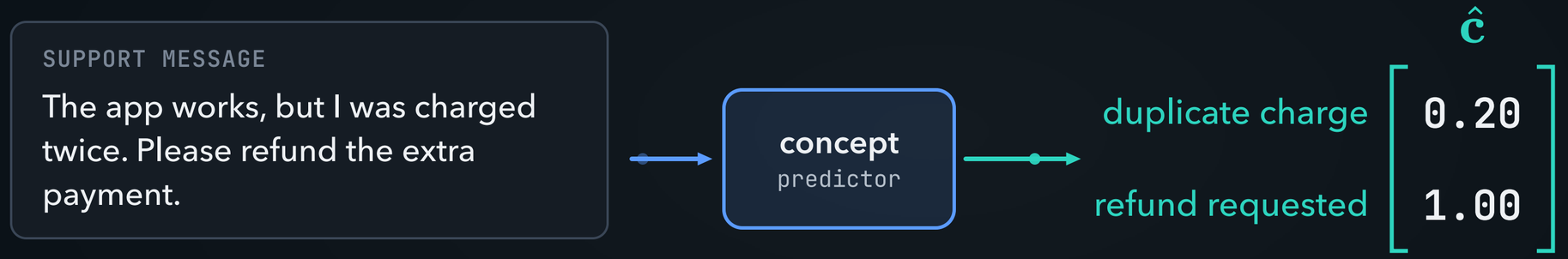
CONSTRUCTED EXAMPLE · an invented message

PART 7

A correct edit can still hurt

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

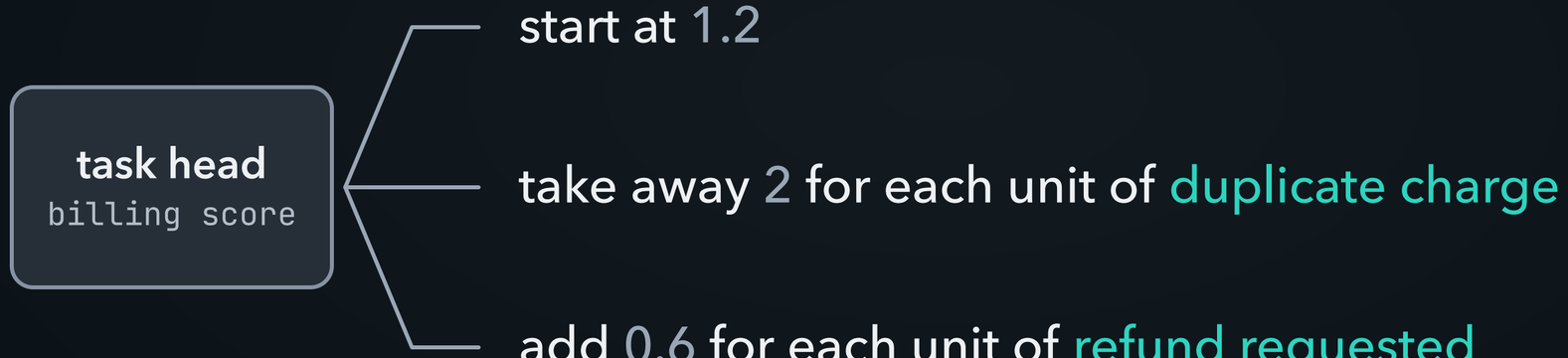
The concept predictor reads two concepts from the message



ILLUSTRATIVE VALUES · hand-written concept values

A correct edit can still hurt

The task head, in words



CONSTRUCTED EXAMPLE · a hand-written head

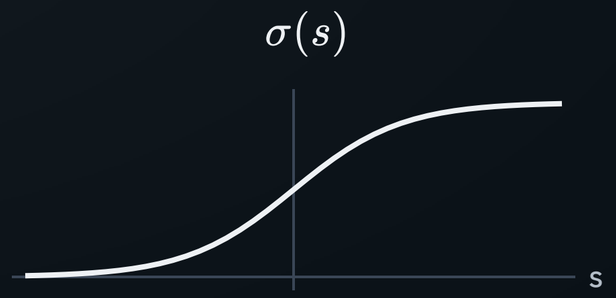
A correct edit can still hurt

The same head in symbols

score start duplicate charge refund requested

$$s = 1.2 - 2\,\hat{c}_{\text{dup}} + 0.6\,\hat{c}_{\text{refund}}$$

a sigmoid turns the score s into the **billing share**



PART 7

A correct edit can still hurt

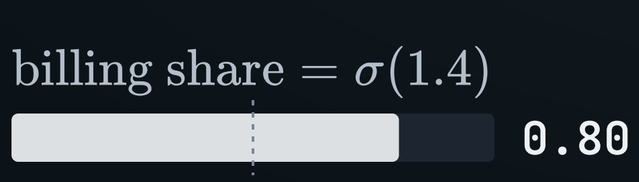
x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

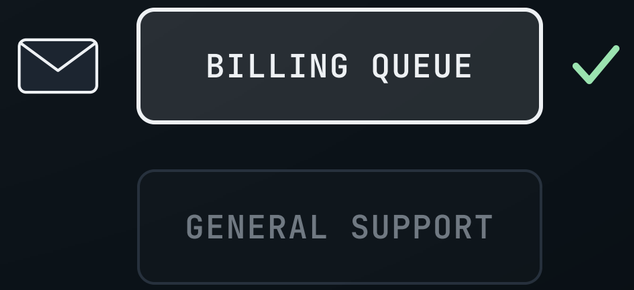
Before the edit, the message goes to billing

$$s = 1.2 - 0.4 + 0.6 = 1.4$$

-2×0.20

0.6×1.00





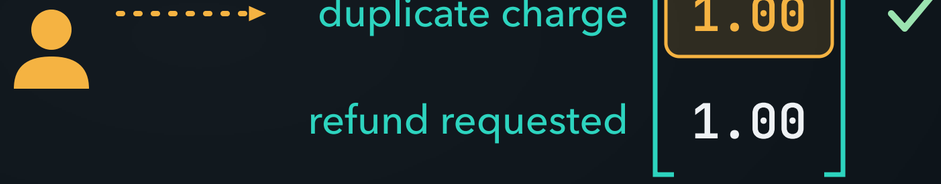
CONSTRUCTED EXAMPLE · hand-written head

A correct edit can still hurt

A reviewer **correctly** sets duplicate charge to 1

SUPPORT MESSAGE

The app works, but I was charged twice. Please refund the extra payment.



matches the message: charged twice

PART 7

A correct edit can still hurt

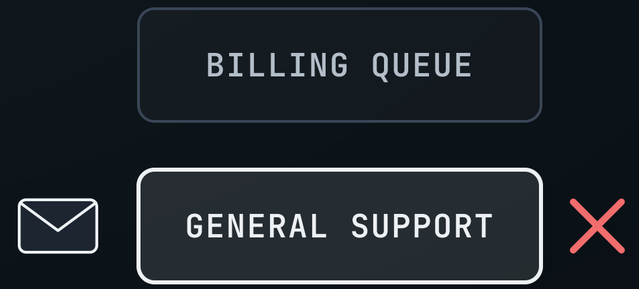
After the edit, the message leaves billing

$$s = 1.2 \quad -2 \quad +0.6 \quad = -0.2$$

-2×1.00

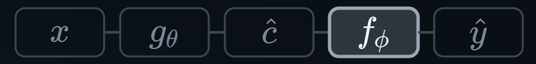
0.6×1.00

$\text{billing share} = \sigma(-0.2)$



CONSTRUCTED EXAMPLE · hand-written head

A correct edit can still hurt



The fault is in the head: a weight with the wrong sign

$$s = 1.2 \; \mathbf{-2}\,\hat{c}_{\text{dup}} + 0.6\,\hat{c}_{\text{refund}}$$

pushes **billing** down as duplicate charge rises

✓ the edit: correct

✗ the weight: wrong sign

CONSTRUCTED EXAMPLE · deliberately faulty head

A correct edit can still hurt

Correcting a concept does not repair the decision rule

measure the effect of an intervention; do not assume it

A correct edit can still hurt

A constructed case: the failure can happen

$$s = 1.2 - 2\,\hat{c}_{\text{dup}} + 0.6\,\hat{c}_{\text{refund}}$$

built faulty on purpose: the failure **can** happen

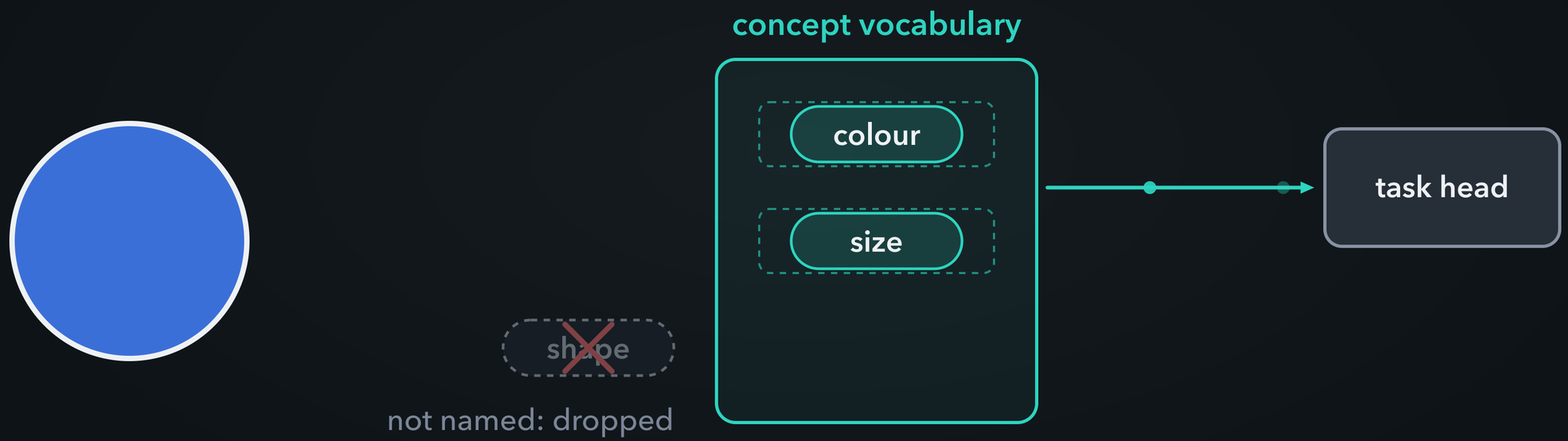
it says nothing about how often it does

CONSTRUCTED EXAMPLE · deliberately faulty head: existence, not frequency

What if the concepts are incomplete?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The bottleneck can pass on only what the vocabulary names



What if the concepts are incomplete?

Two objects from two different classes

fill: the object's own colour

CONSTRUCTED EXAMPLE · two-object example

What if the concepts are incomplete?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The vocabulary names only colour and size

vocabulary: colour, size

both: blue = 1, large = 0

CONSTRUCTED EXAMPLE · two-object example

What if the concepts are incomplete?

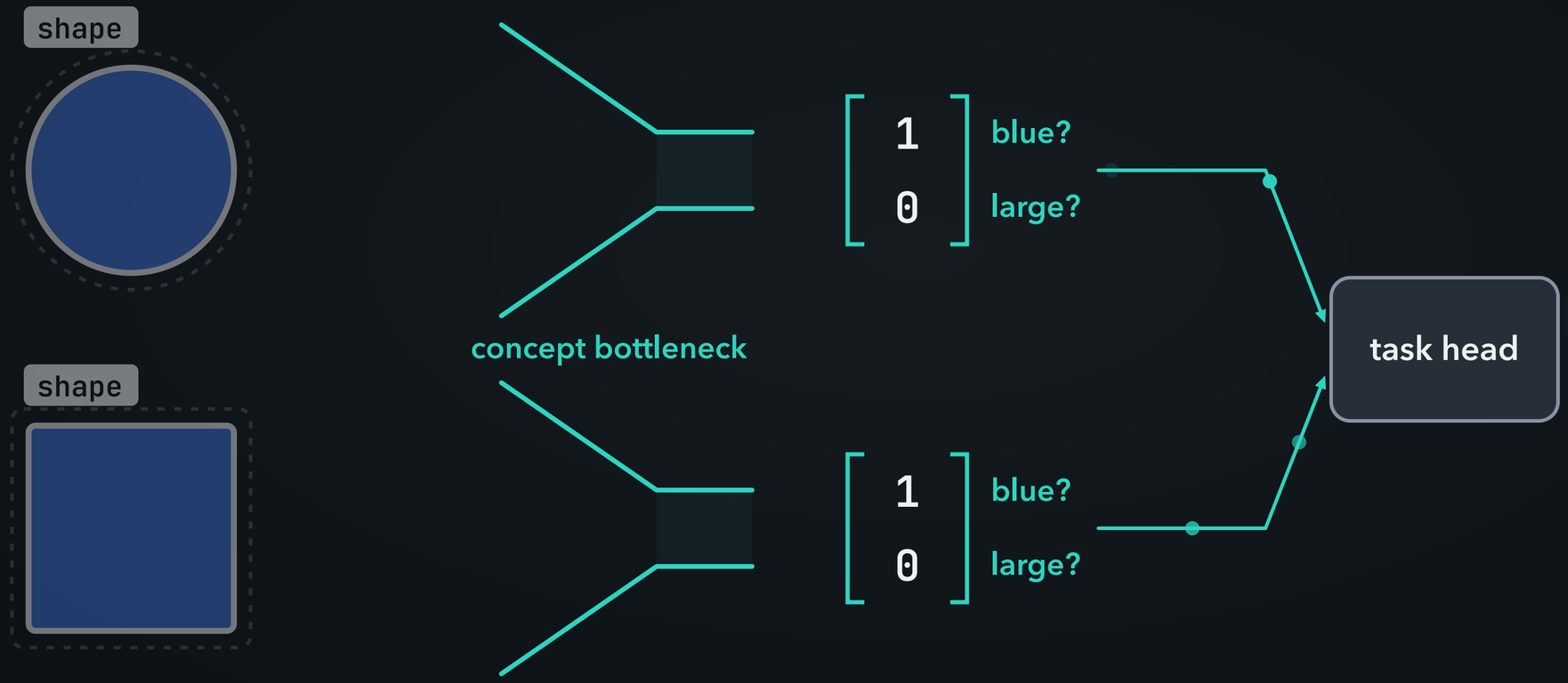
Both objects get the same concept vector

$$c(x_1) = c(x_2), \quad y_1 \neq y_2$$

What if the concepts are incomplete?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Shape never reaches the task head

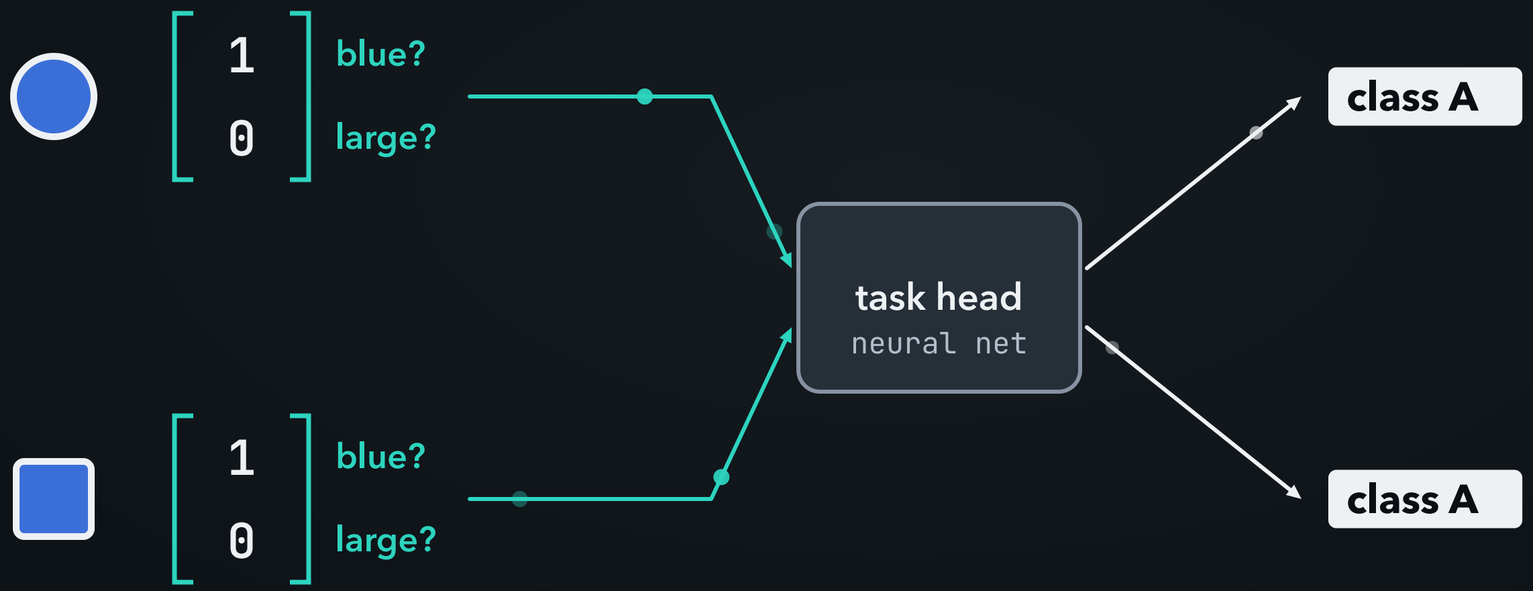


shape: in the image, not in the vector

What if the concepts are incomplete?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

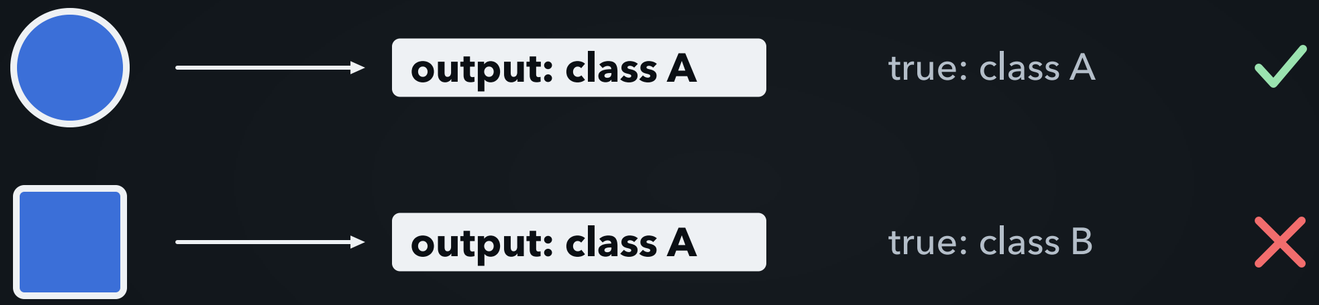
The same input to the head gives the same output



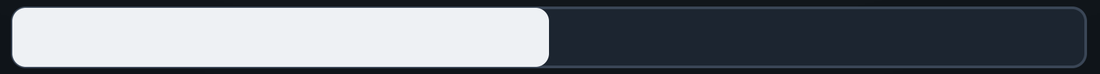
no head can recover what never reached it

What if the concepts are incomplete?

The best possible accuracy is one half



best possible accuracy

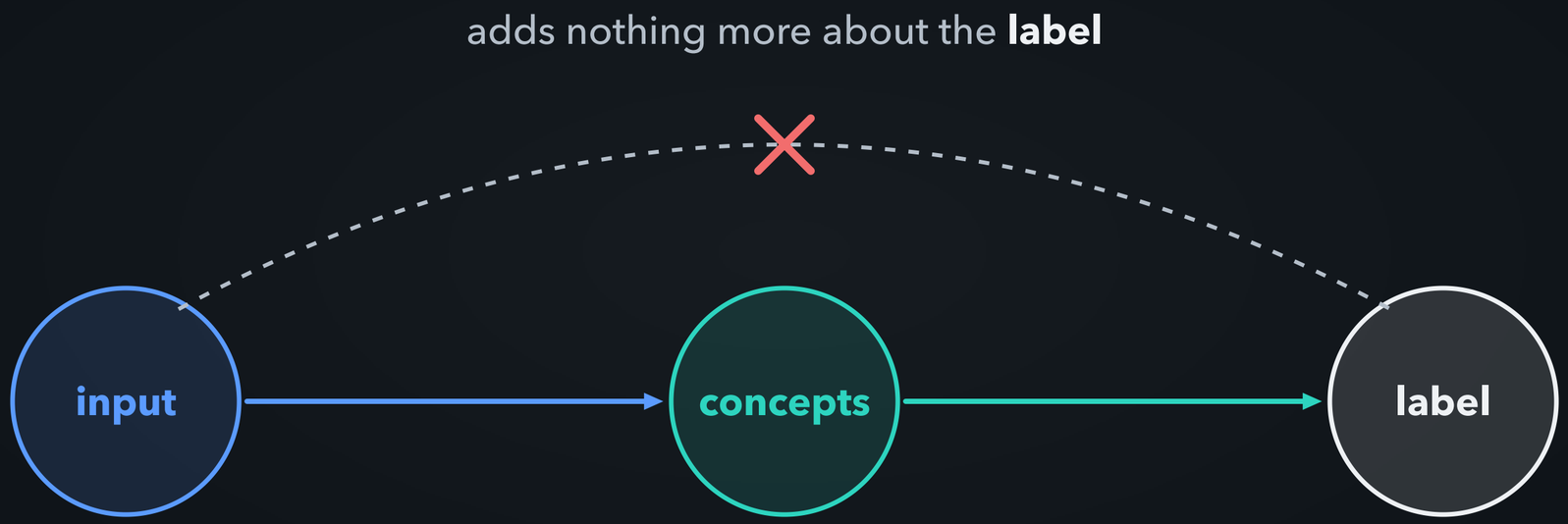


50%

CONSTRUCTED EXAMPLE · two-object example

What if the concepts are incomplete?

Sufficiency: the concepts say all the input says about the label



ideal: **sufficiency**

What if the concepts are incomplete?

Sufficiency in symbols

label independent of input given concepts

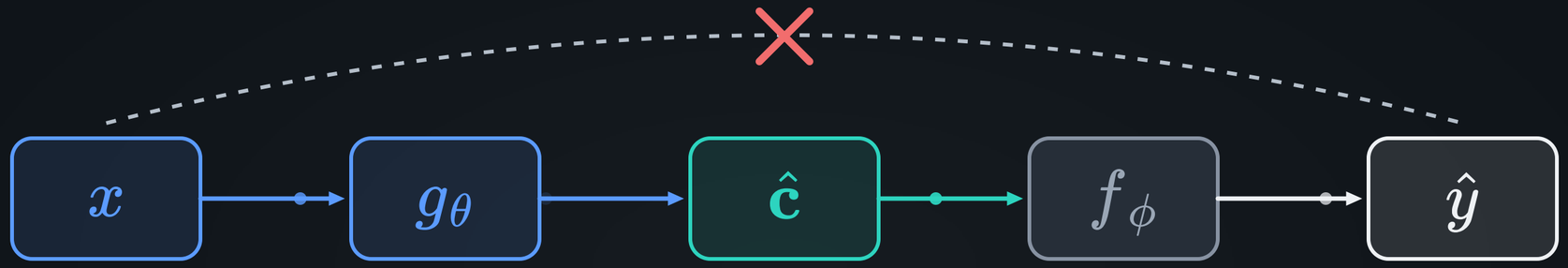
$$Y \not\perp\!\!\!\perp X \mid C$$

here it fails

the **label** depends on shape, which C lacks

What if the concepts are incomplete?

A strict CBM guarantees something weaker



the **prediction** depends on the input only through the concepts

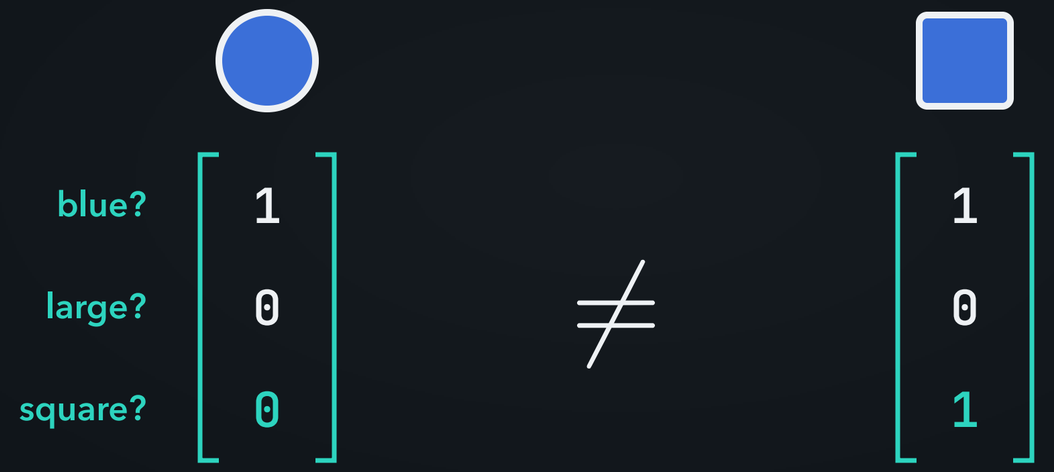
$$\hat{y} = f_\phi(\hat{\mathbf{c}}), \qquad \hat{\mathbf{c}} = g_\theta(x)$$

whether the true **label** does: a property of the vocabulary

What if the concepts are incomplete?

The fix is a richer vocabulary: add shape

vocabulary: colour, size, + shape



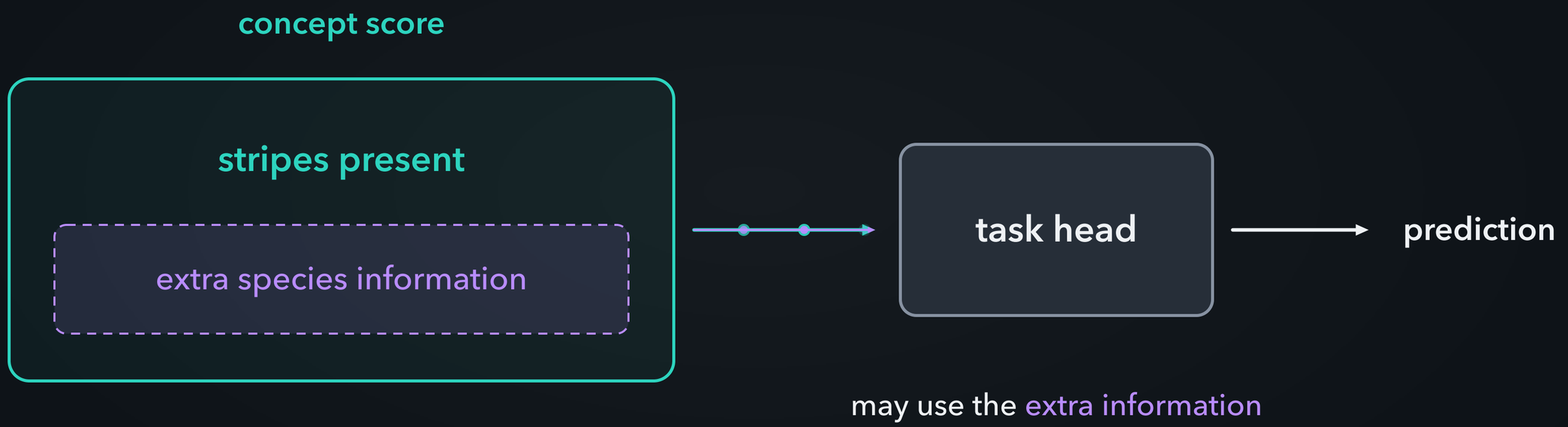
the two vectors now differ

CONSTRUCTED EXAMPLE · two-object example

Concept leakage

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A concept score can carry more than its name says



Test whether correcting a concept has the intended effect

SCHEMATIC · possible information inside a score, not a measured model output

Modern concept bottleneck architectures

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Labelling every concept on every image is the main cost

stripes	1	1	0	0	0	0
horse-like shape	1	1	1	1	0	0
long neck	0	0	0	0	1	1

36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

a person labels every concept on every image

the main cost, which newer methods try to reduce

Modern concept bottleneck architectures

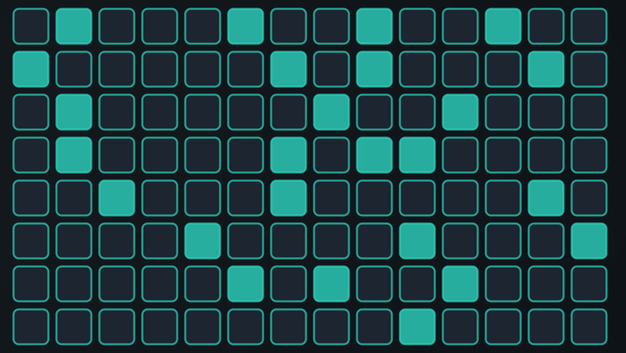
One training record: an image, its concept labels and a class

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

```
{
  "image": "zebra_12.jpg",
  "concepts": {
    "stripes": 1,
    "horse_like_shape": 1,
    "long_neck": 0
  },
  "label": "zebra"
}
```

CONSTRUCTED EXAMPLE · a teaching record

CUB birds, as used by Koh et al. 2020



112 concept labels per photograph

DATASET · Herring Gull's labels in CUB

Modern concept bottleneck architectures

Rung one: images, concept labels and classes

rung 1 · task data

$$\{x,\ c,\ y\}$$

images, concept labels, **class labels**

CBM, concept embedding model (CEM)

rung 2

rung 3

less supervision

Modern concept bottleneck architectures

Rung two: keep class labels, drop per-image concept labels

rung 1 · task data

$\{x, c, y\}$

images, concept labels, **class labels**

CBM, concept embedding model (CEM)

drop

concept labels

rung 2 · task data

$\{x, y\}$

images and **class labels**

post-hoc CBM, label-free CBM, LaBo, VLG-CBM

rung 3

less supervision

Language in a Bottle (LaBo)

Vision-Language-Guided Concept Bottleneck Model (VLG-CBM)

Modern concept bottleneck architectures

Rung three: no training for this task

rung 1 · task data

$\{x, c, y\}$

images, concept labels, **class labels**

CBM, concept embedding model (CEM)

drop

concept labels

rung 2 · task data

$\{x, y\}$

images and **class labels**

post-hoc CBM, label-free CBM, LaBo, VLG-CBM

drop

task training

rung 3 · task data

$\varnothing$

no training for this task

classification by description

less supervision

Language in a Bottle (LaBo)

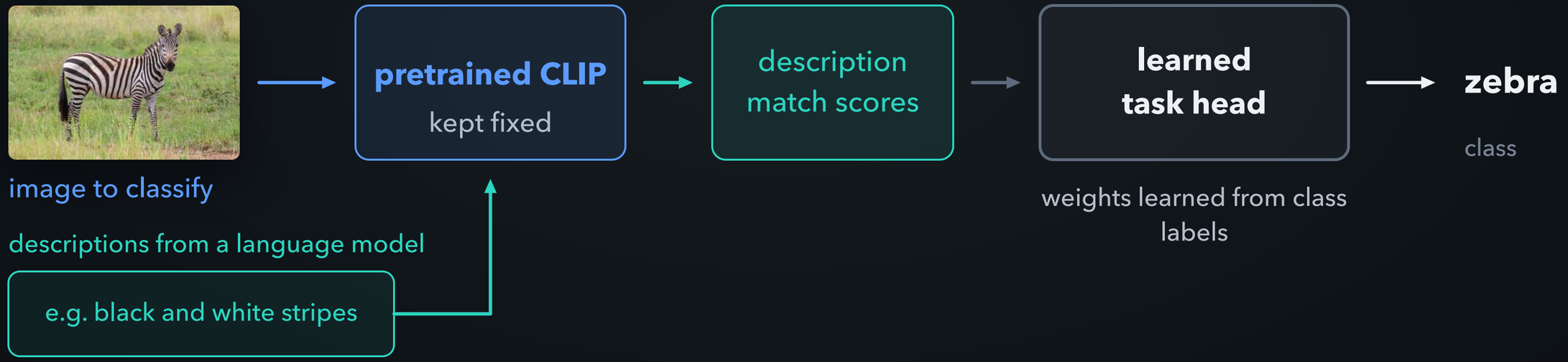
Vision-Language-Guided Concept Bottleneck Model (VLG-CBM)

Modern concept bottleneck architectures

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Rung two: class labels teach the task head

Example: Language in a Bottle (LaBo)



Rung two: **class labels** teach us how to combine the scores.

`SCHEMATIC · example path; no measured scores shown`

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

Modern concept bottleneck architectures

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Rung three: combine the match scores with a fixed rule

Classification by description

no task-labelled examples
no fitting

image to classify

pretrained CLIP
kept fixed

description
match scores

fixed
class averages

zebra

highest average

average the description
scores within each class

descriptions from a language model

e.g. black and white stripes

Rung three: combine the scores with a **fixed average**.

`SCHEMATIC · example path; no measured scores shown`

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

Modern concept bottleneck architectures

Fewer concept labels, so check the concepts another way

	rung 1	rung 2	rung 3
concept labels for each task image		none	none
someone checks each concept against the photo		?	?

without per-image concept labels, check the concepts another way

Modern concept bottleneck architectures

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

For each variant, ask: is the concept really in the image?

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

who checks that a named concept is really visible in the image?

we come back to it later in this part: grounding the concepts

`black and white stripes`

Modern concept bottleneck architectures

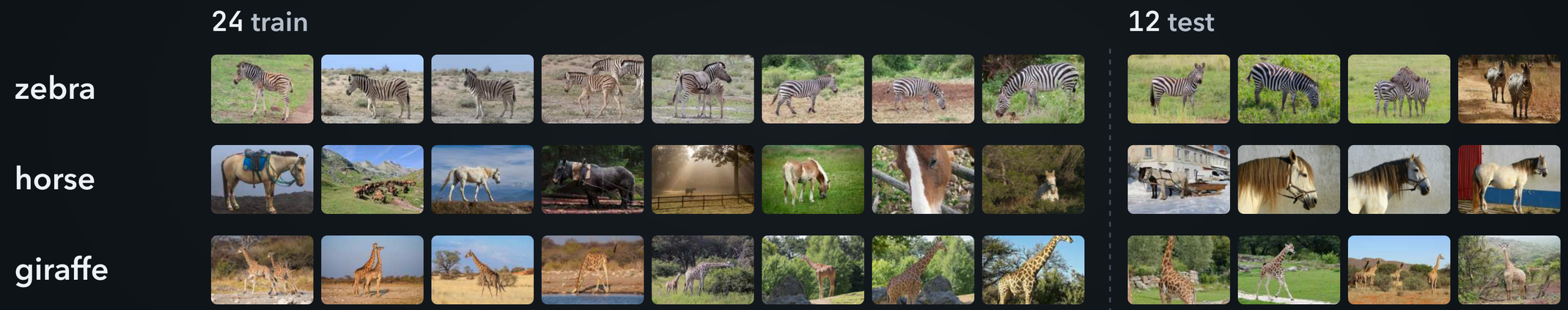
A variant can change four things

what changes	plain CBM
concept source	people label every concept
detector	a trained concept predictor
representation	one number per concept
what reaches the head	only the concept values

Modern concept bottleneck architectures

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

One running example: photographs of zebras, horses and giraffes



36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

the three classes from Part 4

Modern concept bottleneck architectures

For each variant: how the path from image to concepts to class changes

$$x \longrightarrow \hat{c} \longrightarrow \hat{y}$$

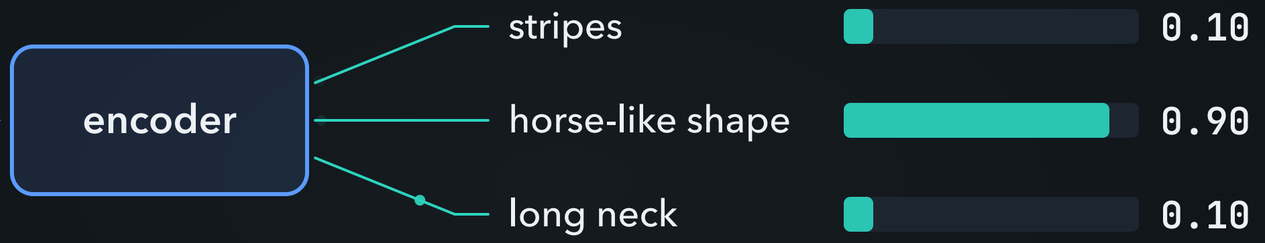
each variant, in symbols		
concept predictor	g_θ	· · ·
what reaches the head	$\hat{c}$	· · ·
task head	f_ϕ	· · ·
what is trained		· · ·
data needed		· · ·

Concept embedding models

One number per concept is a narrow channel

Concept Embedding Model (CEM)

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0



one number per concept: a narrow channel

Part 8: an incomplete vocabulary

`blue 1, large 0 for both`

same values, so best accuracy 0.50

ILLUSTRATIVE VALUES · hand-chosen values

Concept embedding models

What does a prototype mean here?

An embedding is a short list of learned numbers.

stripes present

$$\begin{bmatrix} 0.82 \\ 0.30 \end{bmatrix}$$

stripes absent

$$\begin{bmatrix} 0.22 \\ 0.74 \end{bmatrix}$$

Both vectors come from the current image.

These are not stored reference photos or class averages.

The coordinates are learned features, without individual names.

ILLUSTRATIVE VALUES · hand-chosen vector values

Concept embedding models

Learned layers turn the image into the two vectors

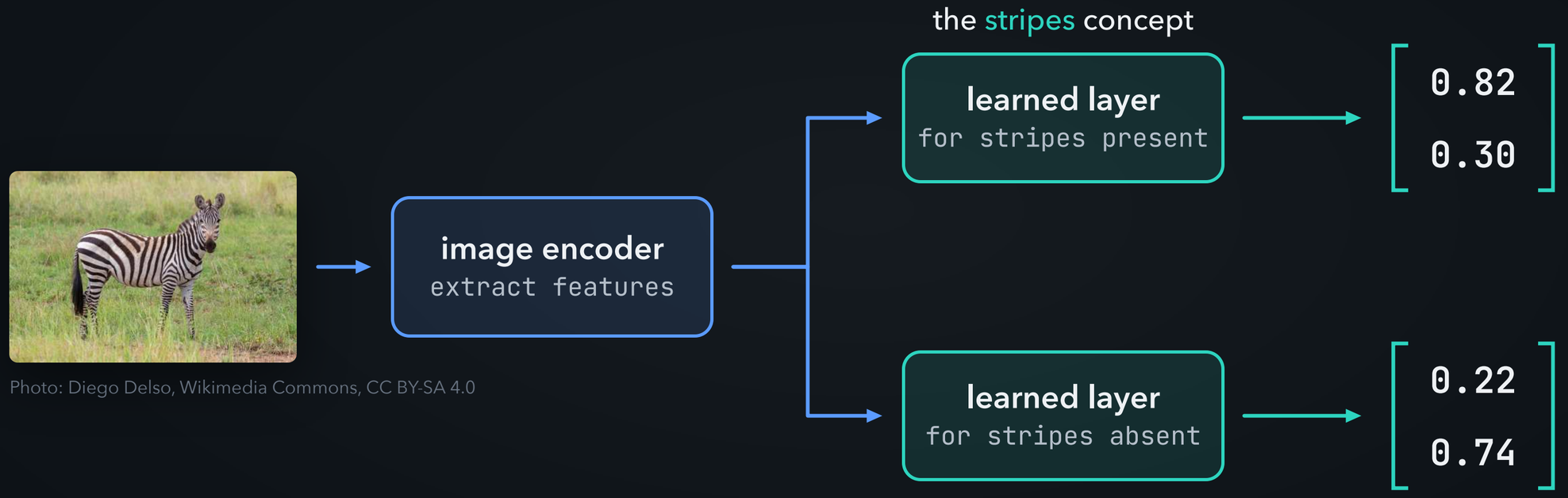


Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

Training learns the **weights** of these layers.

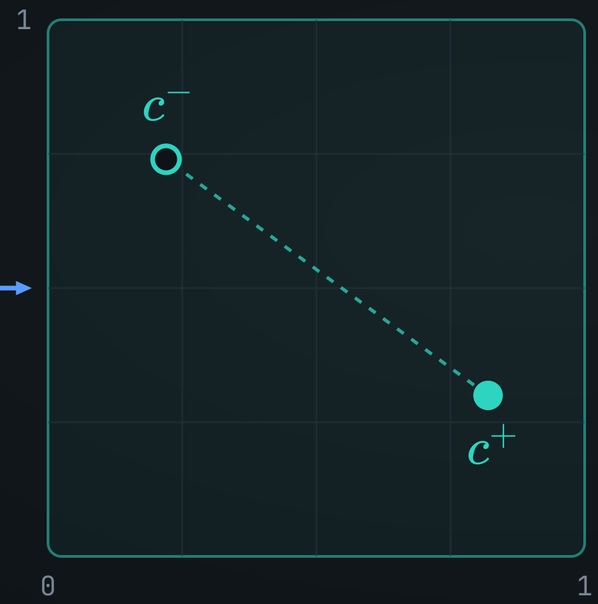
Keep the weights fixed: a new image can still produce different vectors.

ILLUSTRATIVE VALUES · hand-chosen outputs, not a trained run

Concept embedding models

Each concept gets two embeddings: active and inactive

encoder



stripes: two embeddings in one plane

$$c^+ = \begin{bmatrix} 0.82 \\ 0.30 \end{bmatrix}$$ active

$$c^- = \begin{bmatrix} 0.22 \\ 0.74 \end{bmatrix}$$ inactive

one pair for every concept, computed from the image

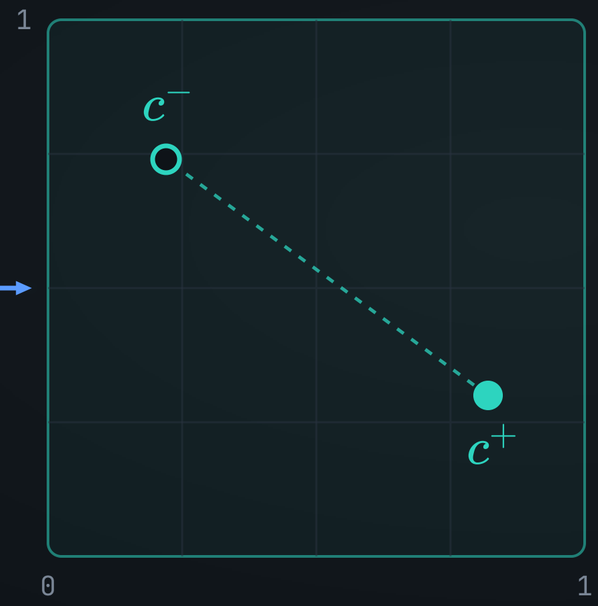
embedding: a short list of numbers (here two)

ILLUSTRATIVE VALUES · hand-chosen 2-D embeddings

Concept embedding models

A learned scorer predicts the probability p

encoder



stripes: two embeddings in one plane

a learned scorer reads both vectors

and predicts the probability of stripes

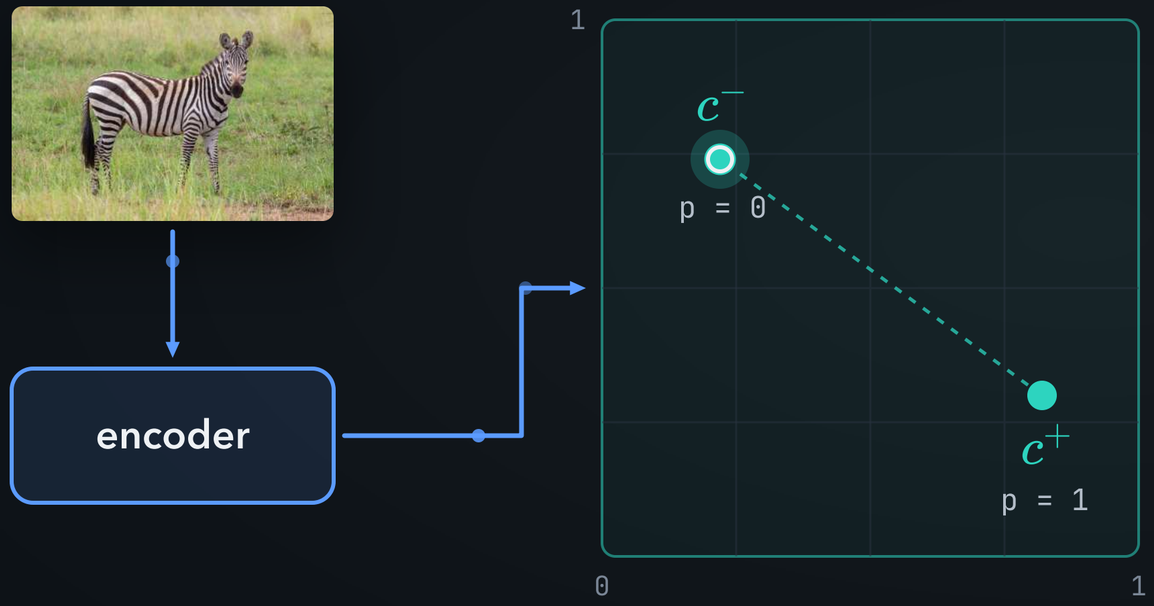
$$p = 0.1$$

suppose it misses the stripes, as in Part 4

ILLUSTRATIVE VALUES · hand-chosen 2-D embeddings

Concept embedding models

The concept embedding mixes the two, weighted by p



stripes: two embeddings in one plane

mixed = p × active + (1 − p) × inactive

$$\hat{c} = p\,c^+ + (1-p)\,c^-$$

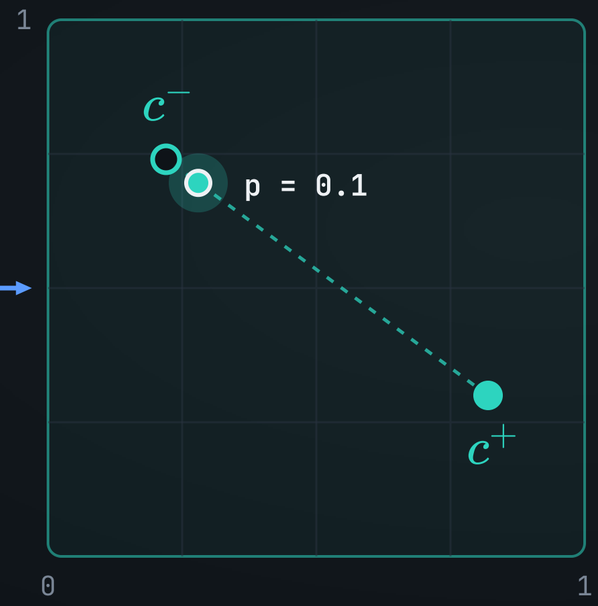
the mixed point lies on the segment from inactive to active

ILLUSTRATIVE VALUES · hand-chosen 2-D embeddings

Concept embedding models

With p = 0.1, the mixed vector sits close to inactive

encoder



stripes: two embeddings in one plane

$\hat{c}$ for stripes, with p = 0.1:

first entry:

$$0.1 \times 0.82 + 0.9 \times 0.22 = 0.28$$

$$0.1 \begin{bmatrix} 0.82 \\ 0.30 \end{bmatrix} + 0.9 \begin{bmatrix} 0.22 \\ 0.74 \end{bmatrix} = \begin{bmatrix} 0.28 \\ 0.70 \end{bmatrix}$$

close to inactive

ILLUSTRATIVE VALUES · hand-chosen 2-D embeddings

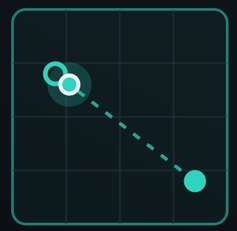
Concept embedding models

At test time, the head reads the mixed vectors

ILLUSTRATIVE VALUES · hand-chosen embeddings and head

stripes
p = 0.10
(0.280, 0.696)

horse-like shape
p = 0.90
(0.705, 0.440)



long neck
p = 0.10
(0.265, 0.650)

6 numbers

0.280
0.696
0.705
0.440
0.265
0.650

task head
then softmax

zebra 0.14
horse 0.72
giraffe 0.13

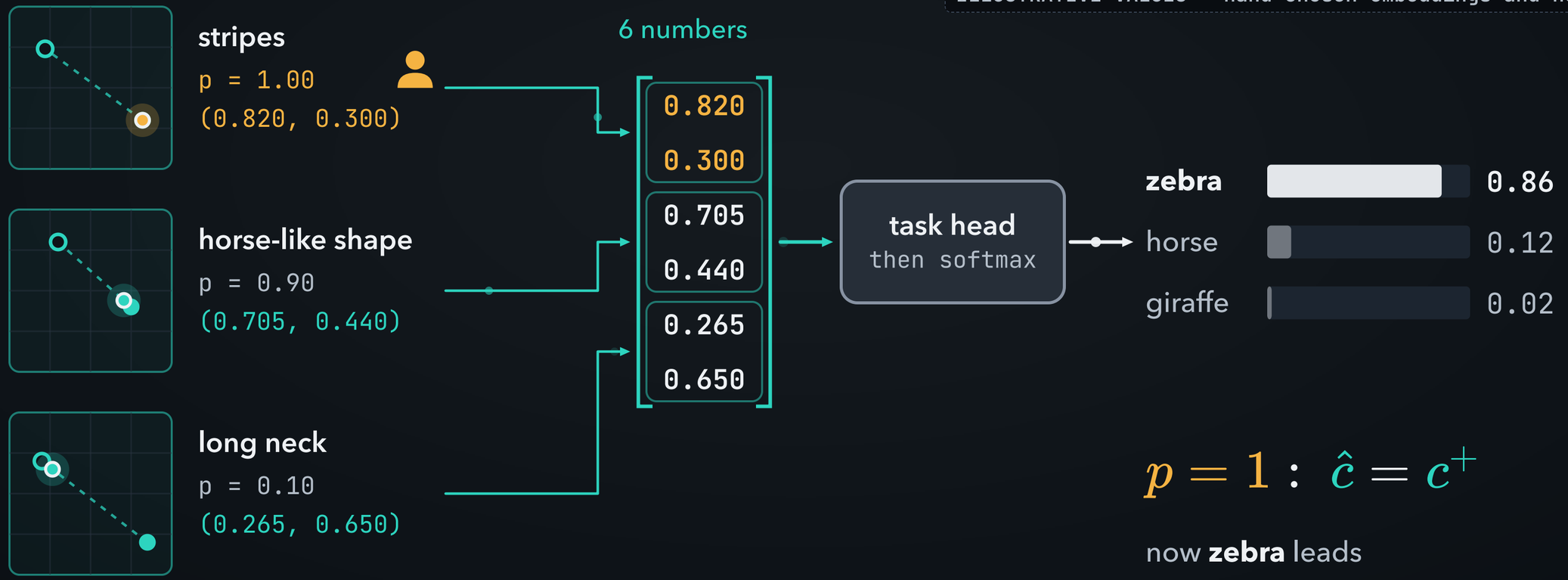
fixed weights; no true labels needed

Concept embedding models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

An optional correction changes p, with weights fixed

ILLUSTRATIVE VALUES · hand-chosen embeddings and head



$p = 1 : \ \hat{c} = c^{+}$

now **zebra** leads

Concept embedding models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Training uses concept labels and class labels

Training example: photo + concept labels + **class label**

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

encoder

vectors and p
learned layers + scorer

mixed vectors

task head
class prediction

class label: **zebra**

concept label: stripes present

concept loss
p versus true concept

task loss
compare with class: zebra

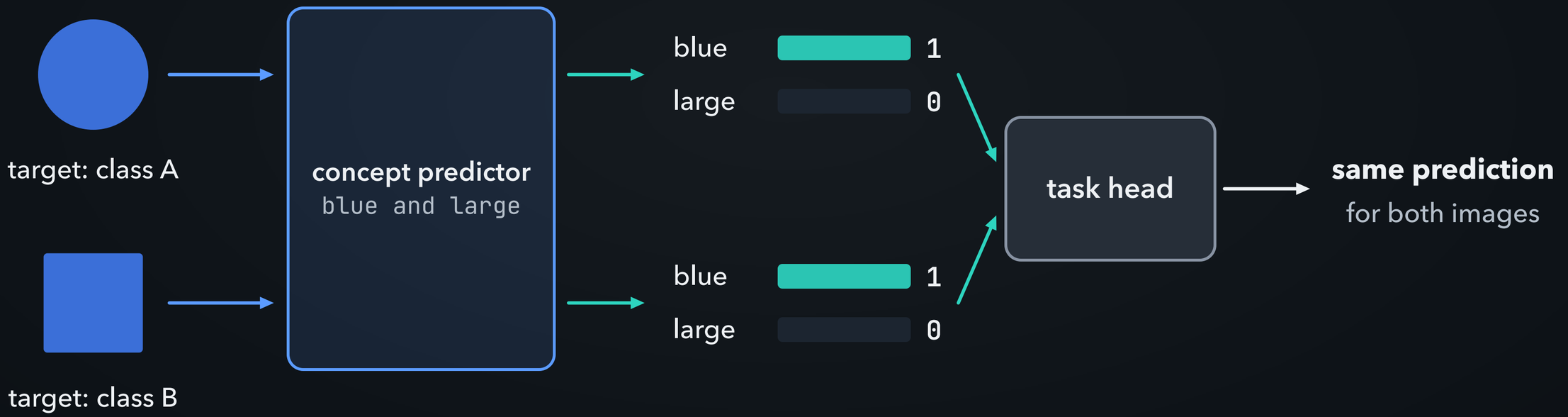
Backpropagation updates the encoder, vector layers, scorer and head.

Concept embedding models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The old problem: the head receives identical values

Part 8: same colour and size, but different classes



Shape is visible in the image, but absent from the head's input.

CONSTRUCTED EXAMPLE · Part 8's two objects; exact concept values

Concept embedding models

Same blue probability, different blue embeddings

Only blue is shown. The large embedding is identical for both images.

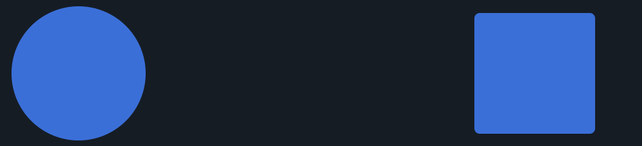
Same answer to ‘is it blue?’ does not mean the same vector.

ILLUSTRATIVE VALUES · hand-chosen vectors; mixing computed; no trained shape CEM

Concept embedding models

Why would training keep the shape information?

Concept labels

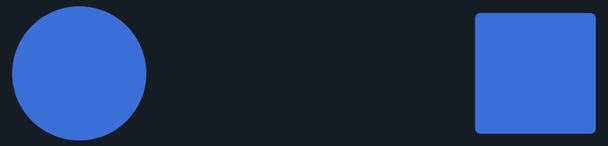


blue, small

blue, small

Concept loss: both should be blue

Class labels



class A

class B

Task loss: answers should differ

The task loss can encourage the vectors to retain shape.

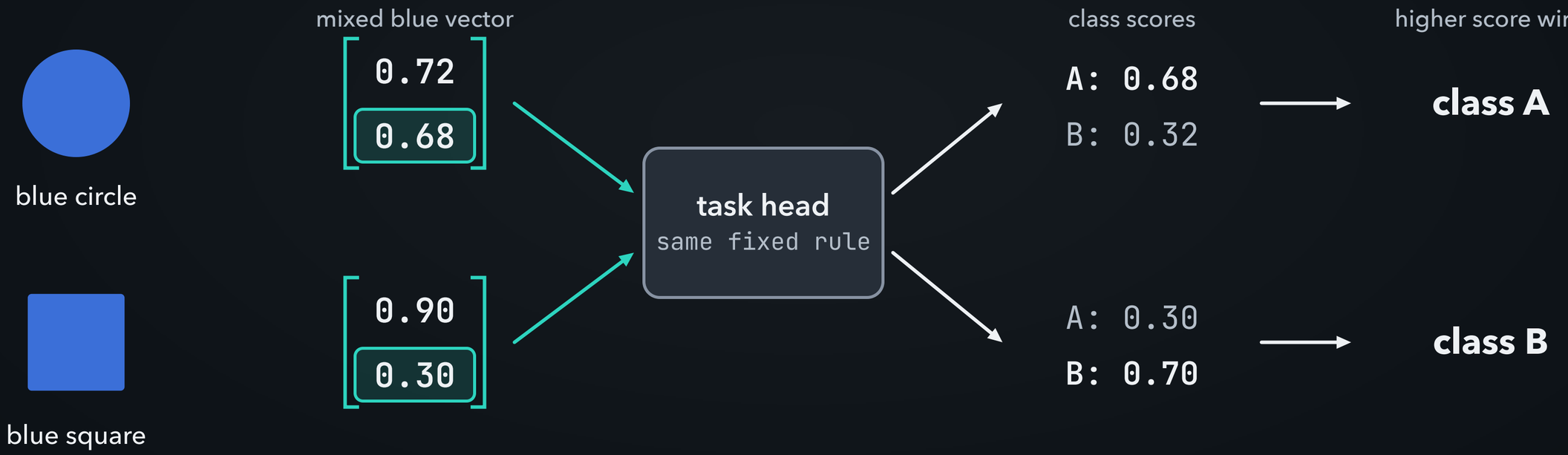
We still have no named shape concept.

Concept embedding models

Different vectors let the same head give different answers

Illustrative head: score A = second entry

score B = 1 − second entry

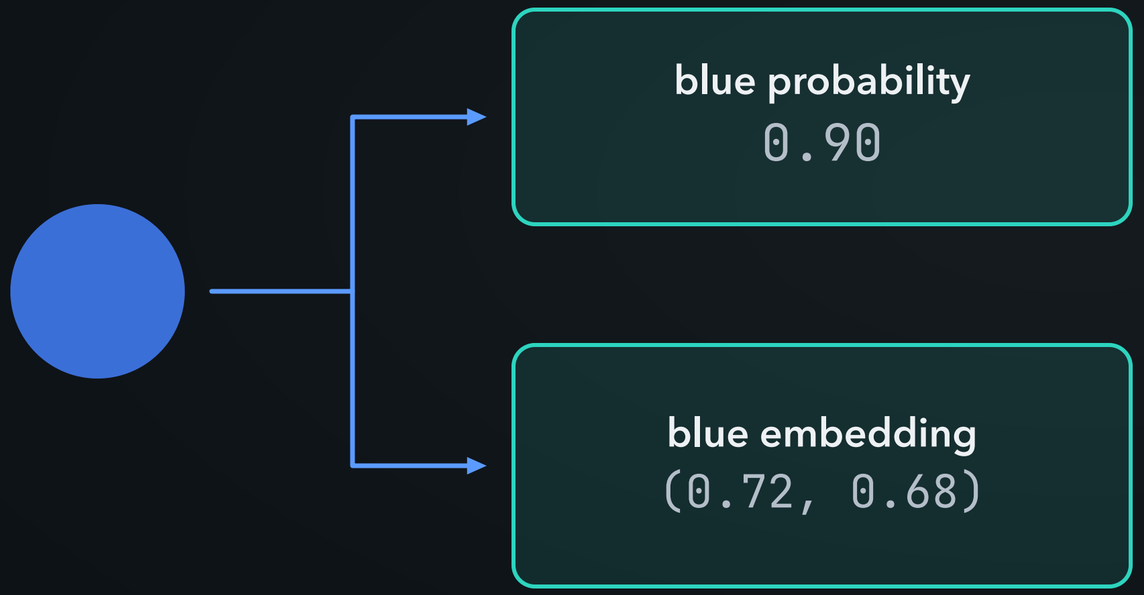


Different inputs to the head remove the forced tie.

ILLUSTRATIVE VALUES · hand-chosen vectors and head; scores computed

Concept embedding models

The extra detail helps, but it has no named concept



Named meaning: how likely is blue?

Extra detail can reach the head.

No coordinate is labelled ‘round’ or ‘square’.

More capacity does not make every detail interpretable.

ILLUSTRATIVE VALUES · same hand-chosen shape example

Concept embedding models

Training learns the weights. Testing uses them

what changes	training	test
input	image, concept labels, class label	new image
network weights	updated using both losses	kept fixed
vectors and probabilities	computed from this image	computed from the new image

Test-time vectors are newly computed; the learned weights stay fixed.

Concept embedding models

Concept embedding models in symbols

Concept embedding models (Espinosa Zarlenga et al., NeurIPS 2022) x → ĉ → ŷ

concept predictor	$g_\theta : x \mapsto (p_j,\ c_j^+,\ c_j^-)$
what reaches the head	$\hat{c}_j = p_j\, c_j^+ + (1 - p_j)\, c_j^- \in \mathbb{R}^m$
task head	$\hat{y} = f_\phi(\hat{c}_1, \ldots, \hat{c}_k)$
what is trained	$g_\theta,\ f_\phi$ (joint training)
data needed	$\{(x,\ c,\ y)\}$ images, concept labels, class labels

j: one concept · k: number of concepts · m: numbers in each embedding

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Reuse a pretrained encoder; build a concept bank

Post-hoc Concept Bottleneck Model (PCBM)

needs concept labels

frozen: never trained here

no need to label every concept on every training image

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Intuition: the encoder already arranges photos along directions

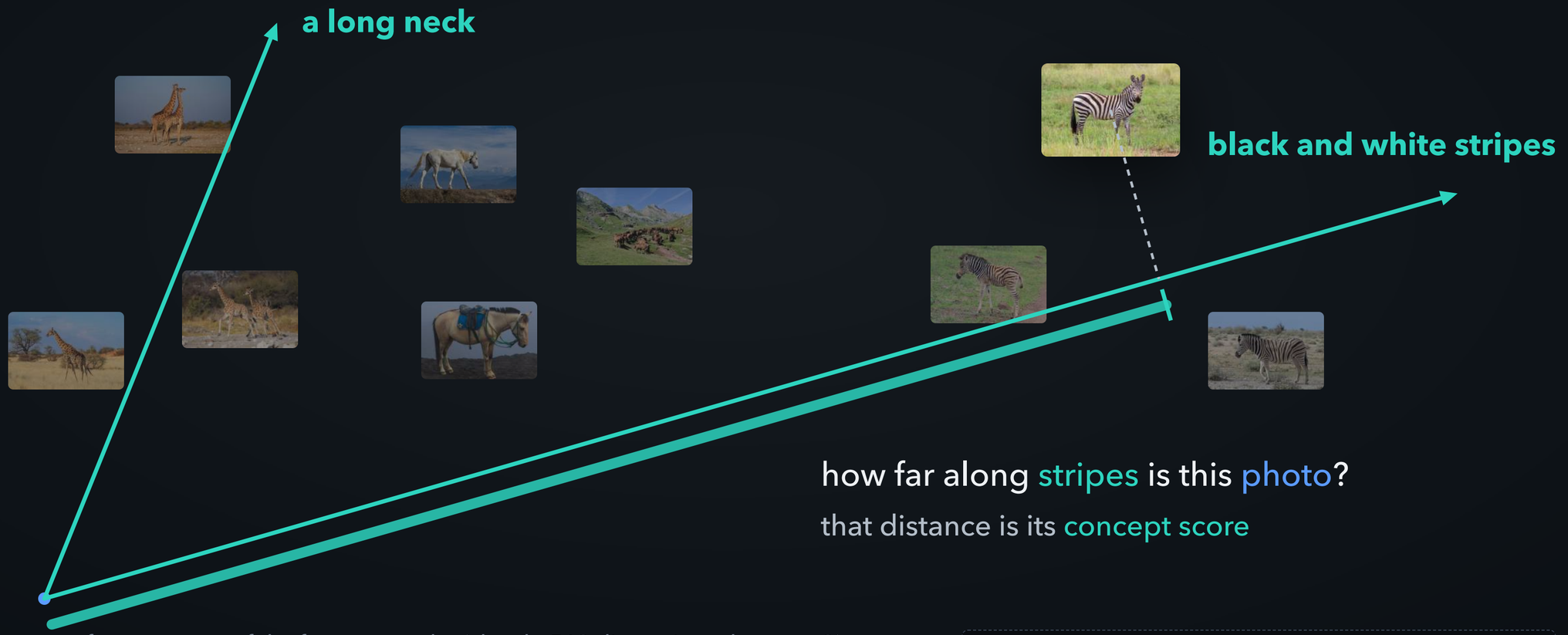
feature space of the frozen encoder (sketch: 512 dimensions drawn as 2)

SCHEMATIC · photo positions drawn for teaching

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A concept bank names directions; a score asks how far along



Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A frozen image encoder turns each photo into 512 numbers

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

image encoder
CLIP ViT-B/32

frozen

image features

1	-0.034
2	-0.031
3	+0.001
⋮	⋮
512	+0.017

$$e(x) \in \mathbb{R}^{512}$$

COMPUTED · REAL MODEL · CLIP ViT-B/32 image feature of this photo (normalised); 4 of 512 entries shown

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Start with examples of one concept: stripes

stripes present

stripes absent

image encoder
frozen

feature vectors
retain concept labels

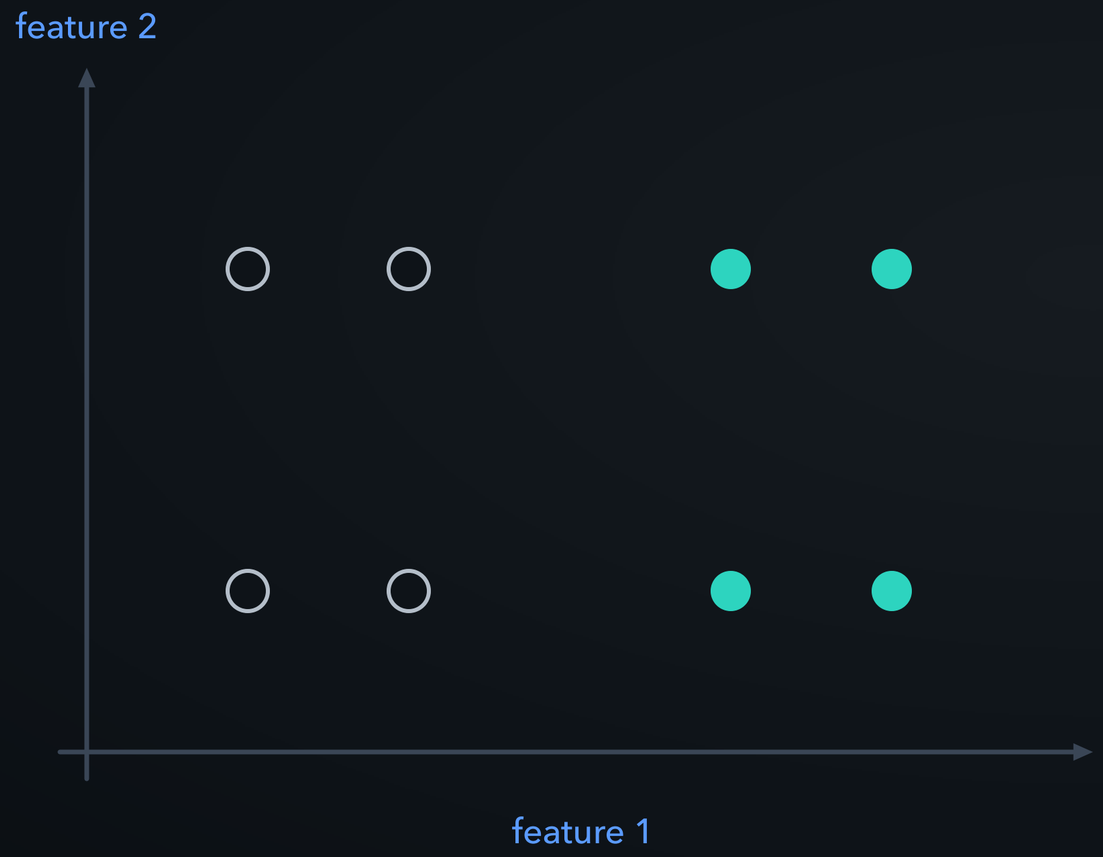
Labels describe the concept, not the animal class.

These examples can come from a separate dataset.

Photo: Charles J. Sharp, Wikimedia Commons, CC BY-SA 4.0; Photo: Giles Laurent, Wikimedia Commons, CC BY-SA 4.0

Post-hoc concept bottleneck models

For intuition, give each image just two features



CONSTRUCTED EXAMPLE · two chosen features; not a plot of CLIP outputs

One point = one image.

Filled points: stripes present

Hollow points: stripes absent

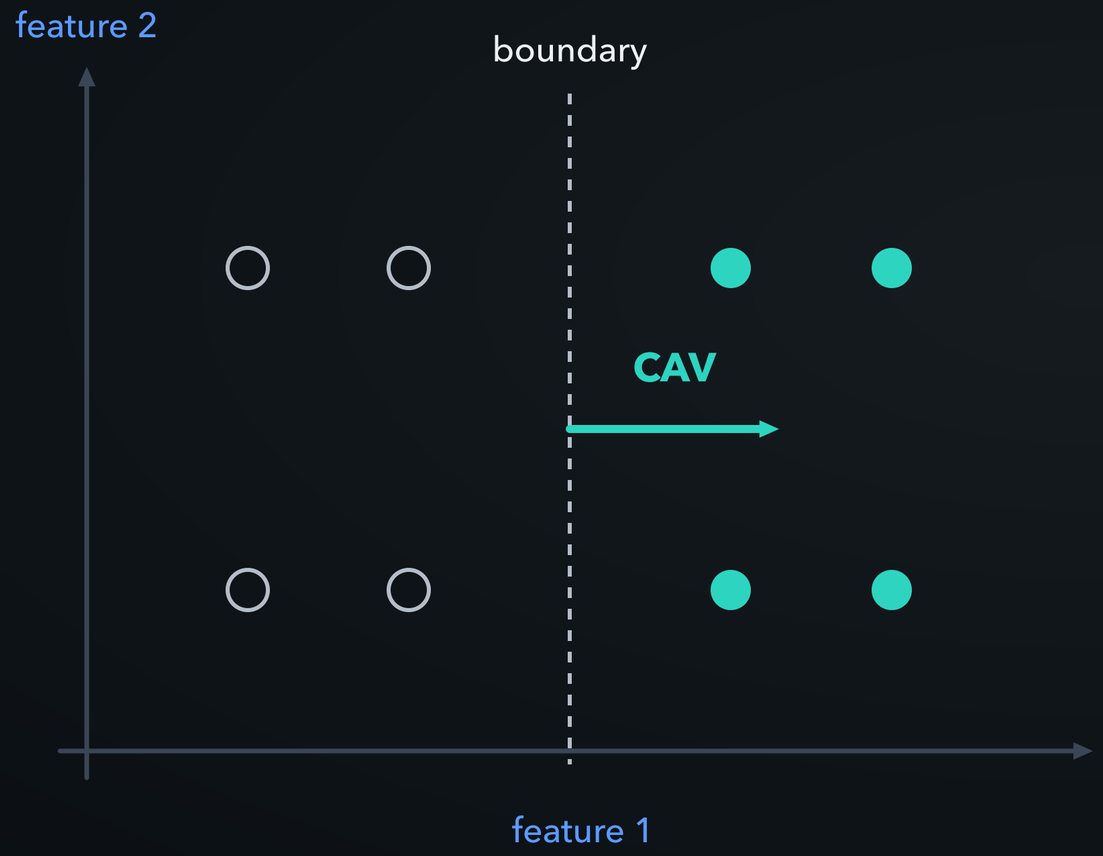
People supply these concept labels.

The encoder stays fixed.

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Fit a boundary, then name the direction across it: CAV



CONSTRUCTED EXAMPLE · chosen features; separating line calculated

Fit a line between the two labelled groups.

The arrow points towards striped examples.

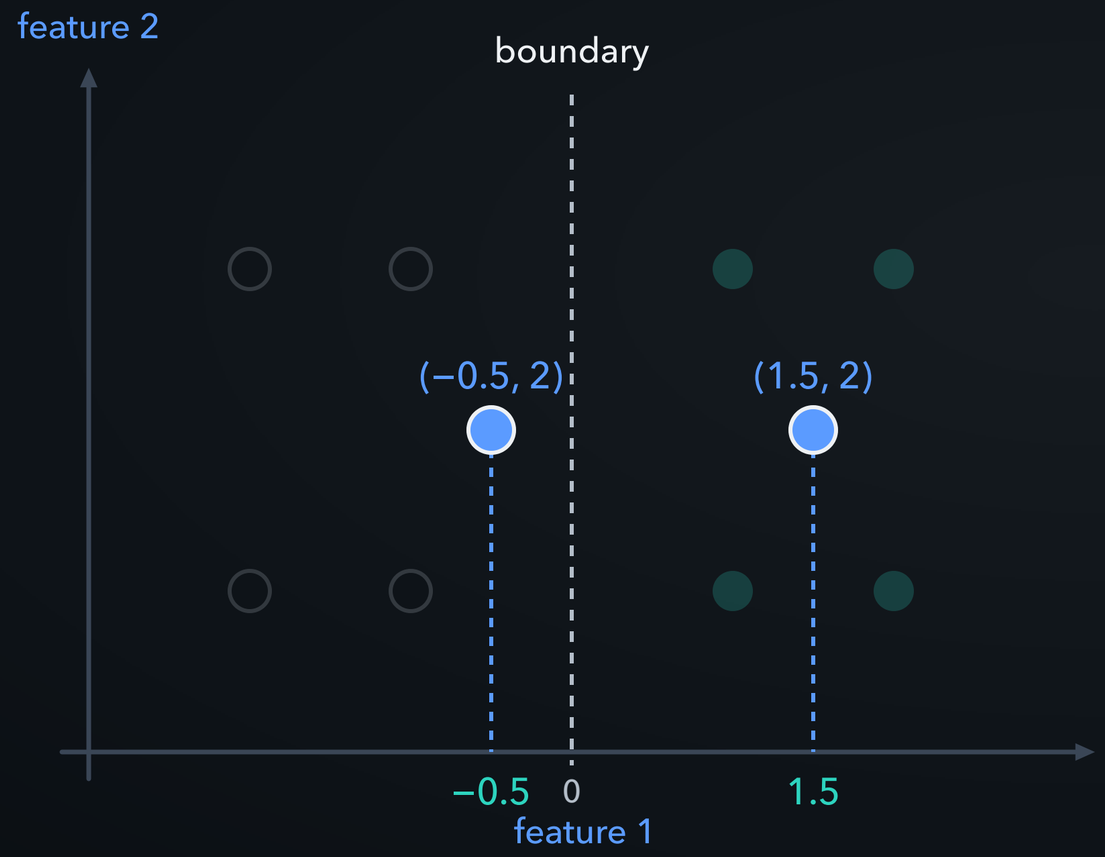
Concept activation vector (CAV)

Save this direction for new images.

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Score a new image along the striped direction



unit CAV = (1, 0)

Here, only the horizontal feature counts.

$1.5 \times 1 + 2 \times 0 = 1.5$

Higher score = further towards striped.

A concept score is not a probability.

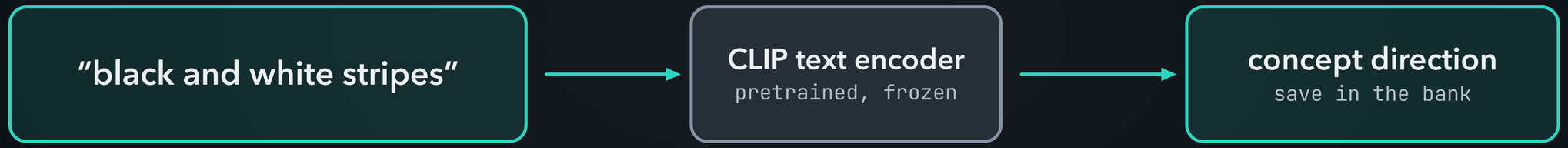
CONSTRUCTED EXAMPLE · chosen features and unit CAV; dot products calculated

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

An alternative: CLIP turns the phrase into a direction

Alternative to labelled concept examples

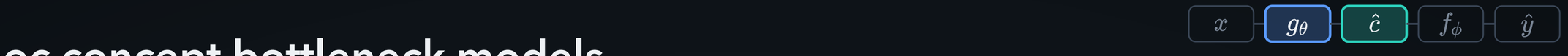


Text and image vectors share the same CLIP feature space.

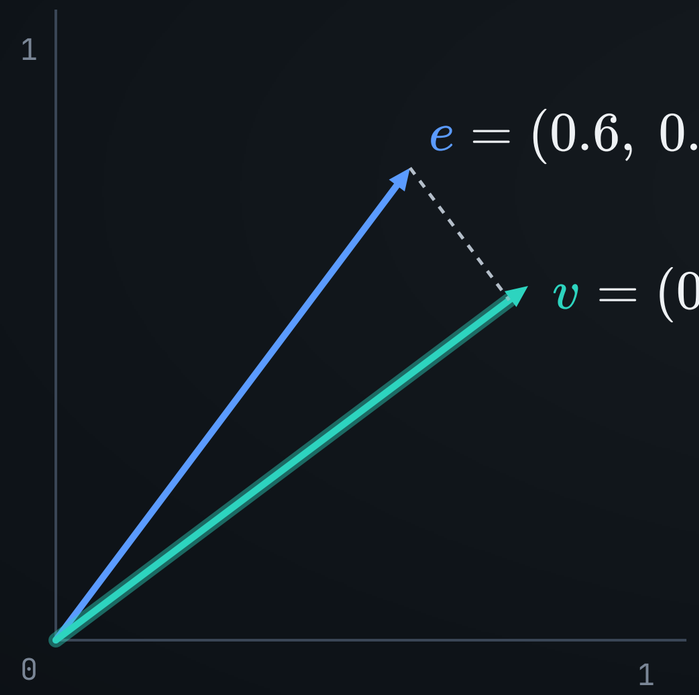
Our animal example uses this route. Both encoders stay frozen.

SCHEMATIC · CLIP text route; no concept classifier is fitted

Post-hoc concept bottleneck models



A projection: multiply matching entries, then add



features · direction = first × first + second × second

$$\langle e,\ v \rangle = e_1 v_1 + e_2 v_2$$

$$0.6 \times 0.8 + 0.8 \times 0.6$$

$$= 0.48 + 0.48 = 0.96$$

the dot product; CLIP does the same over 512 entries

ILLUSTRATIVE VALUES · two entries instead of 512

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Back to 512 features: one score for each phrase

image encoder
512 features

compare with each
saved phrase vector

black and white stripes	0.274
long neck	0.243
flowing mane	0.234
brown patches on a pale coat	0.257
hooves	0.243
saddle	0.223
green grass	0.217
wooden fence	0.211

Unit-length vectors: dot product = cosine similarity

8 named scores; not probabilities

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

`COMPUTED · REAL MODEL · frozen CLIP ViT-B/32; saved phrase vectors`

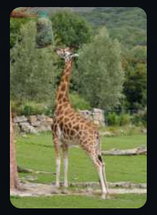
Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Use the same concept prompts for every image

Three held-out photos

raw cosine similarity: three prompts from the same bank

	“black and white stripes”	“a long neck”	“a flowing mane”
zebra	0.274	0.243	0.234
horse	0.168	0.199	0.186
 giraffe	0.208	0.301	0.218

Same encoder. Same prompts. One score per image and concept.

COMPUTED · REAL MODEL · frozen CLIP ViT-B/32; cosine scores, not probabilities

Commons photos: Diego Delso (CC BY-SA 4.0); George Chernilevsky (Public domain); Quartl (CC BY-SA 3.0)

Post-hoc concept bottleneck models

Each prompt gets its own training average and spread

Score all 24 training photos with the same prompts.

	"black and white stripes"	"a long neck"	"a flowing mane"
training average	0.215	0.239	0.227
training spread standard deviation	0.042	0.037	0.023

Calculate down each prompt column, across training images.

The three test photos do not set the scale.

COMPUTED · REAL MODEL · saved training split; each prompt has its own mean and spread

Post-hoc concept bottleneck models

Standardise each score: subtract the mean, divide by the spread

standardised score = (score − training mean) ÷ training spread

$$z = \frac{0.274 - 0.215}{0.042} \approx 1.41$$

COMPUTED · REAL MODEL · stripes scores, mean and spread over the 24 training photos

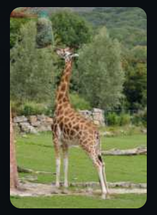
Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The same photos and prompts, now on a common scale

Three held-out photos

raw cosine → standardised score

	“black and white stripes”	“a long neck”	“a flowing mane”
zebra	0.274 → **+1.41**	0.243 → **+0.10**	0.234 → **+0.32**
horse	0.168 → **-1.10**	0.199 → **-1.08**	0.186 → **-1.80**
 giraffe	0.208 → **-0.16**	0.301 → **+1.70**	0.218 → **-0.38**

0 = training average · positive = above · negative = below

Below average does not mean absent.

COMPUTED · REAL MODEL · training statistics reused; values rounded for display

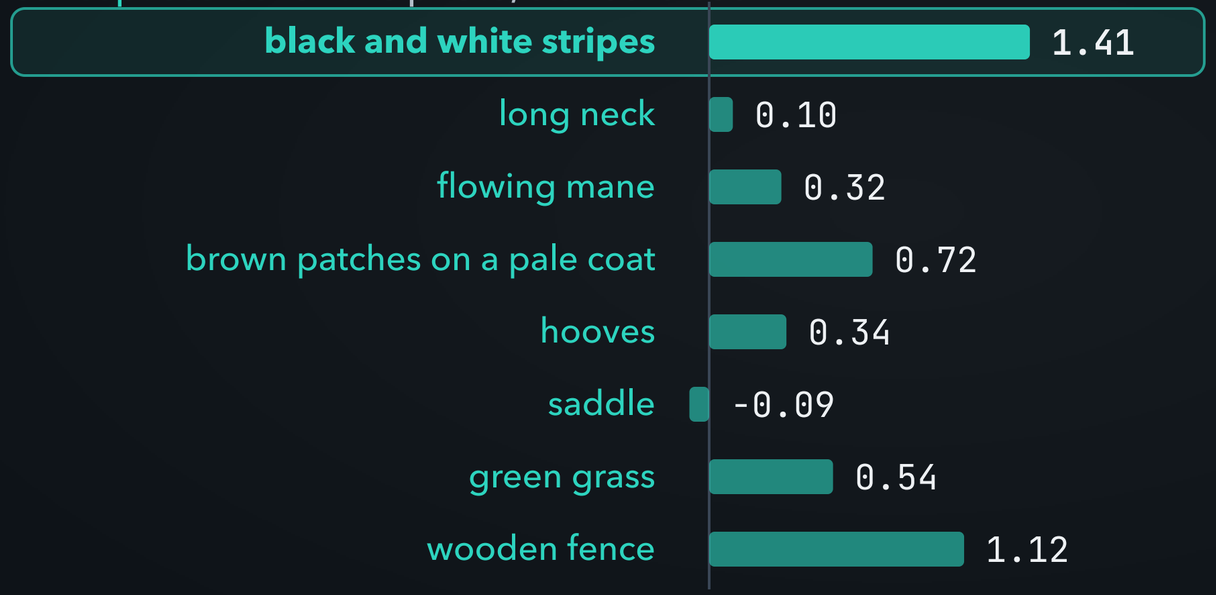
Commons photos: Diego Delso (CC BY-SA 4.0); George Chernilevsky (Public domain); Quartl (CC BY-SA 3.0)

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

For this zebra, stripes scores highest

concept scores for this photo, standardised



each bar: (score − training mean) ÷ training spread

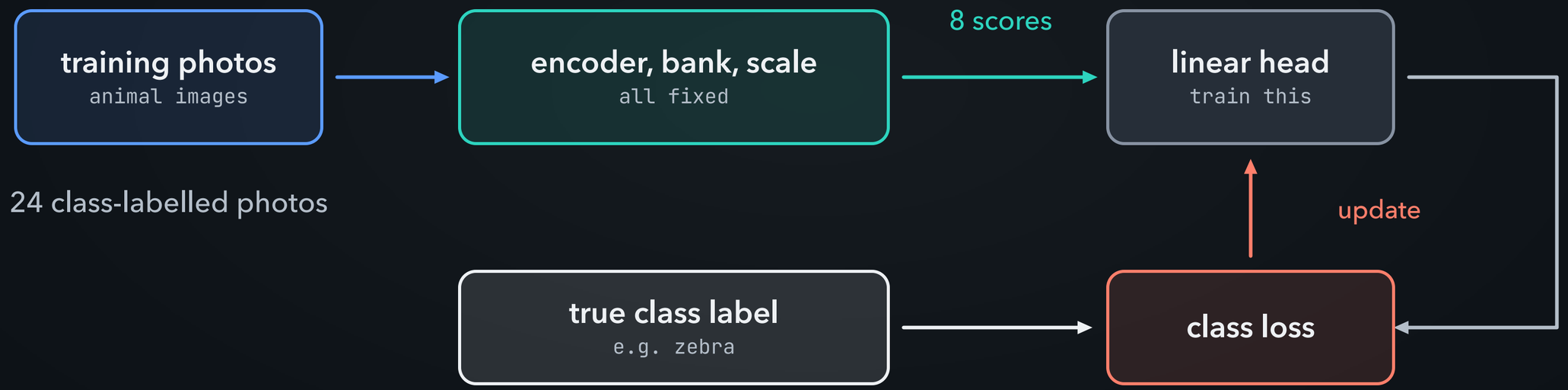
stripes: 1.41 spreads above its average

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Train the task head using concept scores and class labels



Class labels teach the head which concepts predict each animal.

SCHEMATIC · animal example: CLIP phrase bank, then a sparse task head

Post-hoc concept bottleneck models

Sparse: only 4 of 24 weights are non-zero

	zebra	horse	giraffe
black and white stripes	+1.49	0	0
long neck	0	-0.27	+1.02
flowing mane	0	+0.28	0
brown patches on a pale coat	0	0	0
hooves	0	0	0
saddle	0	0	0
green grass	0	0	0
wooden fence	0	0	0

8 concepts × 3 classes = 24 weights

4 / 24

weights are non-zero

L1 penalty: the size of every weight is added to the loss

COMPUTED · REAL MODEL · L1 logistic head fit on 24 photos

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

The zebra score: stripes score times weight, plus a constant

stripes score × weight + constant = **zebra score**

$$1.406 \times 1.486 - 0.054 \approx 2.035$$

class scores

shares, after the softmax

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

At test time: one new image, with all weights fixed

encoder
512 features

bank + scale
8 concept scores

trained head
class scores

softmax

Input: one image.

No concept labels. No class label.

Everything is saved. Nothing is fitted again.

zebra		0.77
horse		0.11
giraffe		0.11

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

COMPUTED · REAL MODEL · same held-out zebra; fixed encoder, bank, scale and head

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

All 12 held-out photos are classified correctly

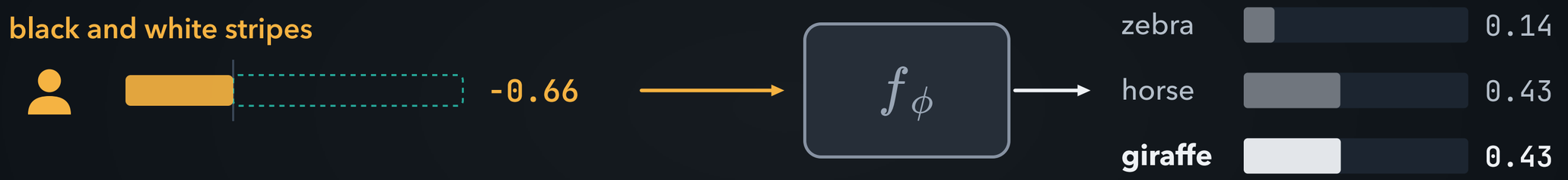
36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

12 of 12 held-out photos correct

a tiny teaching example, not a benchmark

Post-hoc concept bottleneck models

Edit the stripes score, and zebra falls



set to −0.66, its mean on non-zebra training photos

$$-0.659 \times 1.486 - 0.054 \approx -1.034$$

the new zebra logit; the photo and the head stay the same

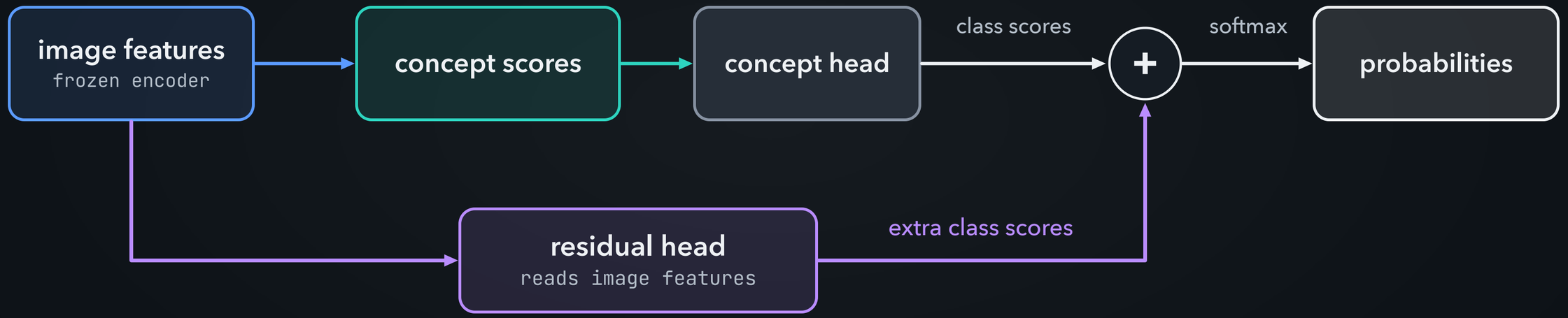
COMPUTED · REAL MODEL · the same head, one score edited

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

PCBM-h adds an extra path for evidence the concepts miss

Hybrid Post-hoc Concept Bottleneck Model (PCBM-h)



Worked photo: zebra class score, before softmax

$$2.03 + 0.31 \approx \mathbf{2.34}$$

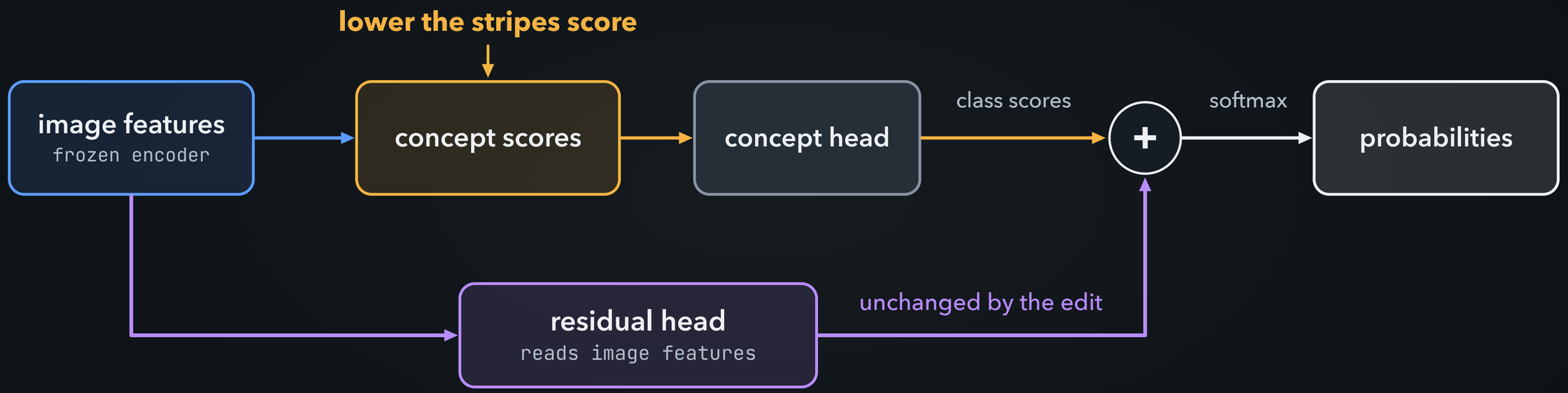
Train the extra head with class labels; keep the concept model fixed.

COMPUTED · REAL MODEL · saved CLIP animal example; scores rounded for display

Post-hoc concept bottleneck models

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

An edit leaves the extra path unchanged



Zebra probability after the same stripes edit

concept model: **0.14**

with residual: **0.20**

The prediction can change, but some evidence bypasses the concepts.

COMPUTED · REAL MODEL · same held-out zebra photo; same edit; residual contribution fixed

Post-hoc concept bottleneck models

Compared with the plain CBM: the concepts come from phrases

what changes	plain CBM	post-hoc CBM
concept source	people label every concept	**a concept bank of phrases**
detector	a trained concept predictor	**a frozen encoder and a projection**
representation	one number per concept	one number per concept
what reaches the head	only the concept values	**concept scores, plus a residual in PCBM-h**

3 of 4 changed; each concept is still one number

Post-hoc concept bottleneck models

Post-hoc concept bottleneck models in symbols

Post-hoc CBM (Yuksekgonul et al., ICLR 2023) x → ĉ → ŷ

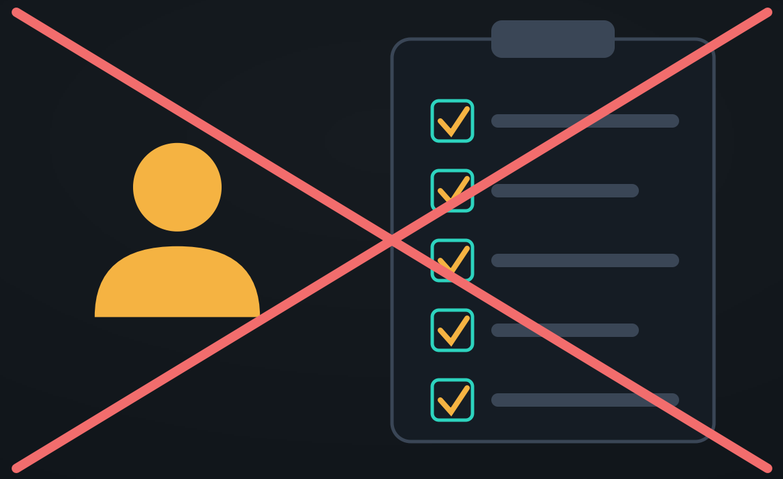
concept predictor	$\hat{c}_j = \langle e(x),\ v_j \rangle$ (e frozen, v: concept bank)
what reaches the head	$\hat{c} = (\hat{c}_1, \ldots, \hat{c}_k)$ concept projections
task head	$\hat{y} = f_\phi(\hat{c})$ sparse linear $+\ r(e(x))$ PCBM-h
what is trained	f_ϕ only (and r in PCBM-h)
data needed	$\{(x, y)\}$ + concept bank

j: one concept · k: number of concepts · r: the residual

Label-free concept bottleneck models

Concept labels are expensive: can a bottleneck do without them?

Label-Free Concept Bottleneck Model (LF-CBM)



CUB: 112 per bird photo

a person labels every concept on every photo

label-free: no human concept labels

Label-free concept bottleneck models

The intuition: a student learns to copy a teacher's scores

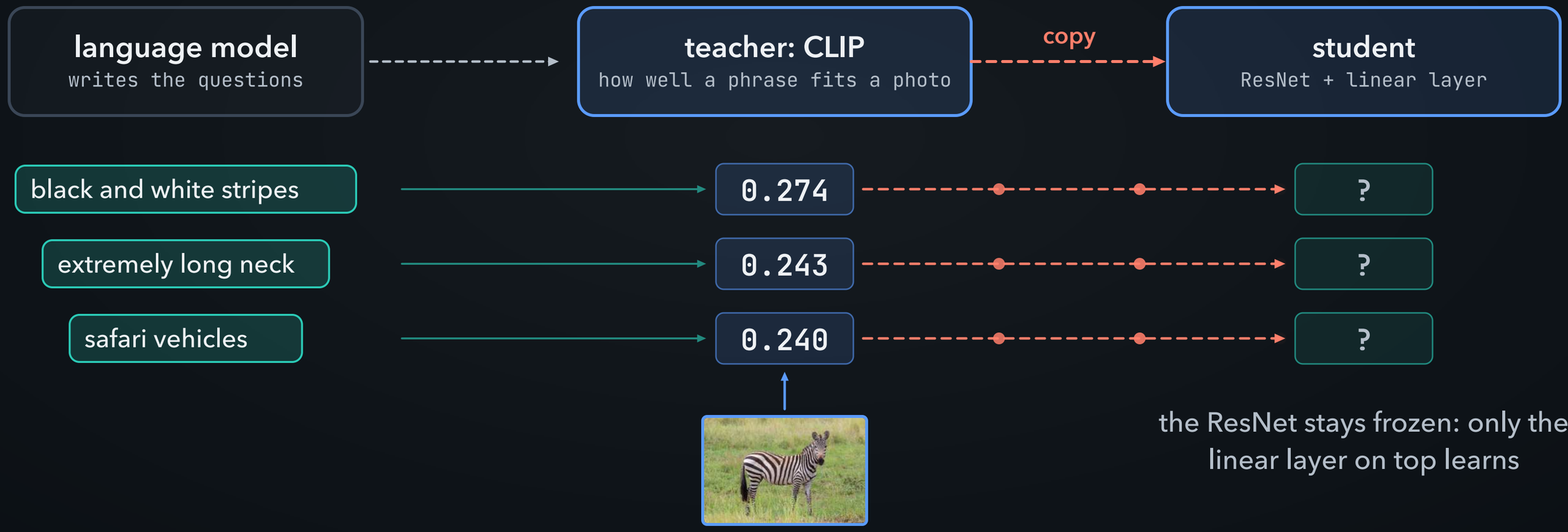


Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

Label-free concept bottleneck models

Five steps, from phrases to a sparse final layer

1	ask a language model	language model
2	filter the phrases	CLIP
3	score every image	the teacher, CLIP
4	learn to copy the scores	the student
5	fit a sparse final layer	**class labels**

only step 5 uses labels, and only the **class labels**

Label-free concept bottleneck models

Step 1: ask a language model about each class

QUESTION 1

List the most important features for recognizing something as a "zebra".

QUESTION 2

List the things most commonly seen around a "zebra".

QUESTION 3

Give superclasses for the word "zebra".

superclasses: the broader groups it belongs to

REPLY TO QUESTION 1 · zebra

```
{"items": [
  "Black and white stripes",
  "Horse-like body",
  "Erect mane",
  "Hooved feet",
  "Long tufted tail",
  "Large rounded ears",
  "Elongated muzzle"
]}
```

COMPUTED · REAL MODEL · real reply · gemini-3.1-pro-preview

Label-free concept bottleneck models

Three questions for three classes give 59 candidate concepts

	features	seen around	superclasses	new phrases
zebra	7	8	8	23
horse	8	8	6	22
giraffe	8	2	4	14
candidates				**59**

a phrase already listed for an earlier class is counted once

COMPUTED · REAL MODEL · counted from the real replies, gemini-3.1-pro-preview

Label-free concept bottleneck models

Step 2: drop class names, duplicates and phrases CLIP barely sees

contains a class name	-3	~~zebras~~ ~~horse-like body~~
near-duplicate of "equine animal"	-4	~~equine~~
CLIP barely sees it its 5 best-matching photos average below 0.24 here 0.232 and 0.222; stripes: 0.277	-18	~~leather saddle~~ ~~ossicones on head~~

$$59 - 3 - 4 - 18 = 34 \text{ concepts}$$

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

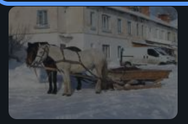
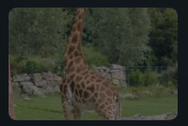
Label-free concept bottleneck models

Step 3: the teacher, CLIP, scores every image against every concept

5 of the 34 concepts

Follow this zebra row

held-out example photo

	black and white stripes	wildebeest herds	safari vehicles	riding bridle	extremely long neck	
	0.274	0.246	0.240	0.195	0.243	
	0.168	0.201	0.236	0.225	0.203	…
	0.208	0.174	0.233	0.190	0.308	

higher score = the phrase fits the image better

the teacher's answers: targets for the student, not labels from a person

36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

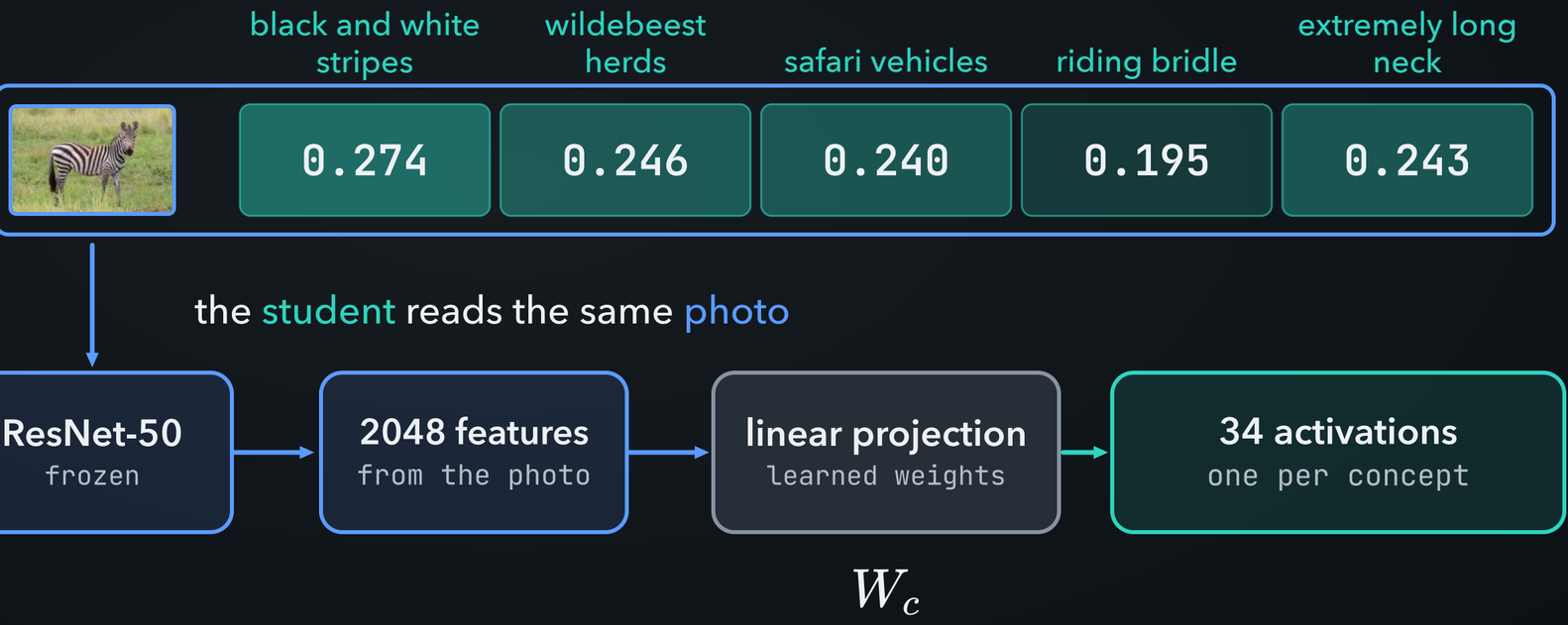
COMPUTED · REAL MODEL · CLIP ViT-B/32, one photo per class; row 1 = worked photo

Label-free concept bottleneck models

Step 4: keep the teacher row, build the student

Same teacher row

raw CLIP scores



Fit the student activations to the teacher's pattern across training images.

Next: put the two outputs on a comparable scale.

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

COMPUTED · REAL MODEL · CLIP scores for the held-out zebra; this photo is not used for fitting

Label-free concept bottleneck models

Follow the stripes entry onto a comparable scale

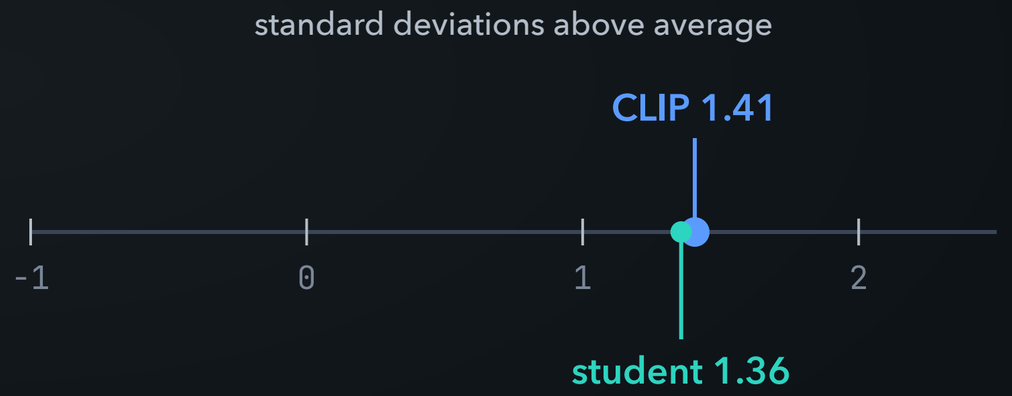
same zebra, first column: black and white stripes

$$z = \frac{\text{value} - \text{average}}{\text{spread}}$$

average and spread (standard deviation) over the training photos

CLIP'S STRIPES SCORE, THIS ZEBRA

$$z = \frac{0.274 - 0.215}{0.042} = 1.41$$

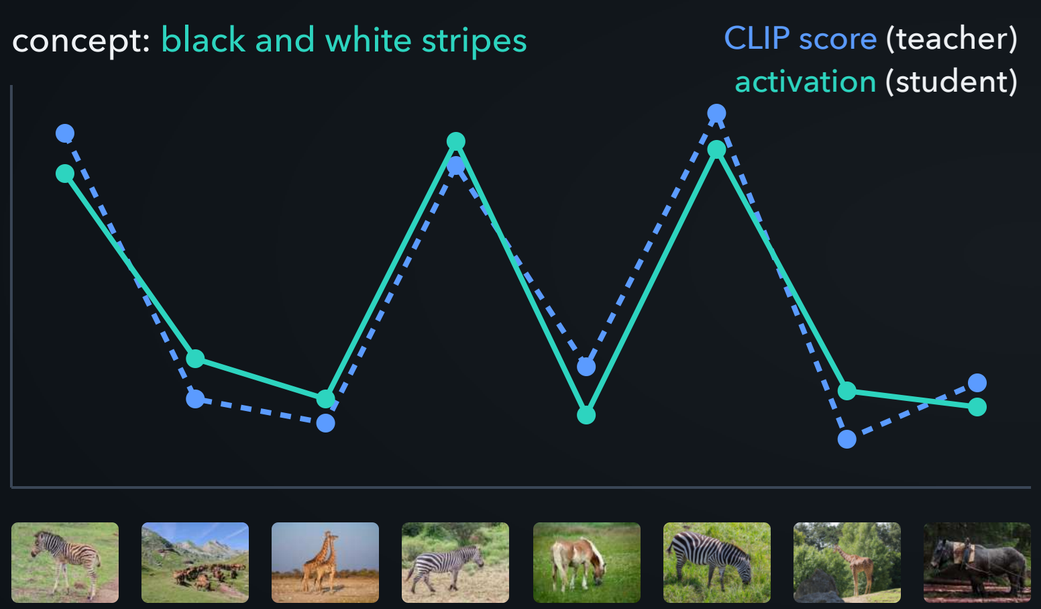


now the teacher and the student can be compared

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example; worked photo, 24 training photos

Label-free concept bottleneck models

Copying means each activation rises and falls with its CLIP score



36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

high where CLIP is high, low where it is low

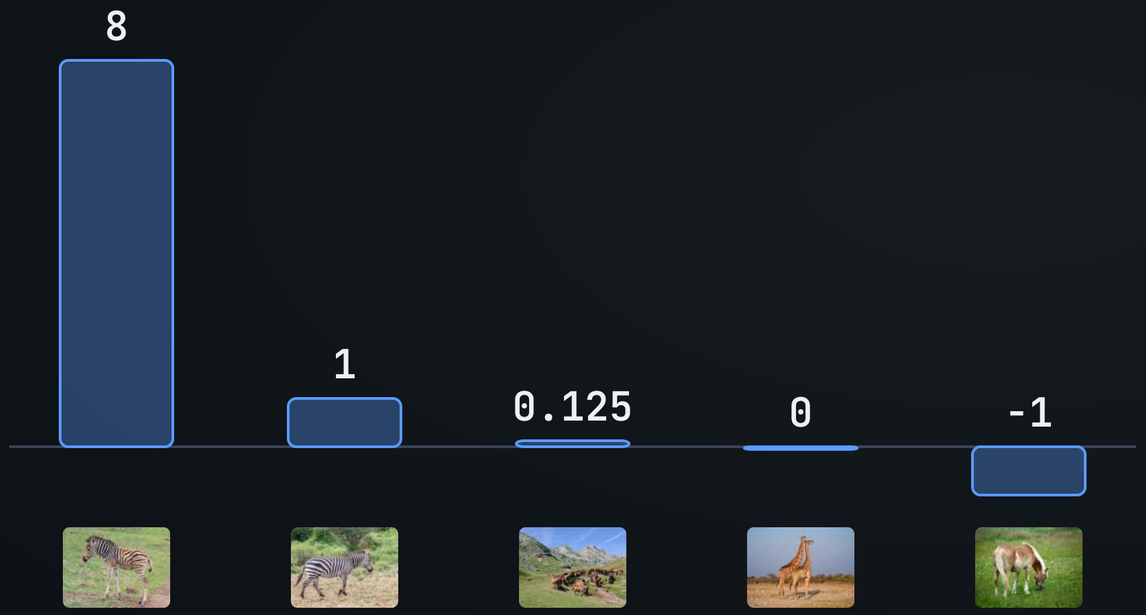
only the pattern across images matters, not the exact values

ILLUSTRATIVE VALUES · curve shapes drawn by hand

Label-free concept bottleneck models

Cubing stretches the images where CLIP scores highest

one concept, five images: CLIP scores, standardised



36 Wikimedia Commons photographs (CC BY-SA / CC BY / public domain)

$$z \longrightarrow z^3 = z \times z \times z$$

z: a standardised value

$$2^3 = 8$$

$$\left(\frac{1}{2}\right)^3 = \frac{1}{8}$$

the images CLIP scores highest count the most

ILLUSTRATIVE VALUES · values chosen to show the effect of cubing

Label-free concept bottleneck models

A cosine asks whether two lists rise and fall together

$$\cos(a, b) = \frac{a_1 b_1 + a_2 b_2 + a_3 b_3}{\text{length}(a) \times \text{length}(b)} \qquad \text{length}(a) = \sqrt{{a_1}^2 + {a_2}^2 + {a_3}^2}$$

$$\cos = \frac{2 \cdot 4 + 1 \cdot 2 + (-1)(-2)}{\sqrt{6} \times \sqrt{24}} = \frac{12}{12} = 1$$

a	b		
2	4	rise and fall together	cos = 1
1	2		
-1	-2		

a	b		
2	1	unrelated	cos = 0
1	-1		
-1	1		

a	b		
2	-2	opposite directions	cos = -1
1	-1		
-1	1		

ILLUSTRATIVE VALUES · lists made up to show the cosine

Label-free concept bottleneck models

The match for one concept: a cosine between cubed scores

	$\bar{q}_j{}^3$	$\bar{P}_j{}^3$
image 1	4.10	8.00
image 2	2.20	1.00
image 3	0.01	0.13
image 4	0.00	0.00
image 5	-1.33	-1.00
	student, cubed	CLIP, cubed

$$\cos = 0.92$$

$\bar{q}_j$: the student's standardised activations for concept j

$\bar{P}_j$: CLIP's standardised scores for concept j

$$\text{match}_j = \cos\left(\bar{q}_j{}^3,\ \bar{P}_j{}^3\right)$$

W_c makes the match large for every concept, with no labels: 60 unlabelled photos × 6 crops

ILLUSTRATIVE VALUES · the vectors on the left are made up

Label-free concept bottleneck models

Step 5: a sparse final layer keeps 6 of 102 weights

	black and white stripes	wildebeest herds	muscular equine build	riding bridle	safari vehicles	extremely long neck	28 other concepts
zebra	+1.46	+0.06	0	0	0	0	0 … 0
horse	0	0	+0.39	+0.16	-0.38	0	0 … 0
giraffe	0	0	0	0	0	+1.18	0 … 0

3 **classes** × 34 concepts = 102 weights; an L1 penalty sets 96 of them to exactly 0

COMPUTED · REAL MODEL · sparse final layer, trained on 24 labelled photos

Label-free concept bottleneck models

One prediction: the weights turn activations into a class score

class score = sum of weight × activation, plus a small constant

$$\text{zebra} = \underbrace{1.46 \times 1.36}_{\text{stripes}} + \underbrace{0.06 \times 1.09}_{\text{wildebeest herds}} - 0.02 = 2.02$$

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

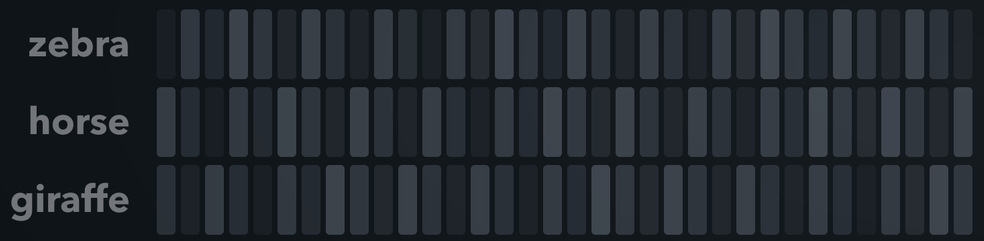
class scores			shares	
zebra	2.02	softmax →	**zebra**	0.82
horse	-0.40		horse	0.07
giraffe	-0.02		giraffe	0.11

COMPUTED · REAL MODEL · sparse final layer on the worked photo

Label-free concept bottleneck models

Why sparse: a person can read 6 weights, not 102

dense: all 102 weights in use



nobody can check 102 numbers per decision

sparse: 6 weights

black and white stripes	→	**zebra**	+1.46
wildebeest herds	→	**zebra**	+0.06
muscular equine build	→	**horse**	+0.39
riding bridle	→	**horse**	+0.16
safari vehicles	→	**horse**	-0.38
extremely long neck	→	**giraffe**	+1.18

each weight is a claim a person can check

SCHEMATIC · dense layer drawn for scale, values not shown

COMPUTED · REAL MODEL · weights of the sparse final layer

Label-free concept bottleneck models

Reading the weights shows a risk: three concepts describe the setting

about the animal

black and white stripes	+1.46	→ **zebra**
muscular equine build	+0.39	→ **horse**
extremely long neck	+1.18	→ **giraffe**

about the setting

riding bridle	+0.16	→ **horse**
safari vehicles	-0.38	→ **horse**
wildebeest herds	+0.06	→ **zebra**

the head uses context as evidence

a horse beside a safari vehicle loses evidence for **horse**

COMPUTED · REAL MODEL · weights of the sparse final layer

Label-free concept bottleneck models

What changed compared with the plain CBM

	plain CBM	**label-free CBM**
who names the concepts	a person	**a language model**
the concept layer learns from	concept labels	**CLIP's scores**
the head learns from	class labels	**class labels, sparse**

label-free is not unsupervised:

language model CLIP **class labels**

Label-free concept bottleneck models

LF-CBM in symbols

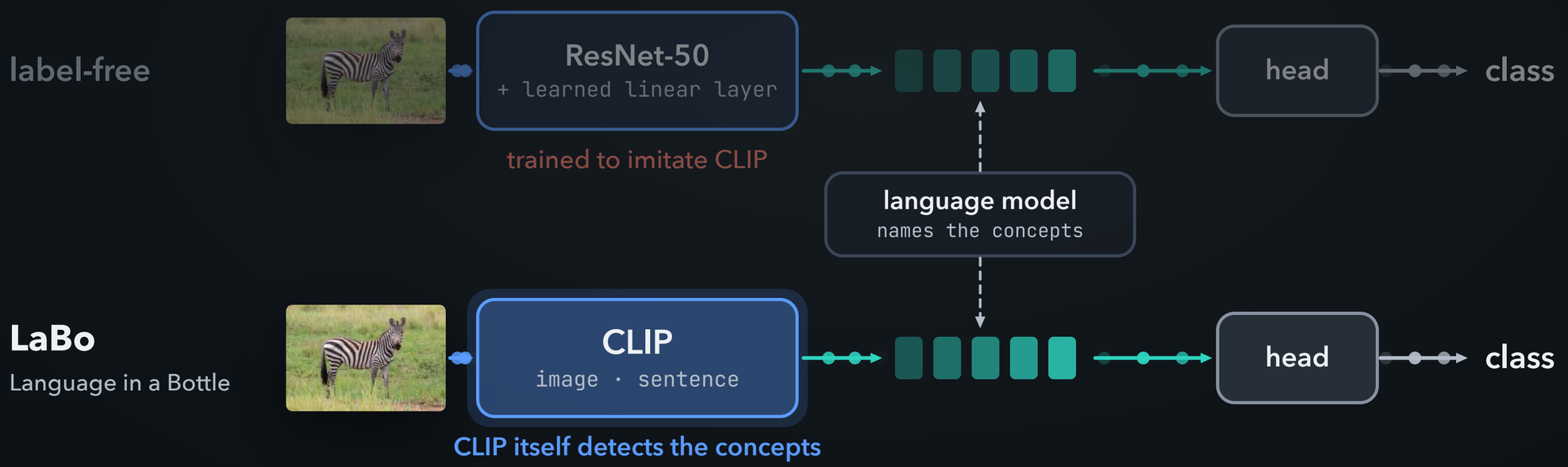
Label-free CBM (Oikarinen et al., ICLR 2023)

x → ĉ → ŷ

concept predictor	$\hat{c} = W_c\, e(x)$ fitted to CLIP scores $(\cos^3)$
what reaches the head	$\hat{c} \in \mathbb{R}^k,\ k = 34$ one per named concept
task head	$\hat{y} = f_\phi(\hat{c})$ sparse linear
what is trained	$W_c,\ f_\phi$ (e frozen)
data needed	$\{(x, y)\},$ no concept labels

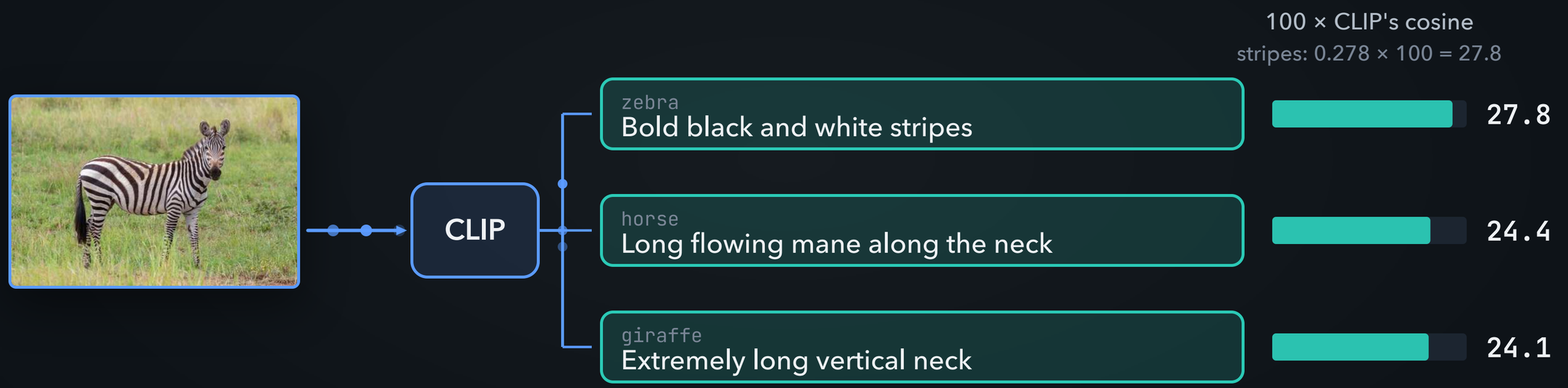
Language in a Bottle

LaBo keeps CLIP itself as the concept detector



Language in a Bottle

The intuition: let CLIP read short sentences about each class



these scores are the concepts: nothing is trained to imitate CLIP

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example; worked photo

Language in a Bottle

Four steps, from sentences to a head that starts from a guess

1	write sentences for each class	language model
2	keep the most useful sentences	a selection rule
3	score every image against them	CLIP
4	fit a linear head started from a guess: the language prior	**class labels**

Language in a Bottle

Step 1: a language model describes each class in sentences

PROMPT · zebra

Describe what a zebra looks like in a photograph, as 8 short visual sentences that would help tell it apart from similar animals.

8 sentences × 3 classes = 24 candidates

```
{"items": [
  "Bold black and white stripes.",
  "Short stiff upright mane.",
  "Stocky horse-like body.",
  "Dark rounded muzzle.",
  "Large striped ears.",
  "Tail with a dark tuft.",
  "Striped legs down to the hooves.",
  "Unique facial stripe patterns."
]}
```

COMPUTED · REAL MODEL · real reply · gemini-3.1-pro-preview

Language in a Bottle

A good set of sentences separates the classes

Bold black and white stripes

Stocky horse-like body

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example; mean CLIP similarity on the training photos

Language in a Bottle

A good set of sentences does not repeat itself

	sentence added to the set	what it adds that is new
1st	Bold black and white **stripes**	0.69
2nd	Unique facial **stripe** patterns	0.10
3rd	Large **striped** ears	0.04

once stripes are covered, another stripes sentence adds little

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example; coverage gains of the zebra picks

Language in a Bottle

Step 2: score each sentence's discriminability

ZEBRA CANDIDATES	discriminability
Bold black and white stripes	1.00
Short stiff upright mane	0.17
Stocky horse-like body	0.00
Dark rounded muzzle	0.26
Large striped ears	0.66
Tail with a dark tuft	0.23
Striped legs down to the hooves	0.63
Unique facial stripe patterns	0.68

discriminability: the margin, rescaled within the class so the best is 1 and the worst is 0

$$\text{large striped ears: } (0.049 - (-0.017)) \,/\, (0.083 - (-0.017)) = 0.66$$

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

A greedy rule balances discriminability and coverage

ZEBRA CANDIDATES	discriminability	coverage gain	score
Bold black and white stripes	1.00		
Short stiff upright mane	0.17		
Stocky horse-like body	0.00		
Dark rounded muzzle	0.26		
Large striped ears	0.66		
Tail with a dark tuft	0.23		
Striped legs down to the hooves	0.63		
Unique facial stripe patterns	0.68		

coverage gain: what a sentence adds beyond the sentences already chosen

$$\text{score} = 0.7 \times \text{discriminability} + 0.3 \times \text{coverage gain}$$

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Language in a Bottle

The greedy picks for zebra, in order

ZEBRA CANDIDATES	discriminability	coverage gain	score	pick
Bold black and white stripes	1.00	0.69	0.91	1
Short stiff upright mane	0.17			
Stocky horse-like body	0.00			
Dark rounded muzzle	0.26			
Large striped ears	0.66	0.04	0.47	3
Tail with a dark tuft	0.23			
Striped legs down to the hooves	0.63			
Unique facial stripe patterns	0.68	0.10	0.51	2

$$\text{round 1, stripes: } 0.7 \times 1.00 + 0.3 \times 0.69 = 0.91$$

$$\text{score} = 0.7 \times \text{discriminability} + 0.3 \times \text{coverage gain}$$

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Language in a Bottle

Three sentences per class, nine in all

zebra	horse	giraffe
1 Bold black and white stripes	1 Long flowing mane along the neck	1 Extremely long vertical neck
2 Unique facial stripe patterns	2 Tail covered entirely in long flowing hair	2 Very long slender legs
3 Large striped ears	3 Smooth short coat in various colors	3 Long tail with dark hair tuft

3 per class, picked greedily: 9 sentences in all

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Language in a Bottle

Step 3: CLIP scores the image against each chosen sentence

Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

$$\hat{c}_j = 100 \cos\big(\phi(x),\ \psi(s_j)\big)$$

ϕ: CLIP's image encoder

ψ: its text encoder; s_j: sentence j

CLIP

- Bold black and white stripes
- Unique facial stripe patterns
- Large striped ears
- Long flowing mane along the neck
- Tail covered entirely in long flowing hair
- Smooth short coat in various colors
- Extremely long vertical neck
- Very long slender legs
- Long tail with dark hair tuft

the bottleneck: 9 numbers

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Step 4: a language prior links each class to its sentences

	zebra sentences			horse sentences			giraffe sentences		
	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9
zebra	1/3	1/3	1/3	0	0	0	0	0	0
horse	0	0	0	1/3	1/3	1/3	0	0	0
giraffe	0	0	0	0	0	0	1/3	1/3	1/3

language prior: each **class** puts 1/3 on each of its own three sentences, 0 elsewhere

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Language in a Bottle

With the prior, a class score is the average of its own sentence scores

class score = average of its own three sentence scores

$$\text{zebra} = \tfrac{1}{3}(27.8 + 28.1 + 28.5) = 28.1$$

$$\text{horse} = \tfrac{1}{3}(24.4 + 24.3 + 20.0) = 22.9$$

$$\text{giraffe} = \tfrac{1}{3}(24.1 + 22.3 + 24.6) = 23.7$$

with the prior alone: **11 of 12** test photos right

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example; worked photo

Why a prior helps when labels are few

start from zero

zebra	0	0	0	0	0	0	0	0	0
horse	0	0	0	0	0	0	0	0	0
giraffe	0	0	0	0	0	0	0	0	0

the **labels** must teach every link

start from the language prior

zebra	1/3	1/3	1/3	0	0	0	0	0	0
horse	0	0	0	1/3	1/3	1/3	0	0	0
giraffe	0	0	0	0	0	0	1/3	1/3	1/3

the links are already there: **11 of 12** right

COMPUTED · REAL MODEL · CLIP ViT-B/32, teaching example

linear probe: a linear layer on CLIP's image features, no concepts

LaBo paper, CUB, one labelled photo per class: LaBo 54.19%, linear probe 47.69%

REPORTED · Yang et al., CVPR 2023, App. Table 10 (CLIP ViT-L/14)

Language in a Bottle

Training with the class labels adjusts the weights

	zebra sentences			horse sentences			giraffe sentences		
	s_1	s_2	s_3	s_4	s_5	s_6	s_7	s_8	s_9
zebra	0.54	0.44	0.47	-0.19	-0.09	-0.12	-0.01	0.05	-0.13
horse	0.03	-0.03	-0.03	0.44	0.43	0.39	-0.14	-0.10	0.04
giraffe	-0.16	-0.01	-0.02	-0.07	-0.09	-0.03	0.53	0.45	0.34

shares, this zebra

zebra	**0.995**
horse	0.004
giraffe	0.001

after training with the **class labels**: small changes everywhere

test photos right: 12 of 12

COMPUTED · REAL MODEL · CLIP ViT-B/32 on 36 Commons photos, teaching example

Language in a Bottle

Do the zebra sentences detect the ears, or stripes anywhere?

the three zebra sentences

- Bold black and white **stripes**
- Unique facial **stripe** patterns
- Large **striped** ears

all three are about stripes

The scores alone cannot tell.

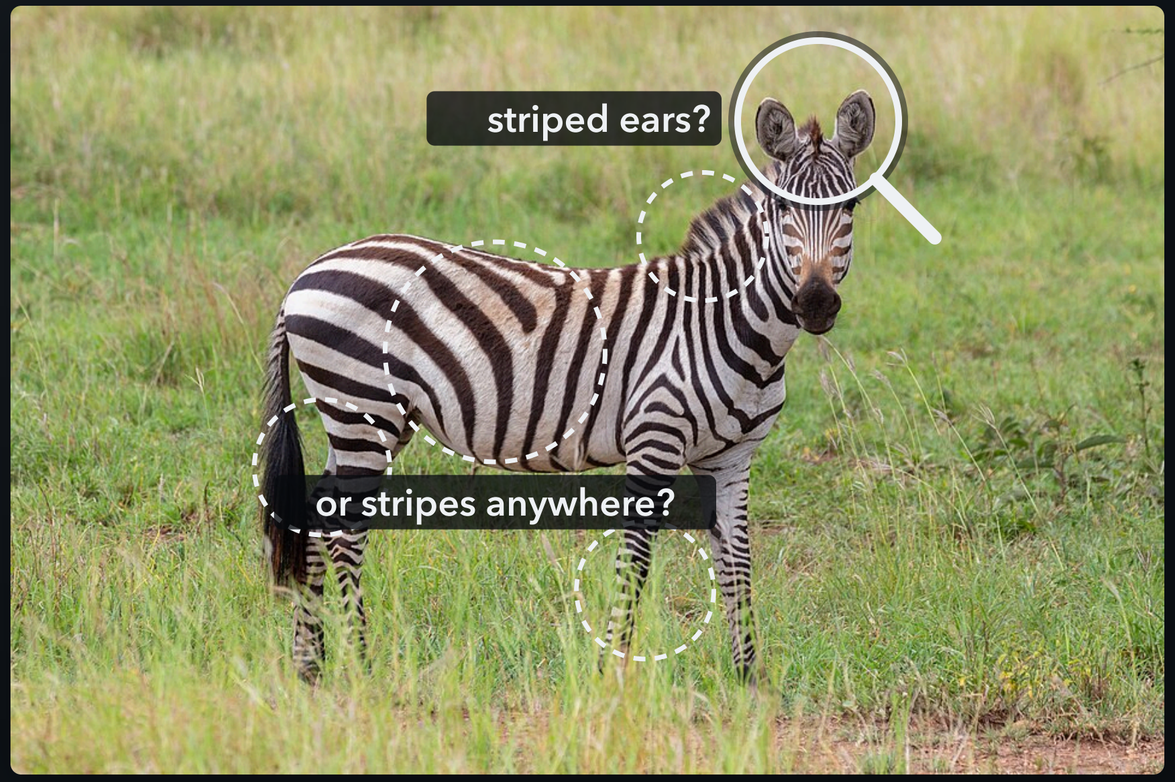


Photo: Diego Delso, Wikimedia Commons, CC BY-SA 4.0

Language in a Bottle

What changed compared with the plain CBM

	plain CBM	**LaBo**
who names the concepts	a person	**a language model, as sentences**
what detects the concepts	a trained concept layer	**CLIP, unchanged**
the head starts from	scratch	**the language prior**

only the head is trained, with the **class labels**

Language in a Bottle

LaBo in symbols

LaBo (Yang et al., CVPR 2023) x → ĉ → ŷ

concept predictor	$\hat{c}_j = 100 \cos(\phi(x),\ \psi(s_j))$ (CLIP, frozen)
what reaches the head	$\hat{c} \in \mathbb{R}^k,\ k = 9$ one per chosen sentence
task head	$\hat{y} = f_\phi(\hat{c})$ linear
what is trained	f_ϕ only, initialised from the language prior
data needed	$\{(x, y)\}$

Grounding the concepts

Nothing checks that a concept is really in the image

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

Nothing checks that the concept is visible in the image.

Grounding the concepts

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Naming a concept is not the same as pointing at it

naming

yellow bill

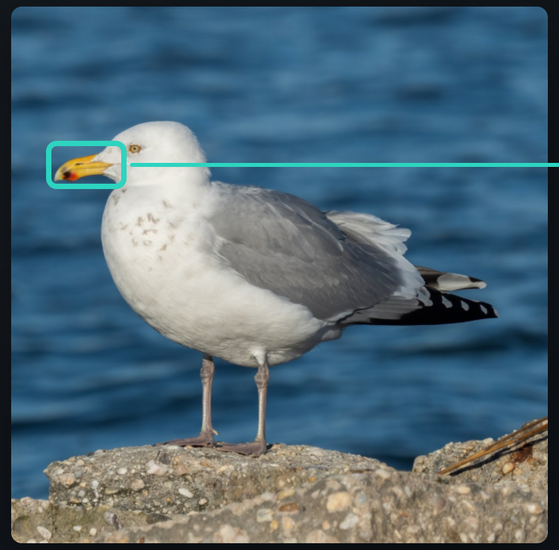
?

where?

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

a name, with no place in the image

pointing



yellow bill

✓ yellow bill box found: counts

✗ loud laughing call no box: not counted

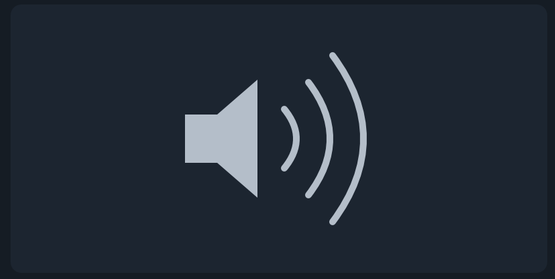
ILLUSTRATIVE VALUES · box drawn for teaching

Grounding the concepts

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

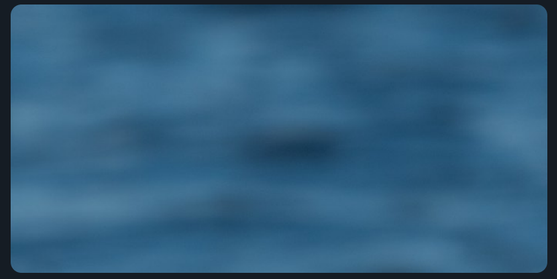
Three ways a concept can fail

1 `loud laughing call`



not visual

2 `near the sea`



the setting, not the bird

3 `black head`

high score, yet absent

but this gull's head is white

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

`ILLUSTRATIVE VALUES · example concepts chosen for teaching`

Grounding the concepts

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

High accuracy does not prove meaningful concepts

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

With many outputs, this baseline can be **almost as accurate as concept models**.

Useful image information can survive without named concept meanings.

Check: does a high yellow-bill score really mean a yellow bill is present?

REPORTED · Srivastava et al. 2024, Section 4 and Figure 3; prediction path schematic

Grounding the concepts

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

VLG-CBM asks a detector to find each concept

Vision-Language-Guided Concept Bottleneck Model (VLG-CBM)

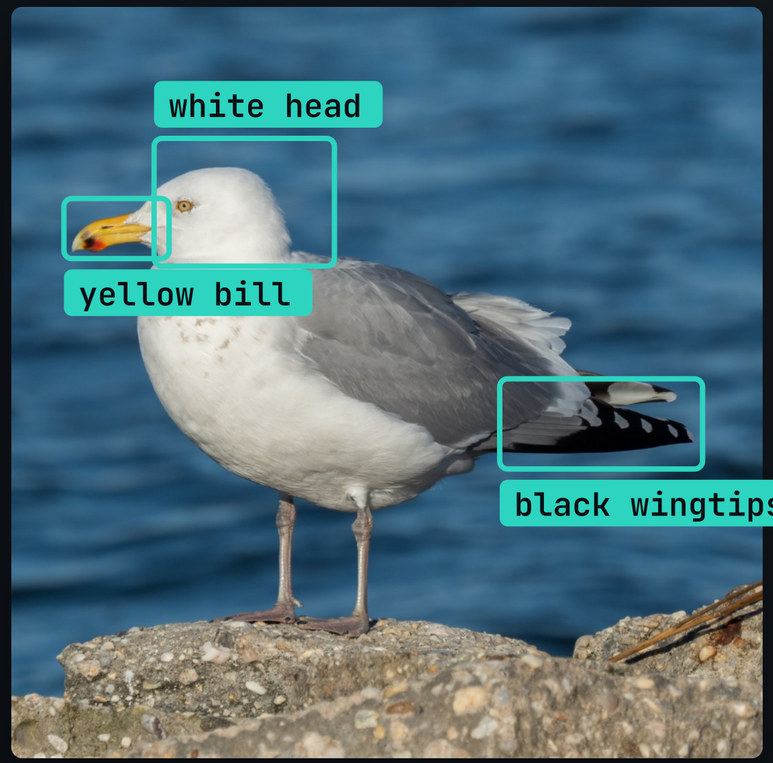


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

each training image

concept names

open-vocabulary detector
one image at a time

a box where it is found

open-vocabulary: it searches for any phrase you give it, not only a fixed list of objects

ILLUSTRATIVE VALUES · boxes drawn for teaching, not a detector run

Grounding the concepts

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

A box means 1, no box means 0

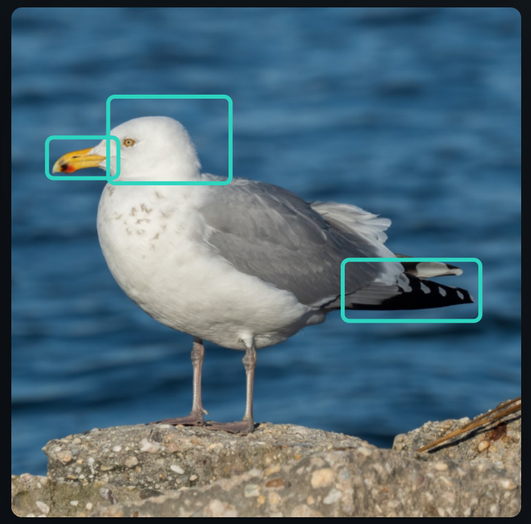


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

CONCEPT	BOX?
yellow bill	✓
white head	✓
black wingtips	✓
loud laughing call	none

grounded labels

$$\mathbf{c}^{\text{det}} = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}$$

concept labels from the detector

ILLUSTRATIVE VALUES · boxes drawn for teaching

Grounding the concepts

The grounded labels train the concept layer with a loss

concept layer: backbone features → one score per concept

concept loss — sum over concepts — binary cross-entropy — detector label — predicted concept

$$\mathcal{L}_C = \sum_j \mathrm{BCE}\left(c_j^{\mathrm{det}}, \ \hat{c}_j \right)$$

as in the original CBM, but no person draws the labels

Grounding the concepts

One term of the loss, worked out

$$\mathrm{BCE}(c, \hat{c}) = -\log \hat{c} \ \text{ if } c = 1, \qquad -\log(1 - \hat{c}) \ \text{ if } c = 0$$

CONCEPT	BOX?	LABEL	PREDICTED	TERM
black wingtips	✓	1	0.8	$-\log 0.8 \approx 0.22$
loud laughing call	none	0	0.2	$-\log(1 - 0.2) \approx 0.22$

a prediction close to its label costs little

ILLUSTRATIVE VALUES · predictions made up for teaching

Grounding the concepts

Count the non-zero weights in each class's row

the head: one weight per concept

one row per class	grey back	black wingtips	yellow bill	red bill spot	dark back	pink legs	non-zero
Herring Gull	1.2	0.8	0	0.9	0	0	3
California Gull	0.7	0	1.1	0	0	0	2
Western Gull	0	0.5	0	0.4	1.3	0.6	4

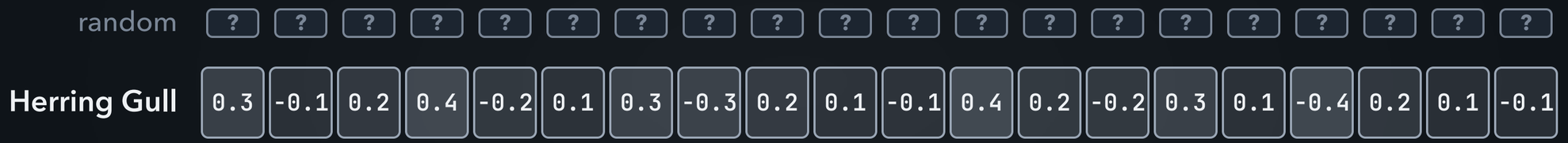
number of effective concepts (NEC): the average count

$$\text{NEC} = \frac{3 + 2 + 4}{3} = 3$$

ILLUSTRATIVE VALUES · weights drawn for teaching

Grounding the concepts

A head that uses every concept explains nothing



20 of 20 weights non-zero

Even random, untrained concepts, given enough of them, can approximate any linear classifier.

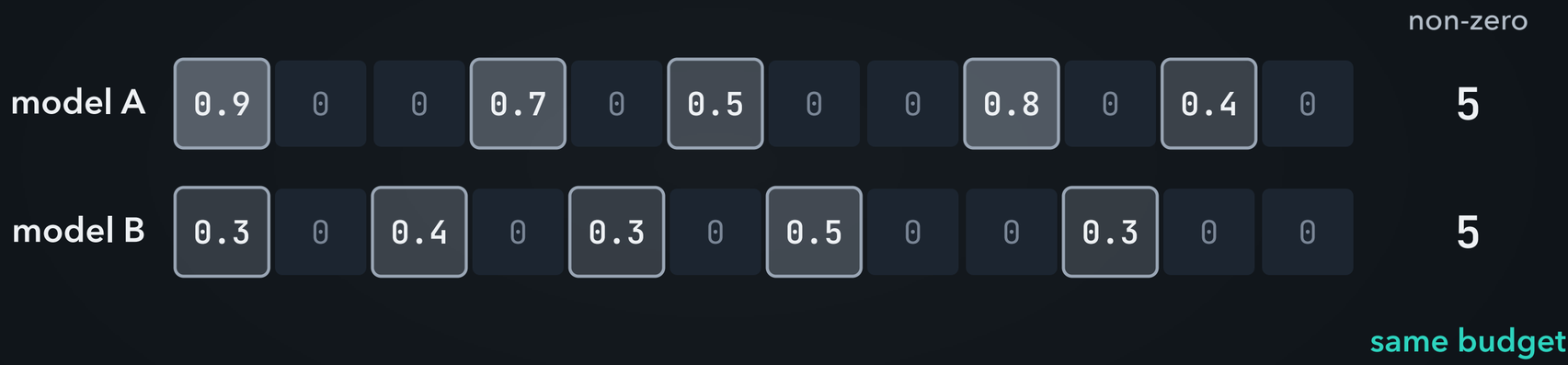
Accurate, and still nothing to read.

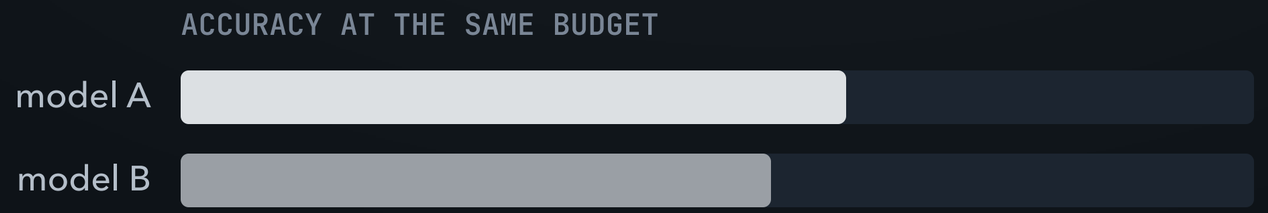
REPORTED · Theorem 4.1, Srivastava et al. 2024

ILLUSTRATIVE VALUES · weights drawn for teaching

Grounding the concepts

A fair comparison: the same number of concepts per class





the gap now says something about the concepts

ILLUSTRATIVE VALUES · weights and bars drawn for teaching

Grounding the concepts

The same count in symbols

average over the **K classes**

non-zero weights in row k

$$\mathrm{NEC} = \frac{1}{K} \sum_{k=1}^{K} \|W_k\|_0$$

Herring Gull

1.2	0.8	0	0.9	0	0

$$\|W_{\mathrm{Herring}}\|_0 = 3$$

$\|\cdot\|_0$ counts the entries that are not zero

ILLUSTRATIVE VALUES · weights drawn for teaching (the row from the count slide)

Grounding the concepts

Five concepts per class can actually be read

$$\mathrm{NEC} = 5$$

each class: a handful of named concepts

ILLUSTRATIVE VALUES · weights drawn for teaching

Grounding the concepts

What changed compared with label-free models

	label-free models		VLG-CBM
the concept layer learns from	copying CLIP scores	→	**detector boxes**
the head is judged at	its own sparsity level	→	**a fixed number per class**

Grounding the concepts

VLG-CBM in symbols

VLG-CBM (Srivastava et al., NeurIPS 2024) x → ĉ → ŷ

concept predictor	$\hat{c} = W_c\, e(x),$ trained on detector labels c^{det}
what reaches the head	concept scores $\hat{c}_1, \ldots, \hat{c}_k$
task head	$\hat{y} = f_\phi(\hat{c}) = W\hat{c} + b,$ sparse, NEC fixed
what is trained	$W_c,\ f_\phi$
data needed	$\{(x, y)\}$ + an open-vocabulary detector

Classification by description

The last rung: no training data at all

DATA EACH VARIANT NEEDS

$\{x,\ c,\ y\}$ CBM, CEM → $\{x,\ y\}$ PCBM, LF-CBM, LaBo, VLG-CBM → $\varnothing$ **by description**

Classification by description (CBD)

Menon and Vondrick, ICLR 2023

no concept labels, no **class labels**, no training

Classification by description

Each class gets a checklist, and CLIP ticks it softly

photo

Herring Gull

☑ a white body with black wingtips	0.9
☑ a yellow beak and legs	0.8
☑ a red spot on its beak	0.7
average	0.80

Ivory Gull

☑ a white bird	0.8
☑ a black bill	0.2
☑ pink legs	0.3
average	0.43

Ring-billed Gull

☑ red eyes	0.2
☑ webbed feet	0.6
☑ a long, black tail	0.4
average	0.40

best average: **wins**

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

ILLUSTRATIVE VALUES · real descriptors; tick strengths drawn for teaching

Classification by description

A language model describes every class

language model
writes descriptors

```
{ "Herring Gull": [
    "a large bird with a wingspan of up to 2.3 meters",
    "a white body with black wingtips",
    "a yellow beak and legs",
    "a red spot on its beak",
    "webbed feet"
]}
```

descriptor: a short phrase about what the **class** looks like

DATASET · Menon and Vondrick's released CUB descriptors

Classification by description

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

CLIP scores the photo against each descriptor

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

each score: a soft tick

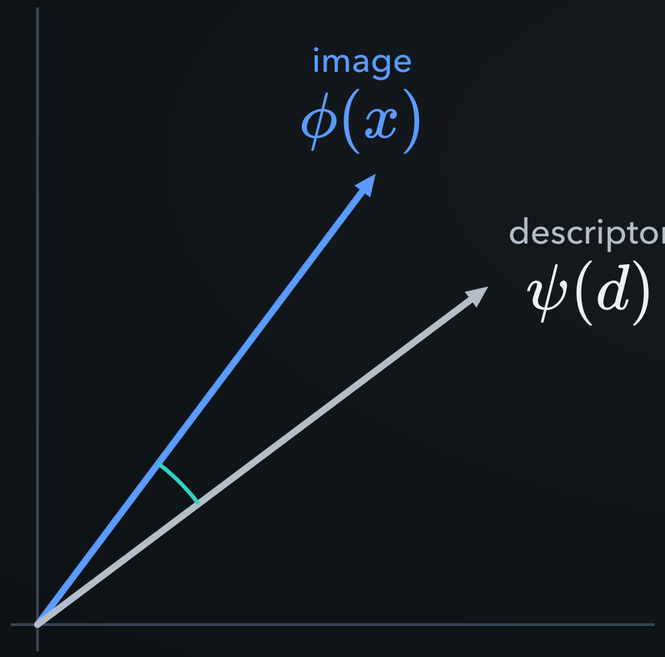
- a large bird with a wingspan of up to 2.3 meters
- a white body with black wingtips
- a yellow beak and legs
- a red spot on its beak
- webbed feet

class score: the average

The **highest average** wins.

Classification by description

One descriptor score: a cosine between two vectors



entries (0.6, 0.8) and (0.8, 0.6)

$$\hat{c}_d = \cos\big(\phi(x),\ \psi(d)\big)$$

$$= 0.6 \times 0.8 + 0.8 \times 0.6 = 0.96$$

both vectors have length 1, so the cosine is the sum of products

near 1: they point the same way; smaller: they point apart

ILLUSTRATIVE VALUES · two-entry vectors for teaching; CLIP's have 512 entries

Classification by description

The class score averages its descriptor scores

class score — average over the class's descriptors — descriptor score

$$\hat{y}_k = \frac{1}{|D_k|} \sum_{d \in D_k} \hat{c}_d$$

D_k: the list of descriptors of class k, and $|D_k|$ how many there are

Nothing is fitted to the target data.

Classification by description

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Every score comes with its reasons

bars start at 0.30

$$\hat{y}_{\text{Herring}} = \tfrac{1}{5}\big(0.324 + 0.339 + 0.339 + 0.340 + 0.328\big) = 0.334$$

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

COMPUTED · REAL MODEL · CLIP ViT-B/32 on our gull photo, released CUB descriptors

Classification by description

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Slaty-backed gull edges ahead

class scores: **Slaty-backed 0.337** · Herring 0.334 · Glaucous-winged 0.328 · Western 0.326 · California 0.324

Slaty-backed Gull 0.337

a large gull with a wingspan of up to 2.3 meters 0.335

are dark grey and the underparts are white 0.339

and neck are white with a black cap 0.337

is yellow with a red spot near the tip 0.342

and feet are yellow 0.339

have a black trailing edge 0.331

Herring Gull 0.334

a large bird with a wingspan of up to 2.3 meters 0.324

a white body with black wingtips 0.339

a yellow beak and legs 0.339

a red spot on its beak 0.340

webbed feet 0.328

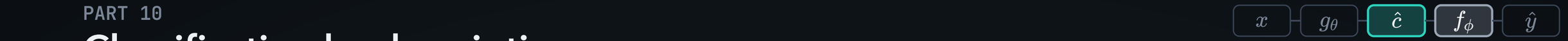
Slaty-backed Gull edges ahead by 0.003

bars start at 0.30; dashed line: the average

Our photo shows a **herring gull**: a near miss.

COMPUTED · REAL MODEL · CLIP ViT-B/32 on our gull photo, released CUB descriptors

Classification by description



Attractive when there are no labels at all

no **labelled images**

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

classes we already score: Slaty-backed, Herring, Glaucous-winged, …

+ one more class

```
{ "Ivory Gull": [
    "a white bird",
    "black wingtips",
    "a black bill",
    "pink legs",
    "yellow eyes"
]}
```

its checklist is scored at once

no retraining

DATASET · Menon and Vondrick's released CUB descriptors

Classification by description

A descriptor can be wrong

The language model says:

```
{ "California Gull": [
    "a white bird with a black head and bill",
    ...
]}
```

It does not have a black head.

The wrong descriptor is still averaged into the score, one of 7.

DATASET · Menon and Vondrick's released CUB descriptors

Classification by description

A match score is not a fact check

Our herring gull photo

White head, yellow bill

From the California Gull checklist:

"a white bird with a black head and bill"

These words do not match this photo.

CLIP

Match score

0.325

Similarity, not a probability

California Gull

Class average

still includes this score

Check the words

Correct and visible in the photo?

Check the score

Does changing that feature change its score?

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

COMPUTED · REAL MODEL · CLIP ViT-B/32 on this photo; released CUB descriptor

Classification by description

What changed compared with LaBo: the head is no longer learned

	LaBo		by description
the task head	a linear layer	→	**a fixed average**
what is trained	the head	→	**nothing**
data needed	images and **class labels**	→	**none**

Classification by description

Classification by description in symbols

Classification by description (Menon and Vondrick, ICLR 2023) x → ĉ → ŷ

concept predictor	$\text{CLIP: image encoder } \phi,\ \text{text encoder } \psi$
what reaches the head	$\hat{c}_d = \cos\big(\phi(x),\ \psi(d)\big),\ \ d \in D_k$
task head	$\hat{y}_k = \frac{1}{\lvert D_k \rvert} \sum_{d \in D_k} \hat{c}_d \ \ \text{(fixed)}$
what is trained	nothing
data needed	$\text{none: a language model writes the descriptors } D_k$

The cost: nothing guarantees that the descriptors are correct, or grounded.

Comparing the variants

One table: one row per variant, one column at a time

variant	data needed	concepts from	what the head reads	trained
CBM Koh et al. 2020				
CEM Espinosa Zarlenga et al. 2022				
PCBM Yuksekgonul et al. 2023				
LF-CBM Oikarinen et al. 2023				
LaBo Yang et al. 2023				
VLG-CBM Srivastava et al. 2024				
CBD (by description) Menon and Vondrick 2023				

Comparing the variants

Column one: the supervision each variant needs

key: x an image, c its concept labels, y its class label

variant	data needed	concepts from	what the head reads	trained
CBM Koh et al. 2020	$\{(x, c, y)\}$			
CEM Espinosa Zarlenga et al. 2022	$\{(x, c, y)\}$			
PCBM Yuksekgonul et al. 2023				
LF-CBM Oikarinen et al. 2023				
LaBo Yang et al. 2023				
VLG-CBM Srivastava et al. 2024				
CBD (by description) Menon and Vondrick 2023				

Comparing the variants

Most variants drop the concept labels

key: x an image, c its concept labels, y its class label

variant	data needed	concepts from	what the head reads	trained
CBM Koh et al. 2020	$\{(x, c, y)\}$			
CEM Espinosa Zarlenga et al. 2022	$\{(x, c, y)\}$			
PCBM Yuksekgonul et al. 2023	$\{(x, y)\}$ + a concept bank			
LF-CBM Oikarinen et al. 2023	$\{(x, y)\}$			
LaBo Yang et al. 2023	$\{(x, y)\}$			
VLG-CBM Srivastava et al. 2024	$\{(x, y)\}$ + a detector			
CBD (by description) Menon and Vondrick 2023	**none**			

Comparing the variants

Column two: where the concepts come from

key: x an image, c its concept labels, y its class label

variant	data needed	concepts from	what the head reads	trained
CBM Koh et al. 2020	$\{(x, c, y)\}$	human labels		
CEM Espinosa Zarlenga et al. 2022	$\{(x, c, y)\}$	human labels		
PCBM Yuksekgonul et al. 2023	$\{(x, y)\}$ + a concept bank	a concept bank		
LF-CBM Oikarinen et al. 2023	$\{(x, y)\}$	a language model		
LaBo Yang et al. 2023	$\{(x, y)\}$	a language model, CLIP		
VLG-CBM Srivastava et al. 2024	$\{(x, y)\}$ + a detector	a language model, a detector		
CBD (by description) Menon and Vondrick 2023	**none**	a language model		

Comparing the variants

Column three: what the task head reads

key: x an image, c its concept labels, y its class label

variant	data needed	concepts from	what the head reads	trained
CBM Koh et al. 2020	$\{(x,\ c,\ y)\}$	human labels	$\hat{c}_j \in [0, 1]$ one number per concept	
CEM Espinosa Zarlenga et al. 2022	$\{(x,\ c,\ y)\}$	human labels	$\hat{c}_j \in \mathbb{R}^m$ a vector per concept	
PCBM Yuksekgonul et al. 2023	$\{(x,\ y)\}$ + a concept bank	a concept bank	$\langle e(x),\ v_j \rangle\ +\ r$ projections + a residual	
LF-CBM Oikarinen et al. 2023	$\{(x,\ y)\}$	a language model	$\hat{c} = W_c\, e(x)$ a learned concept layer	
LaBo Yang et al. 2023	$\{(x,\ y)\}$	a language model, CLIP	$\cos(\phi(x),\ \psi(s_j))$ raw CLIP scores	
VLG-CBM Srivastava et al. 2024	$\{(x,\ y)\}$ + a detector	a language model, a detector	$\hat{c} = W_c\, e(x)$ a learned concept layer	
CBD (by description) Menon and Vondrick 2023	**none**	a language model	$\cos(\phi(x),\ \psi(d))$ raw CLIP scores	

Comparing the variants

Column four: what is trained

key: x an image, c its concept labels, y its class label

variant	data needed	concepts from	what the head reads	trained
CBM Koh et al. 2020	$\{(x,\ c,\ y)\}$	human labels	$\hat{c}_j \in [0,1]$ one number per concept	$g_\theta,\ f_\phi$ predictor + head
CEM Espinosa Zarlenga et al. 2022	$\{(x,\ c,\ y)\}$	human labels	$\hat{c}_j \in \mathbb{R}^m$ a vector per concept	$g_\theta,\ f_\phi$ predictor + head
PCBM Yuksekgonul et al. 2023	$\{(x,\ y)\}$ + a concept bank	a concept bank	$\langle e(x),\ v_j \rangle\ +\ r$ projections + a residual	f_ϕ head only
LF-CBM Oikarinen et al. 2023	$\{(x,\ y)\}$	a language model	$\hat{c} = W_c\, e(x)$ a learned concept layer	$W_c,\ f_\phi$ concept layer + head
LaBo Yang et al. 2023	$\{(x,\ y)\}$	a language model, CLIP	$\cos(\phi(x),\ \psi(s_j))$ raw CLIP scores	f_ϕ head only
VLG-CBM Srivastava et al. 2024	$\{(x,\ y)\}$ + a detector	a language model, a detector	$\hat{c} = W_c\, e(x)$ a learned concept layer	$W_c,\ f_\phi$ concept layer + head
CBD (by description) Menon and Vondrick 2023	**none**	a language model	$\cos(\phi(x),\ \psi(d))$ raw CLIP scores	**nothing**

Comparing the variants

Naming a concept does not ground it

Each change: cheaper concepts, or more accurate **predictions**.

None makes grounding automatic.

a phrase from a language model, or a high $\cos(\phi(x), \psi(s))$

does not show that

the model sees that property in the image

check it

Comparing the variants

Who supplies the concepts, and who checks them?

variant	who supplies the concepts	who checks them
CBM, CEM	people label them	**people**
PCBM	a concept bank	whoever built the bank
LF-CBM, LaBo	a language model	a CLIP similarity only
VLG-CBM	a language model	a detector
CBD	a language model	**nobody**

Comparing the variants

Each rung up the ladder removes a cost and a check

$\{x,\ c,\ y\}$ CBM, CEM

$\{x,\ y\}$ PCBM, LF-CBM, LaBo, VLG-CBM

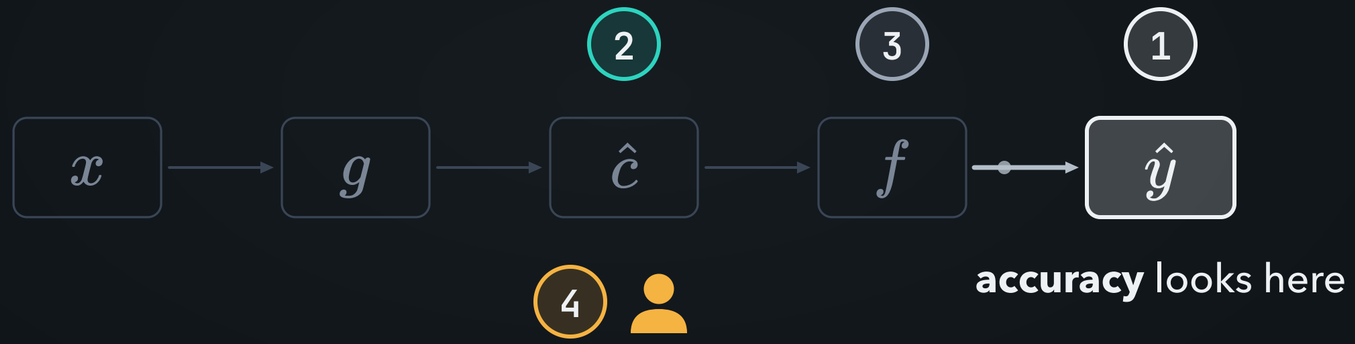
$\varnothing$ by description

Labour saved on labels has to be spent later, on checking.

SCHEMATIC · bar levels drawn for teaching

How should we evaluate a CBM?

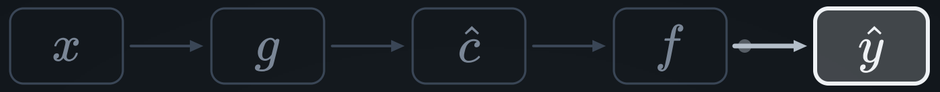
Accuracy alone hides most of what a concept bottleneck promises



four questions, each about a different part of the model

How should we evaluate a CBM?

Task accuracy: are the predictions correct?



In everyday terms: is the final **answer** right?

MEASURE

the share of held-out test images classified correctly

held-out: images the model never saw during training

How should we evaluate a CBM?

Task accuracy counts the right answers

$$\text{task accuracy} = \frac{1}{N}\sum_{i=1}^{N} \mathbf{1}\left[\hat{y}_i = y_i\right]$$

N test images; $\mathbf{1}[\cdot]$ is 1 when its statement is true, 0 otherwise

prediction $\hat{y}_i$	true label y_i	$\mathbf{1}[\hat{y}_i = y_i]$
small red circle	small red circle	1
large blue square	large blue square	1
large red square	small red square	0

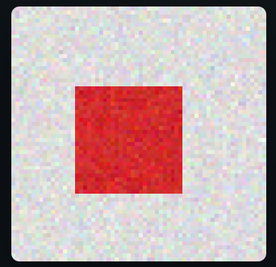
$$\text{task accuracy} = \frac{1+1+0}{3} = \frac{2}{3}$$

ILLUSTRATIVE VALUES · three predictions for teaching

How should we evaluate a CBM?

Concept accuracy: are the concept values right?

In everyday terms: did the model read the evidence correctly?



test image 10
small red square

	score $\hat{c}$	$\text{round}(\hat{c})$	label c	match
red	1.00	1	1	1
blue	0.00	0	0	1
circle	0.04	0	0	1
square	0.99	1	1	1
large	0.78	1	0	0

4 / 5

values right on this image

$$\text{concept acc.} = \frac{1}{Nk} \sum_{i,j} \mathbf{1}\left[\text{round}(\hat{c}_{ij}) = c_{ij}\right]$$

MEASURED · saved model, test image 10

How should we evaluate a CBM?

Grounding: does each concept rest on the right evidence?

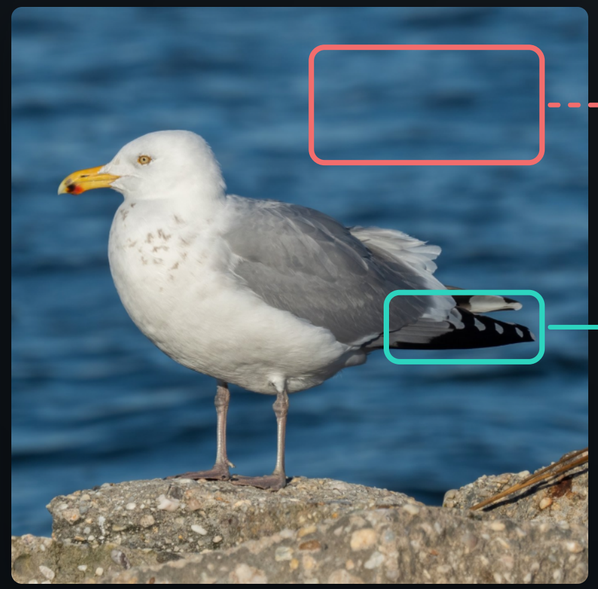


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

not this: water

wingtip black

intended evidence: wingtip

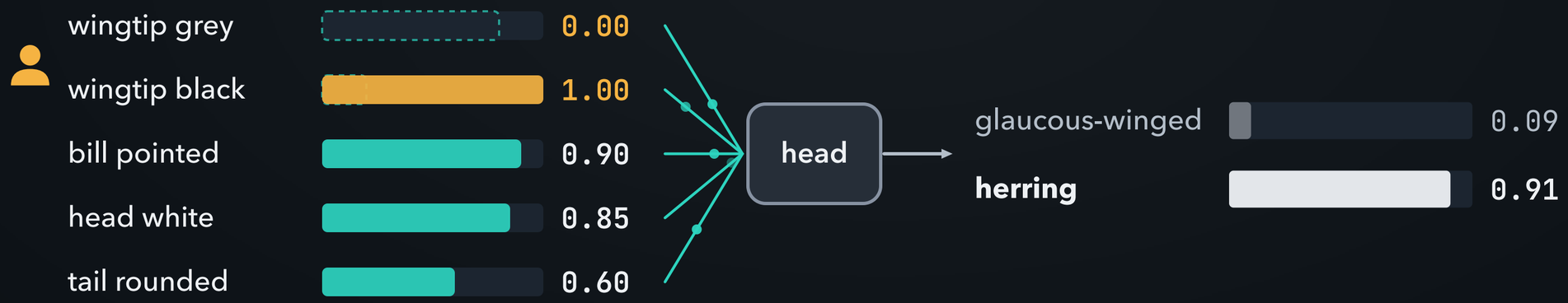
grounding: check where the detector looks

SCHEMATIC · regions drawn by hand

How should we evaluate a CBM?

Faithfulness: does the output really depend on the concepts?

Faithful: the explanation tells the truth about how the **answer** was made

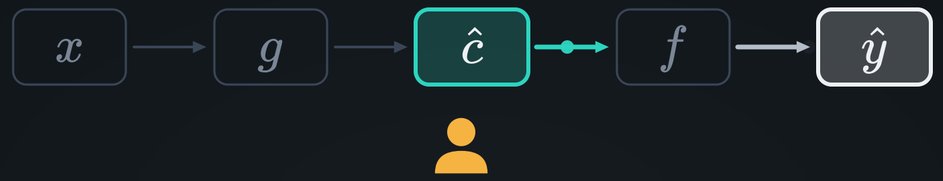


change a concept: the **output** should move as the head says

ILLUSTRATIVE VALUES · the gull head from Part 2

How should we evaluate a CBM?

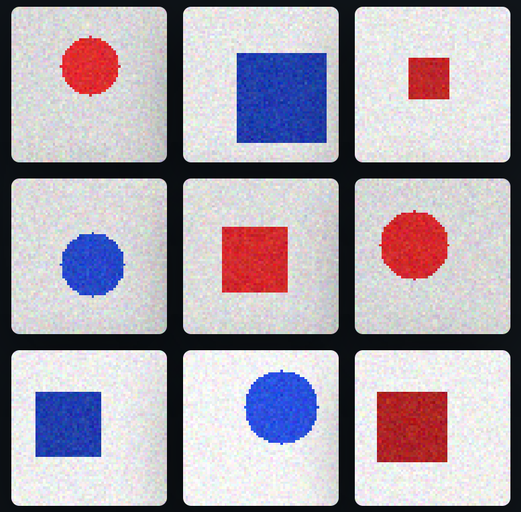
Intervention utility: do corrections help, and at what cost?



when a person fixes a concept, does the **answer** improve, and at what cost in time?

How should we evaluate a CBM?

A controlled test: synthetic shapes with known concepts



real test inputs, 64 × 64 px

red or blue

circle or square

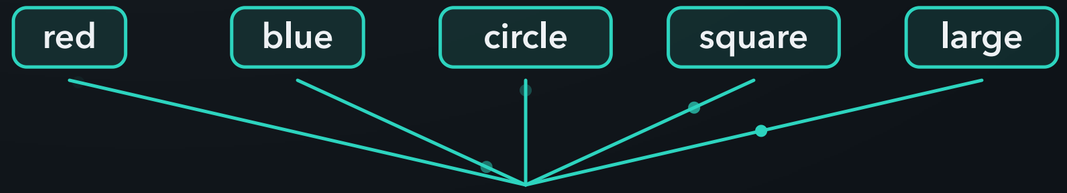
small or large

→ **8 classes**

2 × 2 × 2

5 concepts

each 1 (yes) or 0 (no)



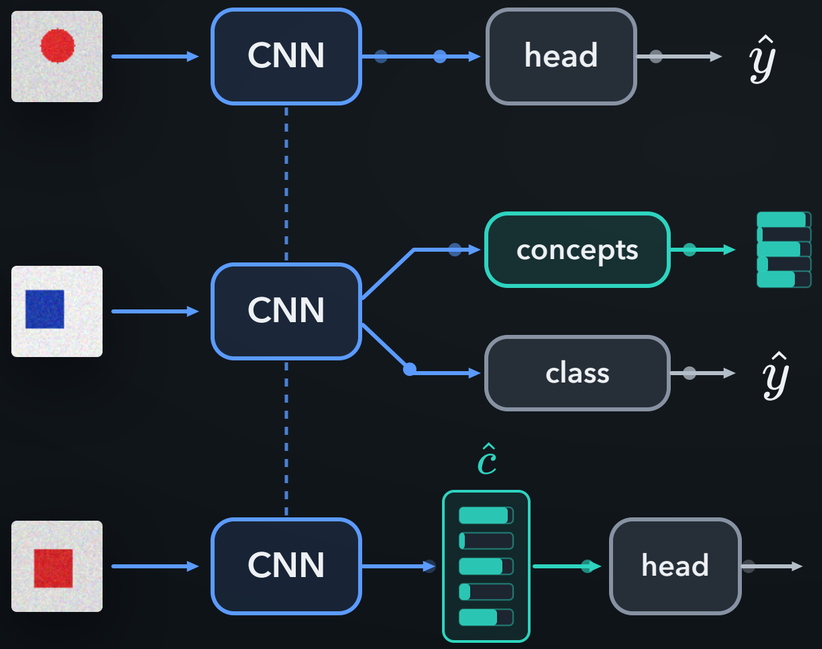
the concepts fully determine the **class**

MEASURED · one seed · 512 synthetic test images · oracle corrections

How should we evaluate a CBM?

x g_θ $\hat{c}$ f_ϕ $\hat{y}$

Three models share one small CNN



same small CNN, same data, 60 epochs

epoch: one pass through the 768 training images

the class head sits beside the concepts

the head reads only the concepts

How should we evaluate a CBM?

Task accuracy: the three models are close

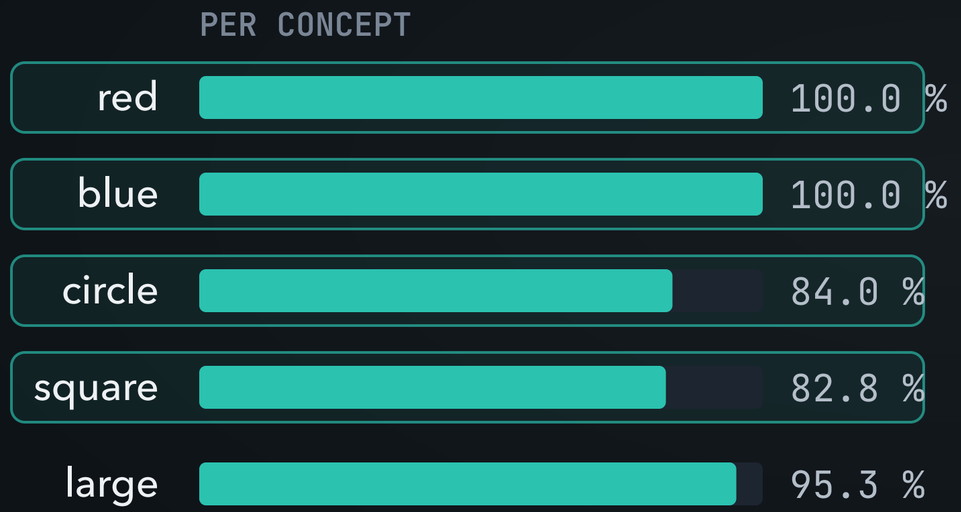
each bar: the share of the 512 test images classified correctly

on this question alone, the three look about the same

MEASURED · one seed · 512 synthetic test images · oracle corrections

How should we evaluate a CBM?

Concept accuracy: most concept values are right



92.4 %

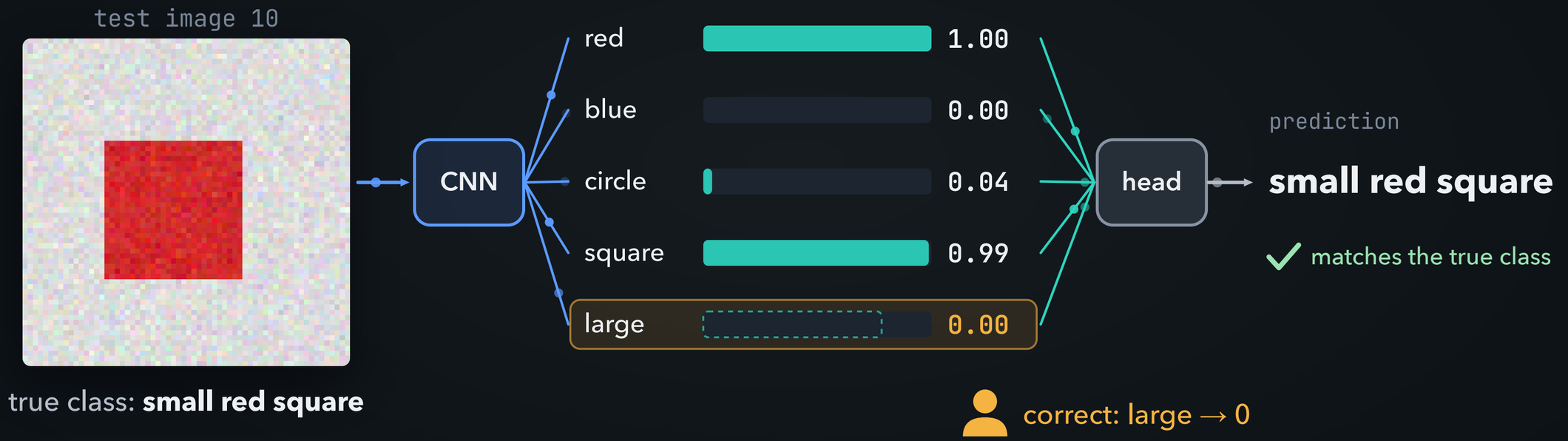
of concept values correct, strict CBM

all 512 images × 5 concepts

MEASURED · one seed · 512 synthetic test images · oracle corrections

How should we evaluate a CBM?

One edit to one concept changes the prediction



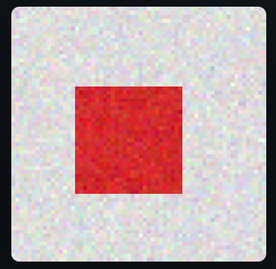
$$\hat{\mathbf{c}} = (1.00,\ 0.00,\ 0.04,\ 0.99,\ 0.78) \rightarrow (1.00,\ 0.00,\ 0.04,\ 0.99,\ 0)$$

MEASURED · saved model, test image 10

How should we evaluate a CBM?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Oracle corrections: fix the worst concept first



test image 10

	score $\hat{c}$	label c	error $\lvert\hat{c} - c\rvert$	
red	1.00	1		0.00
blue	0.00	0		0.00
circle	0.04	0		0.04
square	0.99	1		0.01
large	0.78	0		0.78

fix this one first

oracle: the true labels play a perfect expert

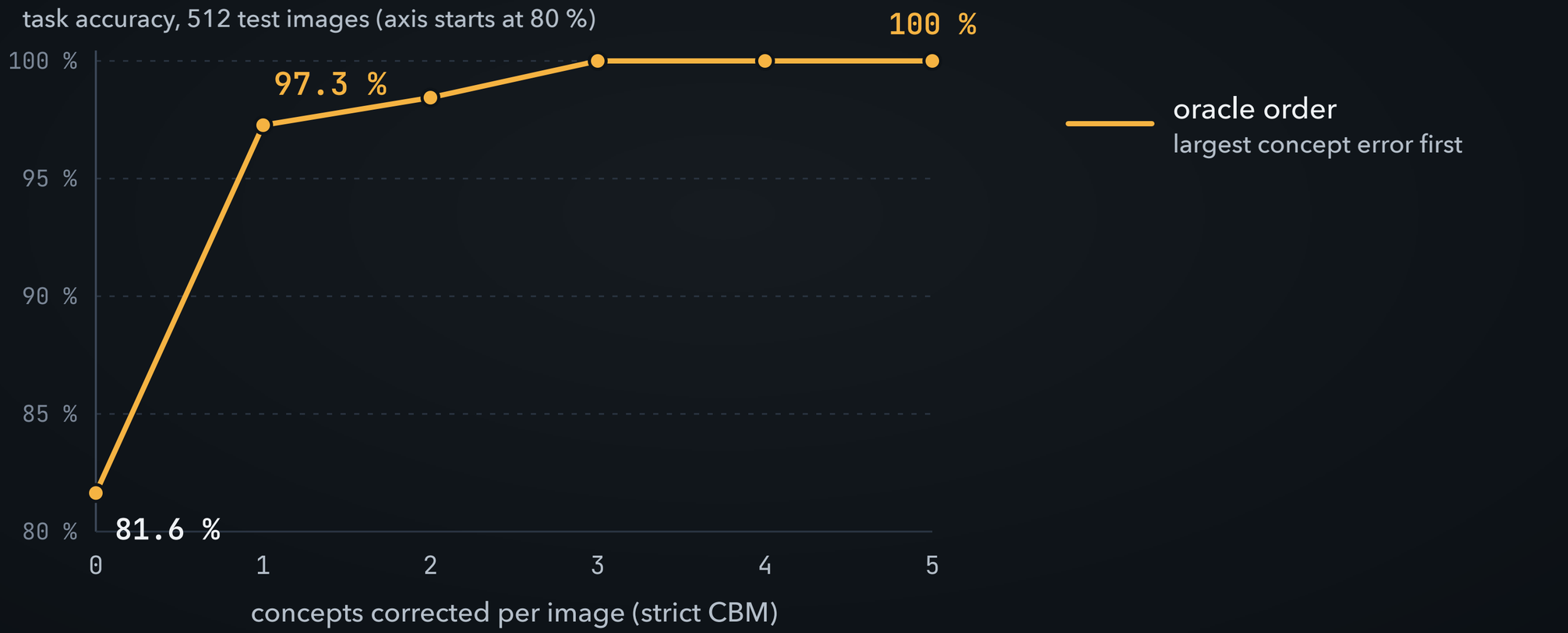
$$\lvert 0.78 - 0 \rvert = 0.78$$

MEASURED · saved model, test image 10

How should we evaluate a CBM?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Each step right is one more **correction** per image

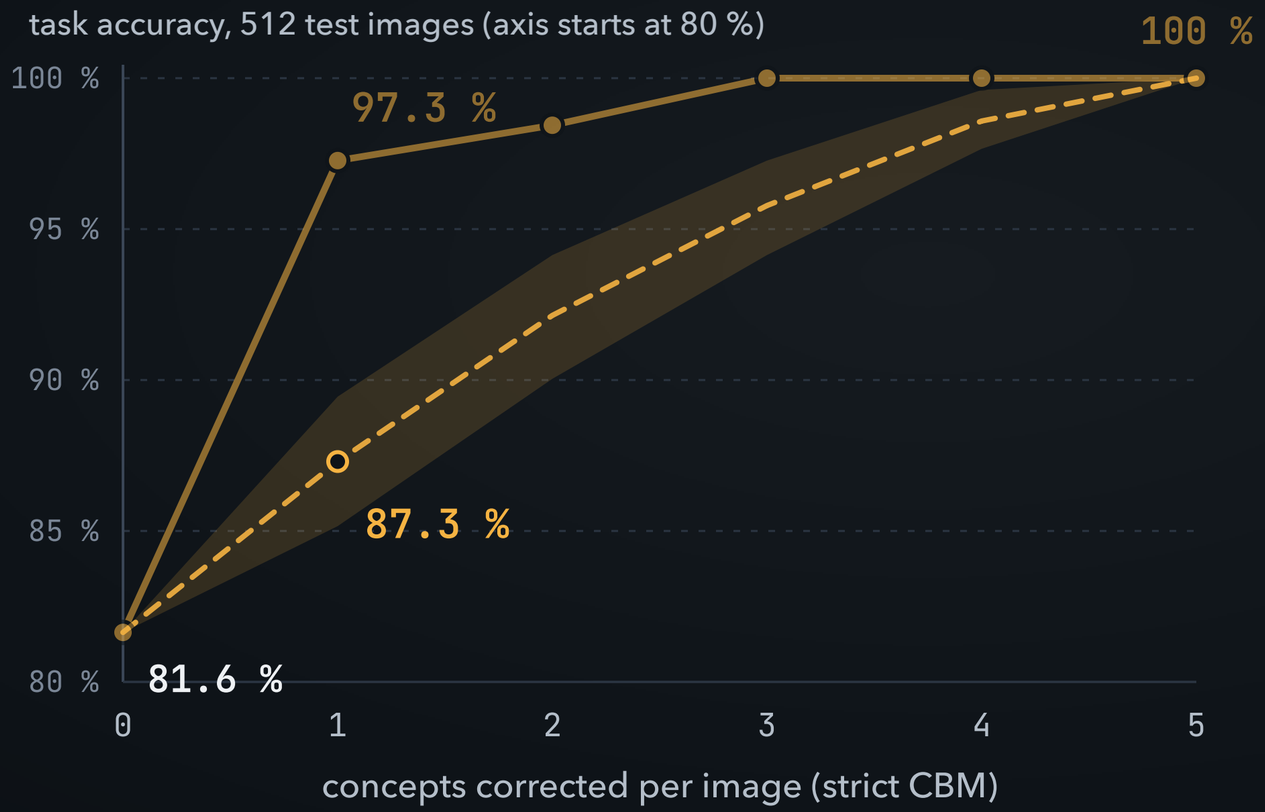


MEASURED · one seed · 512 synthetic test images · oracle corrections

How should we evaluate a CBM?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Random corrections help more slowly



oracle order
largest concept error first

random order
mean over 32 random orders

lowest to highest
of the 32 runs

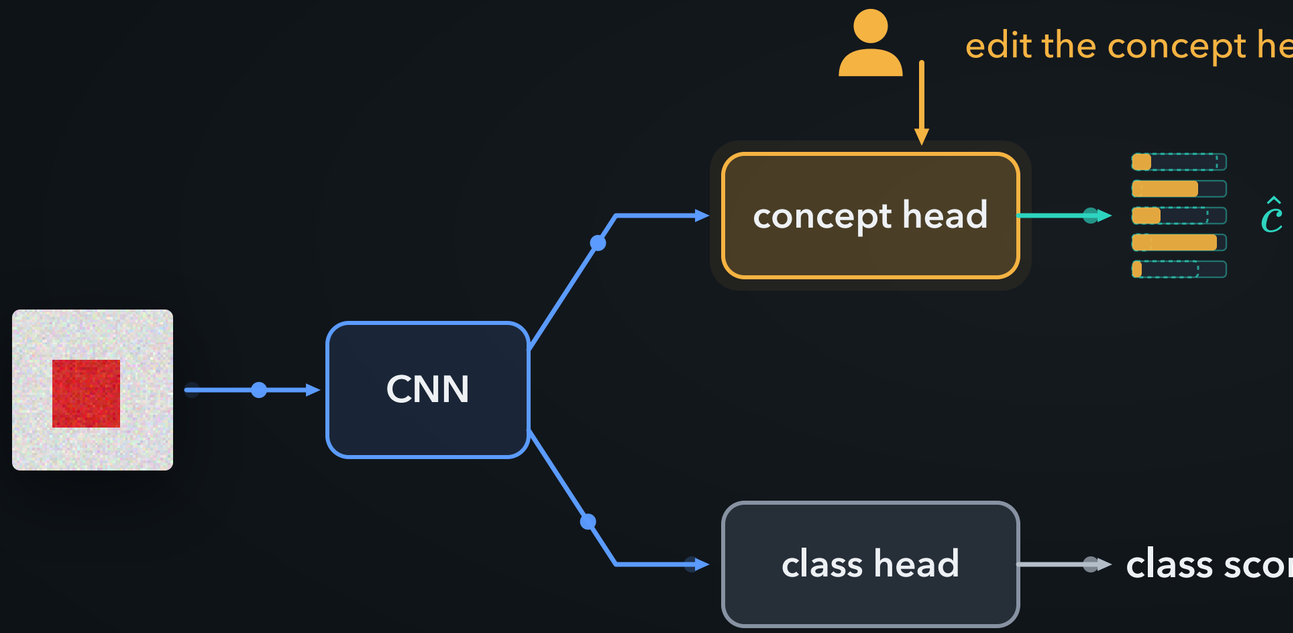
a random pick often fixes a concept that was already right

MEASURED · one seed · 512 synthetic test images · oracle corrections

How should we evaluate a CBM?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

In the multi-task model, a concept edit changes nothing



✓ unchanged (verified exactly)

the class head never reads the concepts

MEASURED · one seed · 512 synthetic test images · oracle corrections

How should we evaluate a CBM?

Read these numbers with care

1. one training seed: a single training run
2. 512 synthetic test images
3. oracle corrections from the true labels
4. no human study: a perfect, effortless expert

---

real experts make mistakes, and their time is costly

How should we evaluate a CBM?

Similar accuracy, very different intervention utility

task accuracy	78.7 % · 82.0 % · 81.6 % black box · multi-task · strict CBM
concept accuracy	92.4 % of concept values (strict CBM) grounding: not measured here
faithfulness	strict CBM: the head reads only the concepts multi-task: the class head ignores them
intervention utility	one correction: 81.6 → 97.3 % (strict CBM), no change (multi-task) human cost: not measured here

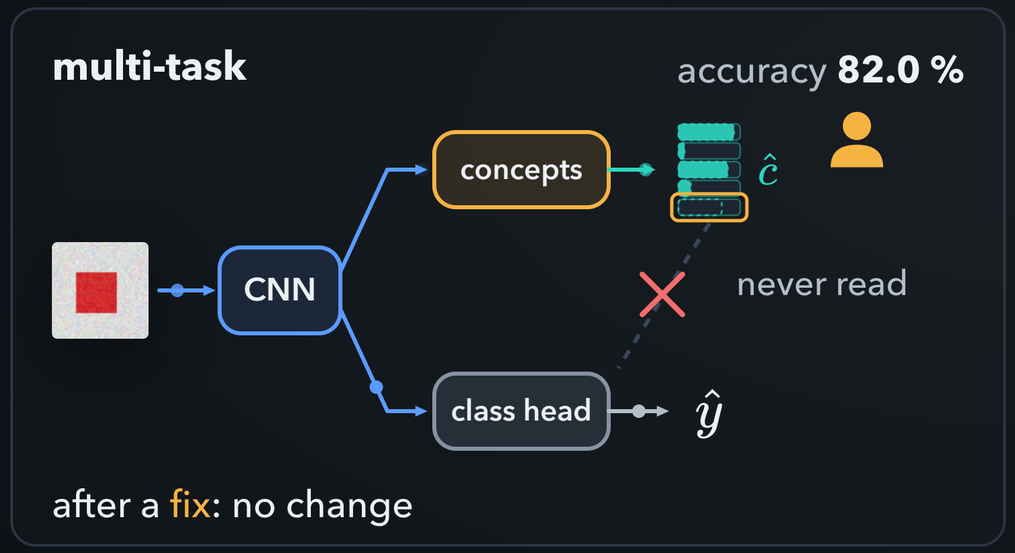
`MEASURED · one seed · 512 synthetic test images · oracle corrections`

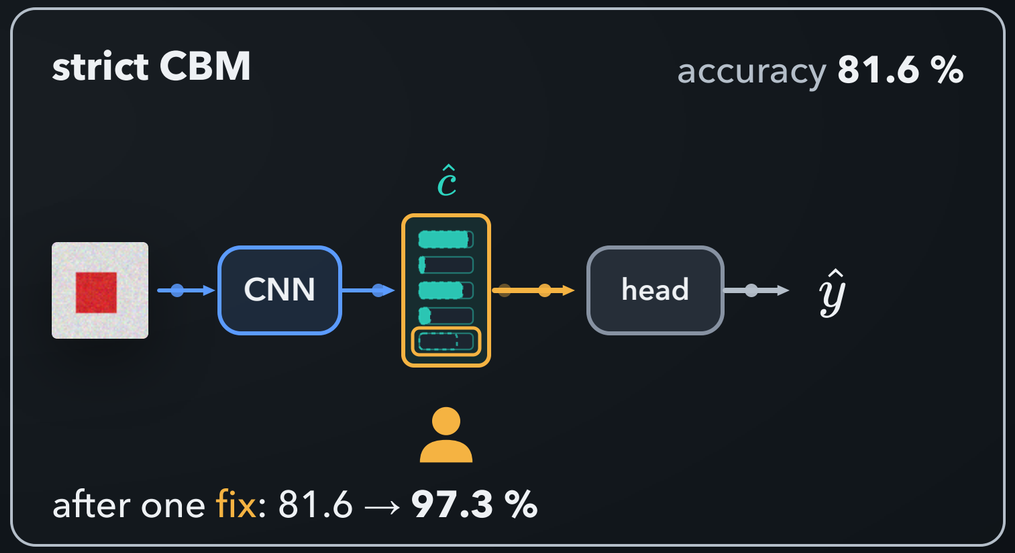
How should we evaluate a CBM?

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Why similar accuracy can hide very different intervention utility

accuracy marks only the final answer, not the route to it





a fix helps only if the head reads the concepts

MEASURED · one seed · 512 synthetic test images · oracle corrections

Beyond images

Where else do concept bottlenecks fit?

SO FAR: PHOTOGRAPHS OF BIRDS

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

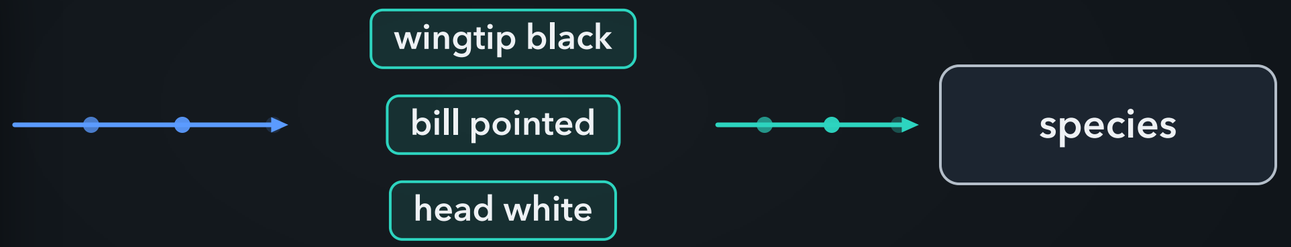


photo x concepts $\hat{\mathbf{c}}$ **species** $\hat{y}$

they fit wherever experts reason with named findings

Beyond images

Medicine: a knee X-ray → clinical findings → a severity grade

input x

named concepts $\hat{\mathbf{c}}$

each finding is a small grade, 0 = none

joint space narrowing	1
bone spurs	2
sclerosis	1
7 more findings	⋮

task $\hat{y}$

osteoarthritis grade

$\hat{y} \in \{0, 1, 2, 3\}$

$\hat{y} = 2$

0 none or doubtful · 3 severe

Knee X-ray: Koh et al. 2020, Fig. 1 (CC BY 4.0), Osteoarthritis Initiative

Koh et al., ICML 2020 · Osteoarthritis Initiative

ILLUSTRATIVE VALUES · values for teaching

Beyond images

A radiologist corrects a finding, and the grade is recomputed

input x

Knee X-ray: Koh et al. 2020, Fig. 1 (CC BY 4.0), Osteoarthritis Initiative

named concepts $\hat{\mathbf{c}}$

each finding is a small grade, 0 = none

joint space narrowing	2
bone spurs	2
sclerosis	1
7 more findings	⋮

radiologist corrects a finding

task $\hat{y}$

osteoarthritis grade

$$\hat{y} \in \{0, 1, 2, 3\}$$

$$\hat{y} = 3$$

0 none or doubtful · 3 severe

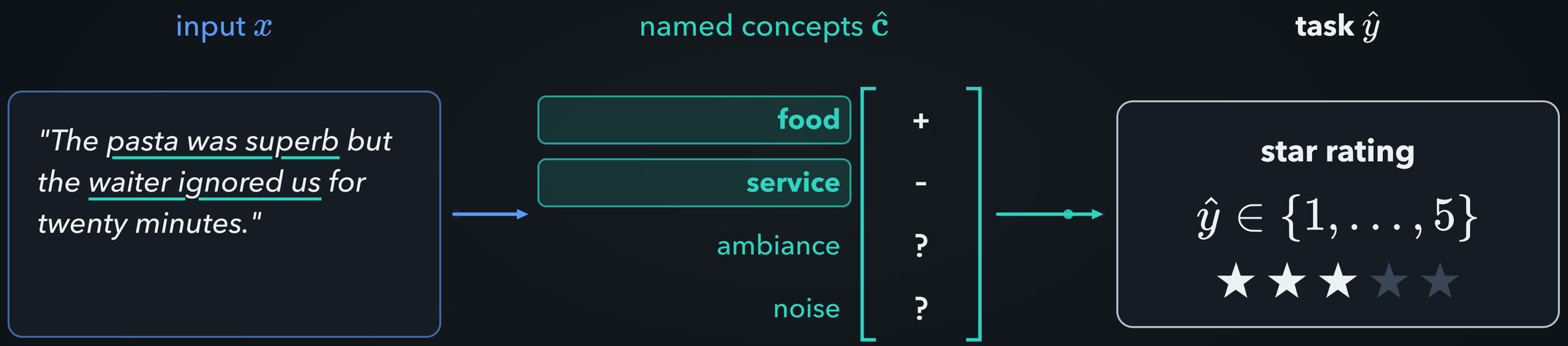
Why it fits: radiologists already write their reports in these terms

Koh et al., ICML 2020 · Osteoarthritis Initiative

ILLUSTRATIVE VALUES · values for teaching

Beyond images

Language: a review → four aspects → a star rating



+ praised · – criticised · ? not mentioned

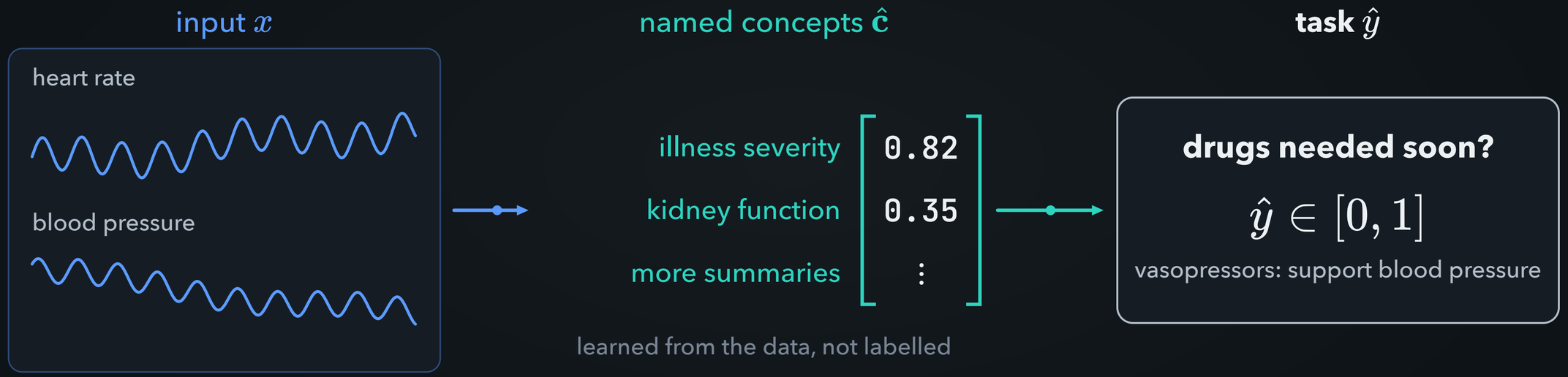
Why it fits: reviewers already write about the food and the service

CEBaB, Abraham et al., NeurIPS 2022 · text bottleneck models, Ludan et al. 2023

ILLUSTRATIVE VALUES · review and values for teaching

Beyond images

Intensive care: vital signs → summary concepts → drugs needed soon?



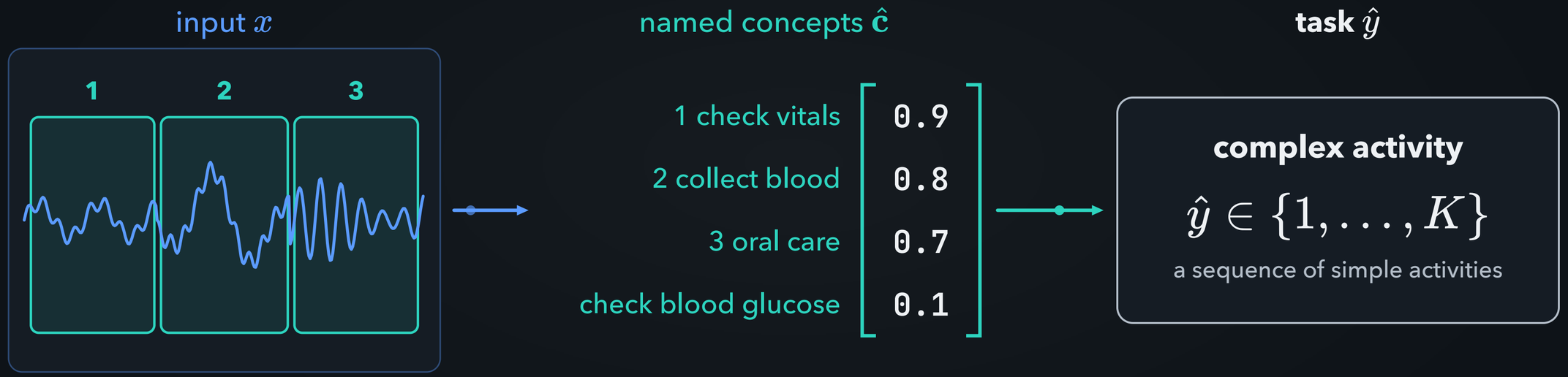
Why it fits: doctors already sum up a patient in terms like illness severity and kidney function

Wu et al., MLHC 2022 · MIMIC-III, 15,552 patients

ILLUSTRATIVE VALUES · schematic traces, values for teaching

Beyond images

Wearables: sensor signal → simple activities → complex activity



Why it fits: people already describe a **complex activity** as its simple steps

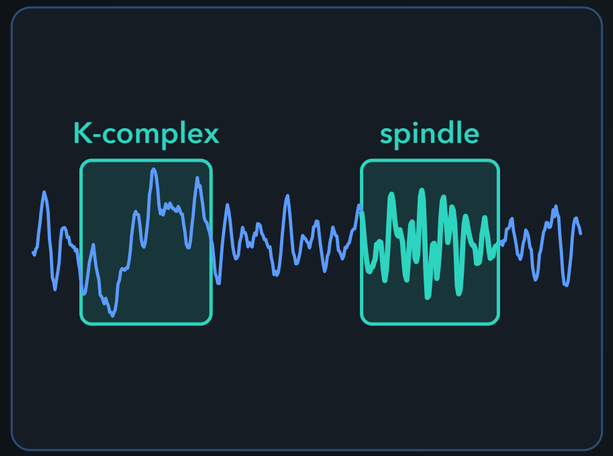
X-CHAR, Jeyakumar et al., IMWUT 2023 · concepts from its nurse data

ILLUSTRATIVE VALUES · schematic trace, values for teaching

Beyond images

Sleep: an EEG → the patterns scorers use → a sleep stage

input x



named concepts $\hat{\mathbf{c}}$

spindle	1.00
K-complex	0.80
slow waves	0.15

task $\hat{y}$

sleep stage

$\hat{y} \in \{\mathrm{W}, \mathrm{N1}, \mathrm{N2}, \mathrm{N3}, \mathrm{R}\}$

N2 candidate: **0.86**

W wake · N1, N2, N3: stages one to three · R: REM

Why it fits: scorers already mark these patterns by hand

patterns from the AASM scoring manual (Berry et al.)

ILLUSTRATIVE VALUES · schematic EEG, the sleep head from Part 6

Beyond images

One recipe in every domain

Medicine	Language	Intensive care	Wearables	Sleep
knee X-ray → grade	review → stars	vital signs → drug need	sensor signal → activity	EEG → sleep stage

THE RECIPE

name the findings an expert checks

$$c_1, \ldots, c_k$$

predict them from the raw signal

$$\hat{\mathbf{c}} = g_\theta(x)$$

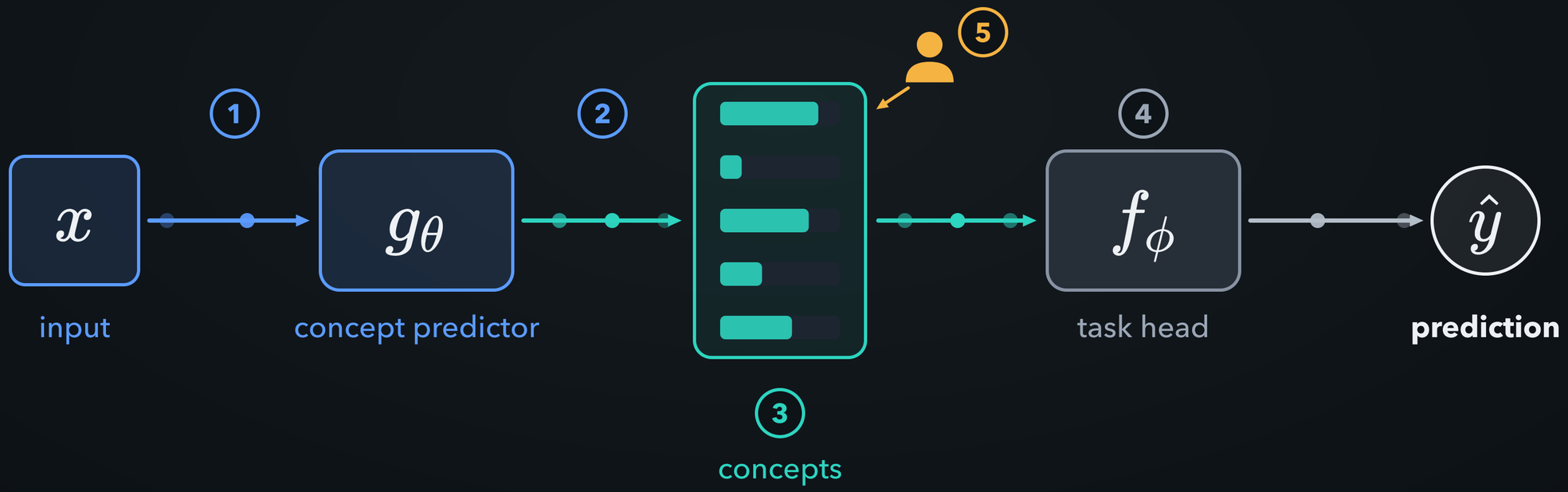
let the **decision** use them alone

$$\hat{y} = f_\phi(\hat{\mathbf{c}})$$

An interpretable interface is not the end of the story

x — g_θ — $\hat{c}$ — f_ϕ — $\hat{y}$

Follow the path from photo to concepts to answer: what can fail?

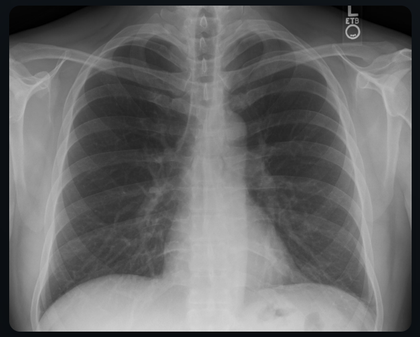


An interpretable interface is not the end of the story

1 · Encoder: distribution shift

new inputs unlike the training inputs

$$p_{\text{new}}(x) \neq p_{\text{train}}(x)$$



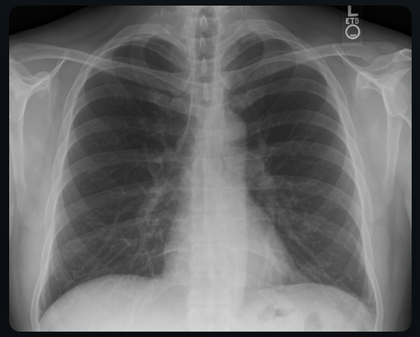
Chest X-ray: Stillwaterising, Wikimedia Commons, CC0

An interpretable interface is not the end of the story

A new hospital: the same model scores lower

new inputs unlike the training inputs

$$p_{\text{new}}(x) \neq p_{\text{train}}(x)$$



Chest X-ray: Stillwaterising, Wikimedia Commons, CC0

pneumonia detection, AUC (1 = perfect ranking, 0.5 = guessing)

same hospitals	0.931
new hospital	0.815

In practice: fine where it was built, worse at the new site

REPORTED · Zech et al., PLOS Medicine 2018 (Part 1)

An interpretable interface is not the end of the story

2 · Concept detector: wrong grounding

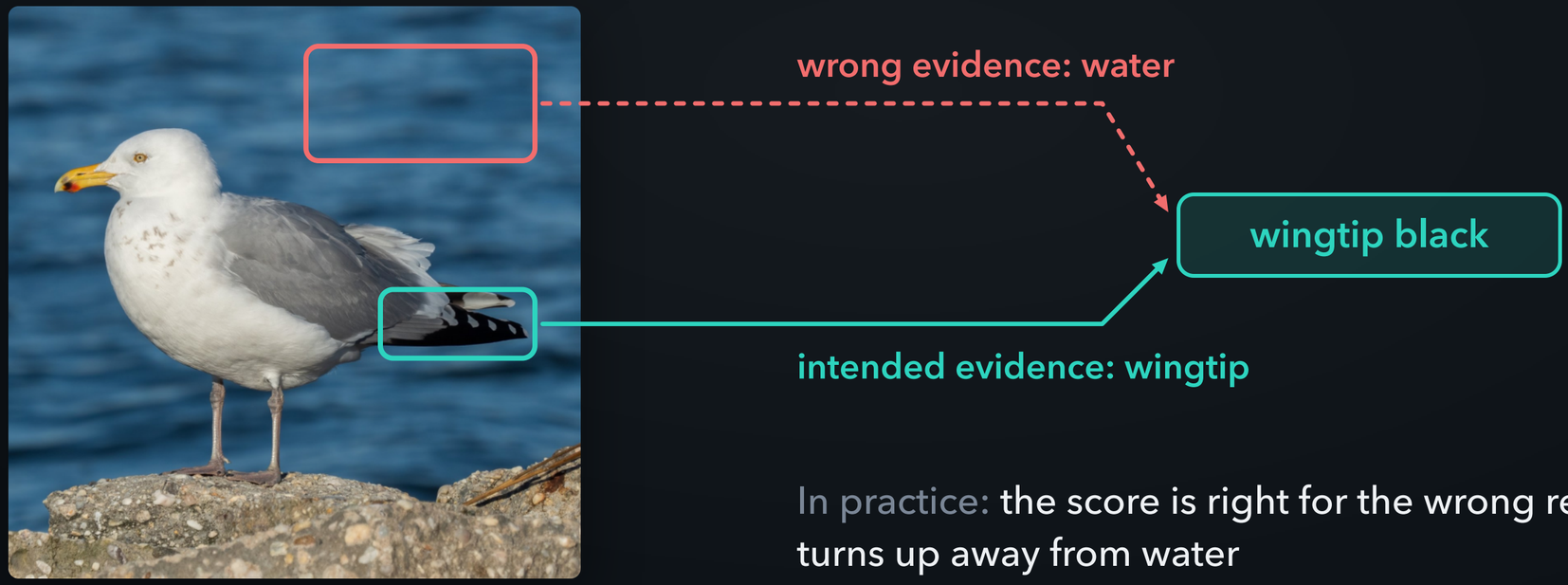


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

In practice: the score is right for the wrong reason, until a gull turns up away from water

SCHEMATIC · regions drawn by hand

An interpretable interface is not the end of the story

3 · Vocabulary: missing information

the same concept vector, different **classes**

$$\mathbf{c}(x_1) = \mathbf{c}(x_2), \quad y_1 \neq y_2$$

shape never reaches the head: best accuracy **50%**

In practice: two cases an expert tells apart look identical to the head

CONSTRUCTED EXAMPLE · the blue circle and square of Part 8

An interpretable interface is not the end of the story

4 · Task head: the wrong relationship

$$\text{score} = 1.2 - 2\,\text{duplicate charge} + 0.6\,\text{refund requested}$$

a correct edit meets a wrong rule: billing falls from 0.80 to 0.45

In practice: the expert's fix is right, yet the decision moves the wrong way

CONSTRUCTED EXAMPLE · head built faulty on purpose, Part 7

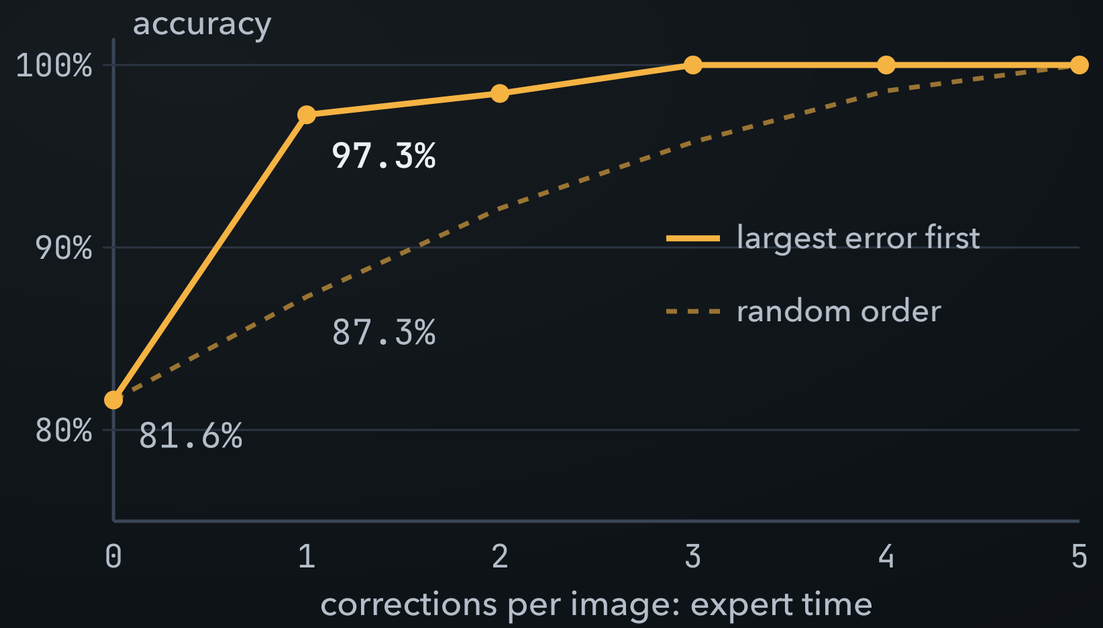
An interpretable interface is not the end of the story

5 · Intervention: wrong, or too costly

size: large 0.00

Part 11 took corrections from true labels:
an expert who never errs

In practice: real experts misjudge too, and every check costs their time



MEASURED · Part 11: one seed, 512 synthetic test images, oracle corrections

An interpretable interface is not the end of the story

Five questions for any concept bottleneck

1. ☐ Are the concepts meaningful?
2. ☐ Are they grounded correctly?
3. ☐ Are they sufficient for the task?
4. ☐ Does the model actually use them?
5. ☐ Do corrections improve decisions?

An interpretable interface is not the end of the story

A 2026 survey maps the field onto the same parts

SURVEY · TMLR 2026

What's in the Bottle? A Survey and Roadmap of Concept Bottleneck Models

P. Knab, D. Steinmann, C. Bartelt, K. Kersting, B. Schiele, T. Seidl, U. Schlegel, W. Stammer

input | concepts | output | training

open challenges include grounding, completeness, leakage and intervention reliability

Final replay

One last time: a black box maps the photo to a species

Glaucous-winged gull 0.71

American herring gull 0.29

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

ILLUSTRATIVE VALUES · hand-written concept values and head

Final replay

Open it: named concepts sit on the path to the prediction

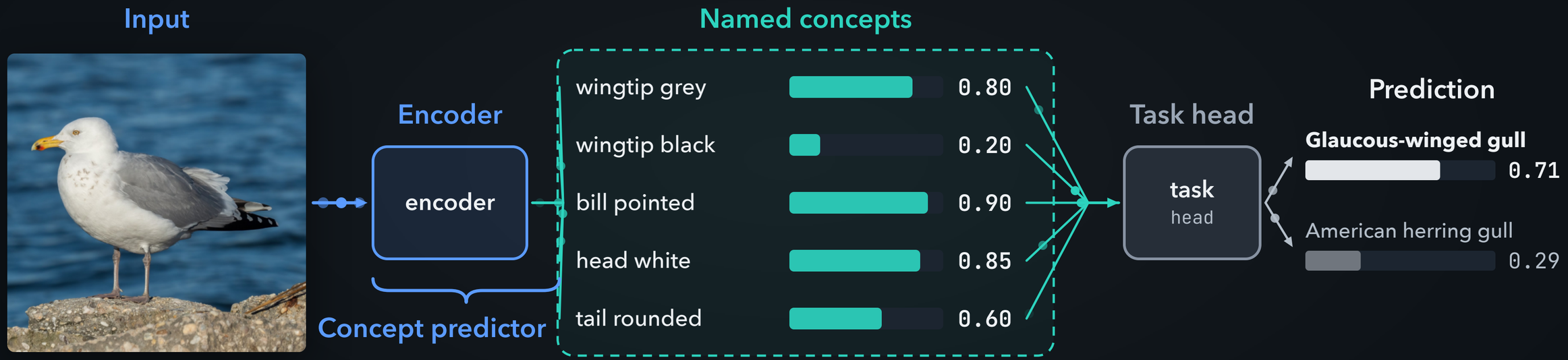


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

ILLUSTRATIVE VALUES · hand-written concept values and head

Final replay

A person corrects one concept, and the prediction follows

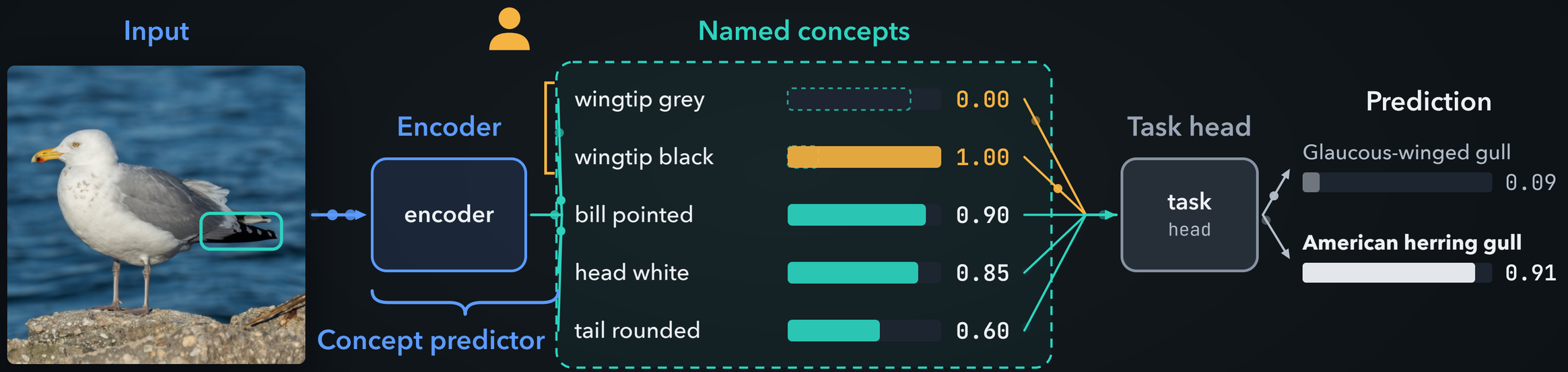


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

ILLUSTRATIVE VALUES · hand-written concept values and head

Final replay

The interface is only as good as its parts

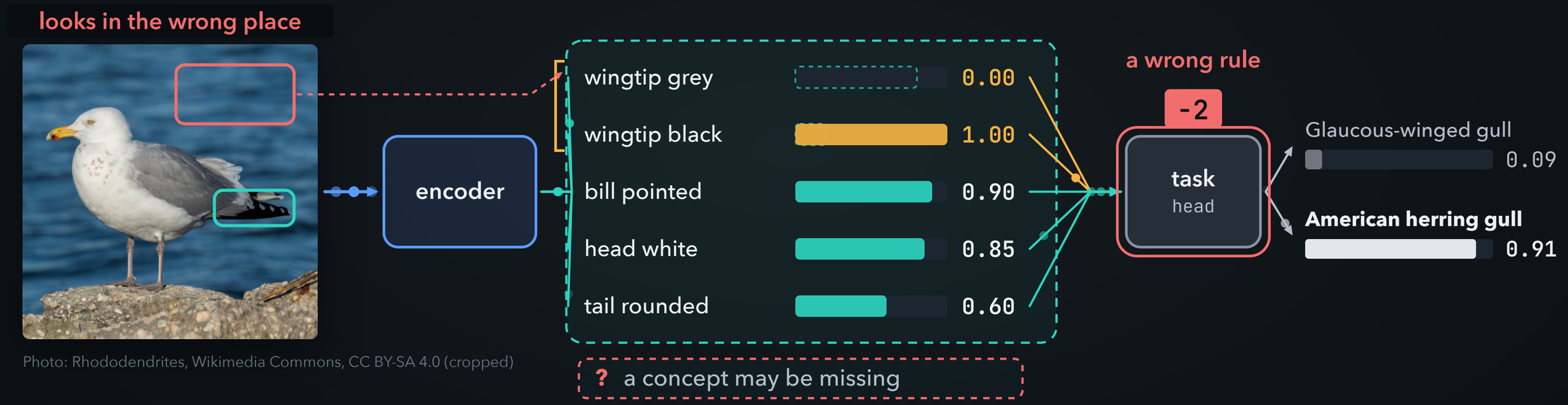


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

ILLUSTRATIVE VALUES · hand-written concept values and head

Final replay

Concepts that can be inspected, evaluated and corrected

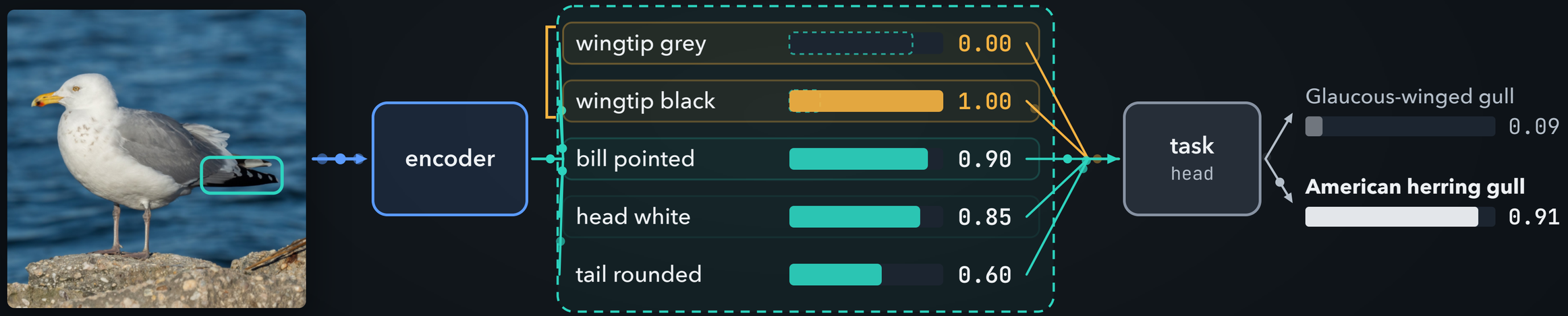


Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

inspected evaluated corrected

ILLUSTRATIVE VALUES · hand-written concept values and head

Final replay

The interface must carry what the prediction depends on

The challenge is to make sure the interface faithfully represents the information on which the prediction depends.

put simply: whatever the **answer** uses should be visible in the concepts

Final replay

What to remember

1. Post-hoc methods explain afterwards; a concept bottleneck makes the **answer** pass through named concepts
2. Fixing a concept helps only if the head reads it the right way
3. A soft score can carry more than its name says: extra information leaks
4. Newer variants need fewer labels, so check each concept looks at the right evidence

Appendix: testing with concept activation vectors

TCAV asks about a named concept: does the model use grey wings when it says herring gull?

Photo: Rhododendrites, Wikimedia Commons, CC BY-SA 4.0 (cropped)

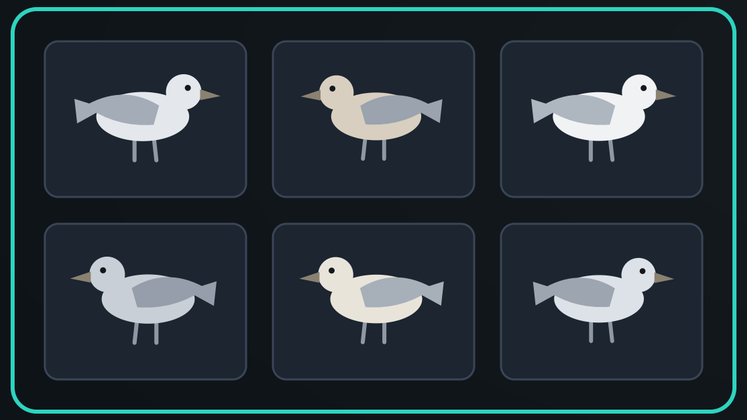
one more post-hoc method: it asks about concepts directly

TCAV: Kim et al., ICML 2018

Appendix: testing with concept activation vectors

Step 1: collect photos with grey wings and photos without

with grey wings



without

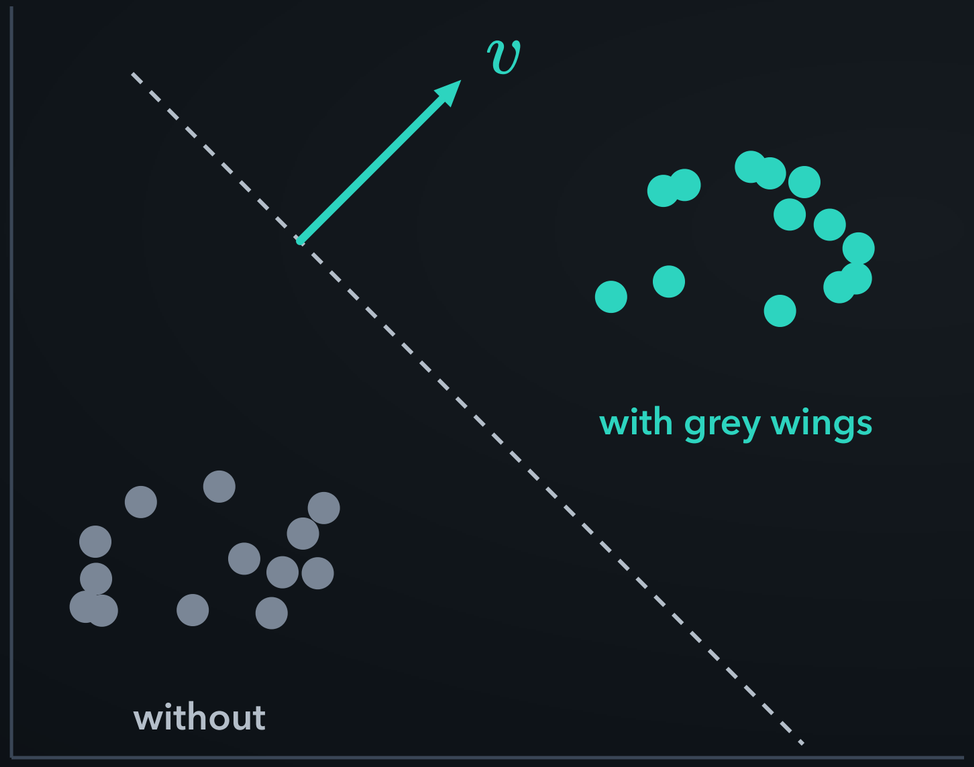
150 photos in each group

none is a **herring gull**

SCHEMATIC · drawings stand in for labelled bird photos (CUB, Part 2)

Appendix: testing with concept activation vectors

Step 2: the concept direction separates them inside the model



SCHEMATIC · 2 of 512 dimensions drawn

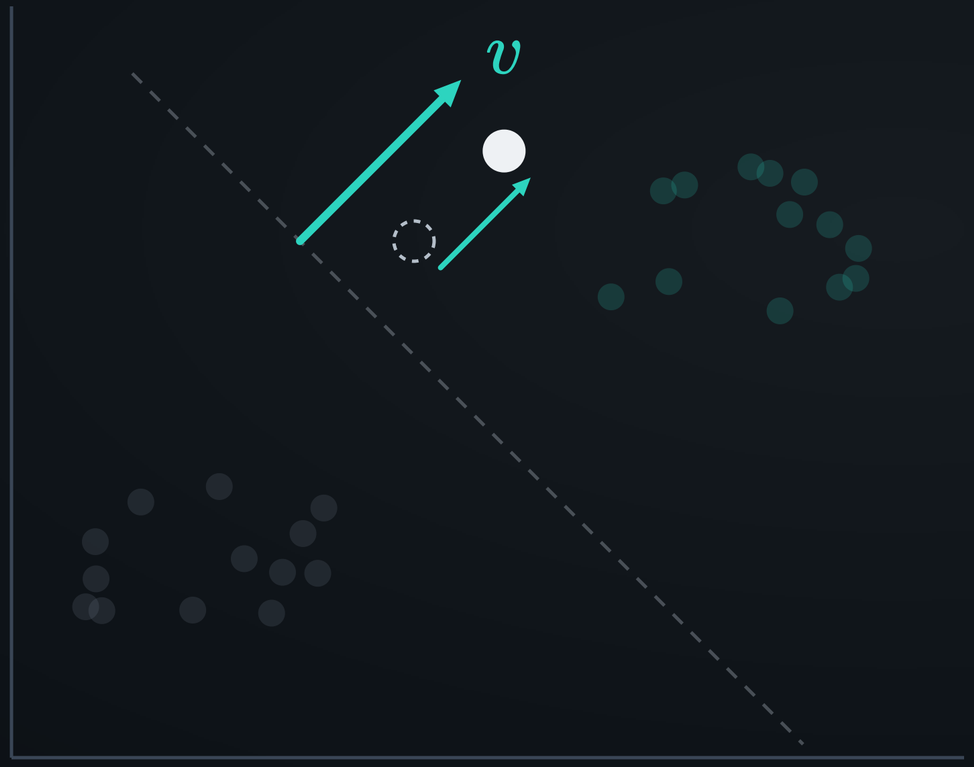
activations: the CLIP features

$$h(x) \in \mathbb{R}^{512}$$

v, the concept activation vector:
moving along it means more grey wing

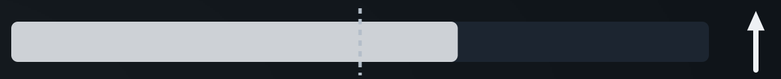
Appendix: testing with concept activation vectors

Step 3: nudge each herring gull photo along that direction



SCHEMATIC · one photo, 2 of 512 dimensions drawn

herring gull score



does the **score** go up?

$$\nabla_h \hat{y}_{\text{herring gull}} \cdot v > 0 \; ?$$

Appendix: testing with concept activation vectors

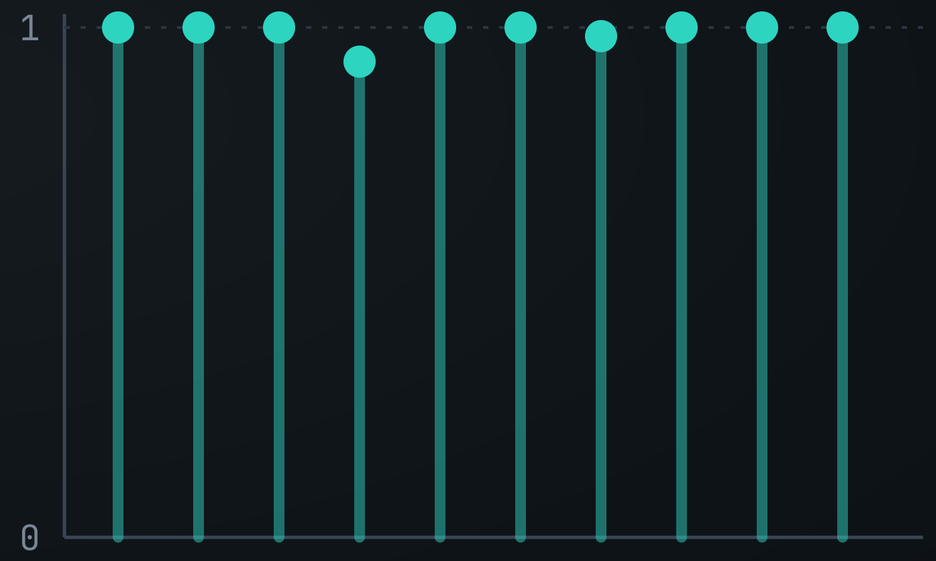
The TCAV score: the share of photos whose score rises

the share of the **60 herring gull photos** whose **score** goes up

$$\mathrm{TCAV} = \frac{\#\{\,\nabla_h \hat{y} \cdot v > 0\,\}}{60}$$

grey wings: between **0.93** and **1.00** in all 10 repeats

grey wings: one score per repeat

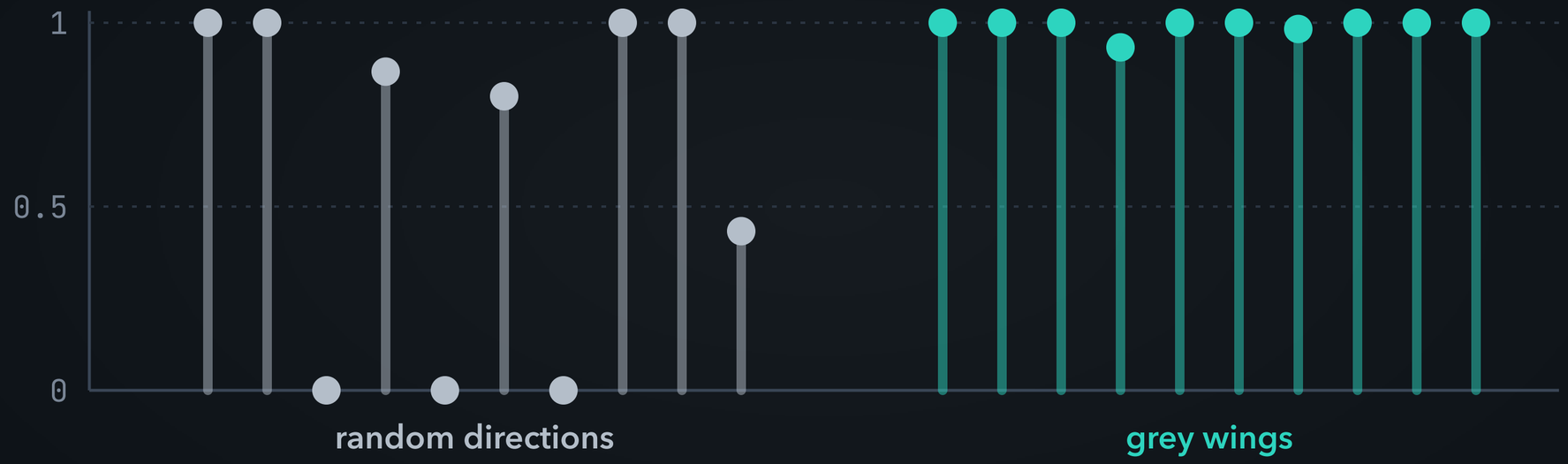


repeats 1 to 10, new example photos each time

COMPUTED · REAL MODEL · CLIP ViT-B/32 zero-shot herring gull score, CUB photos; Kim et al.'s method

Appendix: testing with concept activation vectors

Compare with random directions



a concept counts only if its score is steady and differs from random

it says what the model responds to, not how to fix it

COMPUTED · REAL MODEL · CLIP ViT-B/32 zero-shot herring gull score, CUB photos; Kim et al.'s method

Concept Bottleneck Models

Predicting through interpretable concepts.

Narration: synthetic voice (Gemini TTS). Photographs: Rhododendrites, Yathin S Krishnappa, Gzen92, Dietmar Rabich, Hans Hillewaert, Stillwaterising, and the authors of the 36 Part 10 photographs, via Wikimedia Commons (CC BY-SA, CC BY, CC0, public domain); knee X-ray from Koh et al. 2020 (CC BY 4.0). Satellite imagery: Sentinel-2, SentinelKilnDB. Data: CUB-200-2011. Models: CLIP ViT-B/32, ResNet-50; concept lists: Gemini.
Sources: Koh et al. 2020; Wah et al. 2011; Simonyan et al. 2014; Kim et al. 2018; Adebayo et al. 2018; Zech et al. 2018; Rudin 2019; Chen et al. 2019; Zarlenga et al. 2022; Yuksekgonul et al. 2023; Oikarinen et al. 2023; Yang et al. 2023; Srivastava et al. 2024; Menon and Vondrick 2023; Mahinpei et al. 2021; Havasi et al. 2022; Knab et al. 2026.